

Beyond Recall: Behavioral Specification as an Interpretive Layer for AI Personalization

Abstract

If an AI agent makes decisions on a person’s behalf, those decisions must align with its user. We introduce **representational accuracy** to measure how faithfully a system captures a person’s interpretation. An **interpretive layer** is operationalized as a **Behavioral Specification**. Our reference implementation aggressively compresses a person’s data into interpretive patterns, served as context to a language model. We evaluate the Specification on a prototype benchmark of held-out behavioral predictions scored by a calibrated 5-judge LLM panel. We test it independently and in composition with a range of context conditions: full raw corpus, full extracted facts, and four commercial memory systems (Mem0, Letta, Supermemory, Zep).

Across 14 public-domain autobiographical corpora, the Specification lifts representational accuracy in aggregate and nearly eliminates model hedging. It recovers most of what the raw corpus delivers, at $\sim 25\times$ less context cost. The Specification lifts subjects toward a common predictive level regardless of pretraining baseline; the lift in absolute points is therefore largest where the baseline is lowest, suggesting the population of relevance is anyone not adequately represented in pretraining. Lift is greatest on interpretation-required questions, where providing an interpretive layer enables model behavior that extracted facts or raw corpus do not. Conversely, on recall-required questions, this layer can interfere rather than help.

We conclude that representational accuracy is distinct from recall and that human-AI alignment is dependent on how accurately the user is represented. Representational accuracy makes that alignment testable.

Study repository: github.com/agulaya24/beyond-recall

For AI agents: github.com/agulaya24/beyond-recall/blob/master/AGENTS.md
(repository is fully indexed and vectorized for semantic search).

Pipeline source: github.com/agulaya24/BaseLayer

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Preprint · 2026-05-14 · Manuscript CC-BY-4.0 · Code Apache 2.0

1. Introduction

1.1 Recall is not interpretation. Interpretation can be measured.

AI is moving from a tool a person uses to an agent that acts on a person’s behalf, and that shift changes what “memory” must do for a specific individual. State of the art AI memory has been optimizing for recall as the success metric. The four prominent commercial systems (Zep, Letta, Mem0, and Supermemory) compete on standard recall benchmarks such as LOCOMO and LongMemEval, reporting accuracies in roughly the 70% to 93% range depending on provider, model, and benchmark variant (§2.2). Optimizing further on recall leaves something more fundamental unmeasured. This paper examines how recall is one part of memory, and how the function of memory is dictated by how an individual processes the facts and experiences of their life.

We use **interpretation** to refer to this human-side property: the way a specific person processes facts and experiences into judgments, decisions, and reactions. Viewing situations from different lenses can lead to entirely different interpretations of the same set of facts. This principle holds across domains: the same set of facts can produce different conclusions depending on the interpretive framework the reader brings to them. Memory is therefore personal in a deeper sense than recall: the same facts arrange differently inside different people. For an AI to serve a specific person, it must be given context on the framework that person uses to reason, not just the raw facts or information itself. Throughout this paper we use the term **Behavioral Specification** to refer to a static document that extracts and encodes a person’s behavioral patterns; the operational definition is developed across §3.7. A Behavioral Specification is an artifact that captures this interpretive framework, and is provided to an AI as context.

We introduce **representational accuracy** as the corresponding AI-side property: how well a system’s internal model of a specific person captures their interpretive patterns. It is not recall, preference matching, or persona consistency. It is a distinct property of the AI system, and state of the art memory benchmarks do not isolate it. Prior work closest to this axis (Twin-2K for scaled behavioral prediction, PersonaGym for persona fidelity, AlpsBench for preference alignment) measures related properties but not the transfer of a person’s interpretive patterns to new situations the system has never seen. §2.1 positions each benchmark against what this paper measures, and Appendix F develops the scope differences in detail.

The core hypothesis of this research is that representational accuracy of a person’s interpretation improves an AI system’s behavioral alignment with that person. This is the operational primitive for any AI system meant to act on a person’s behalf: the system’s behavior can only match the user’s reasoning to the extent the system represents that reasoning accurately. The operational test in this paper is behavioral prediction on held-out situations: given a situation drawn from text the model has never seen, the model generates how the subject would respond; the response is scored by a panel of calibrated large language model (LLM) judges against the subject’s own verbatim response in the held-out text on a 1-5 interpretive rubric (§3.3). Accurate prediction on held-out text is evidence that the representation captures the subject’s recurring patterns of reasoning, distinct from the facts and stylistic surface that current extraction pipelines already produce. The design also reduces the risk of sycophancy¹: the answer is checked against the person’s narrative, which the model has never seen, not against anything the user says during the conversation. The held-out test is one operationalization of the hypothesis.

¹Sycophancy: a model adjusting its answer to match what the user appears to want, often by agreeing with or flattering them. See Sharma et al. (2023), Perez et al. 2022, Jain et al. 2025 (§2.4).

We test this hypothesis on the leading state-of-the-art AI memory systems and on a diverse set of 14 autobiographies from authors across the world. For this initial examination we use baselined and calibrated LLM judges to evaluate the performance of each memory system, on its own and in combination with a **Behavioral Specification**: a static document that extracts and encodes a stable representation of a corpus’s behavioral patterns. The Specification captures the recurring patterns in how the subject reasons, drawn from the shape of judgments and reactions across the corpus (for example: “*spiritual integrity over social cost...*”, “*reform through love...*”, “*hierarchical deference...*”). A walked example of the audit chain from such a pattern back to its grounding facts and source passages appears in §2.3.

Defined terms used throughout the paper are collected in [Appendix H](#) for reference.

1.2 What we tested

We tested a Behavioral Specification across 14 historical subjects, each with a public domain autobiography sourced from Project Gutenberg or the Internet Archive (per-subject sources in §3.4 Table 3.1). For every subject we split the source corpus in half: the training half was used to generate the Specification, to seed each memory system, and to provide the retrievable fact pool. The held-out half was used only to produce behavioral prediction questions and was never shown to the **response model**, the language model being asked to predict how the subject would respond. The Behavioral Specification is one of the context conditions the response model receives; the full set of conditions is defined in §3.2. The set of held-out questions for each subject is the **question battery** (size and composition per subject in §3.5). The test was whether each system, under each tested condition, could predict how that specific person would respond in situations drawn from text it had never seen. The evaluation is a prototype benchmark for representational accuracy, not a finished one; §7 flags the work needed to harden it into a standardized instrument.

The Behavioral Specification itself is built from the training-half corpus through an extraction-and-authoring pipeline (§3.7). The pipeline distills the recurring patterns of how the subject reasons into a single structured document, typically around 7,000 tokens (~5,000 words) long. That document is what the response model receives as context when asked to predict how the subject would respond.

Hypotheses. The study tests five claims about how a representation of a person shapes AI behavior on that person’s behalf:

- **H1.** A response model given a Behavioral Specification produces responses that align with the person’s documented behavior more closely than the same model given no context, facts retrieved by a memory system, the full extracted fact list, or the raw source corpus (§4.1).
- **H2.** The Specification’s benefit is inversely proportional to the response model’s pretraining coverage of the person. Its effect is largest on people the model does not already know (§4.1).
- **H3.** The benefit comes from the content of the correct Specification for the correct person, not from the mere presence of a structured prompt. A random other person’s specification, applied in its place, produces a substantially smaller and content-specific effect than the matched Specification (§4.3).
- **H4.** The Specification interacts with memory-system retrieval in a structured way that depends on the type of question being asked. Aggregate effects on each memory system reflect the balance of these per-question patterns and shift with retrieval architecture (§4.4).
- **H5.** A Behavioral Specification’s quality advantage is also a compression advantage: a ~7,000-token (~5,000-word) Specification recovers most of the predictive accuracy of an 80-400K-token (~60-300K-word) raw corpus (§4.2).

Post-hoc analyses surfaced during the work are reported alongside these results.²

Each memory system is tested in both a controlled configuration (identical pre-extracted fact pool) and a native configuration (the provider’s own ingestion pipeline); design detail in §3.2. Running in parallel across both is the Behavioral Specification, tested alone and layered on top of each configuration. Every meaningful combination of inputs is evaluated as its own condition:

Group	Condition	Inputs given to the model	Purpose
Direct context manipulations	No context (C5)	Nothing. The model answers from pretraining alone.	Pretraining baseline. Measures what the model already knows about the subject from public sources.
	Specification alone (C2a)	The Behavioral Specification, with no retrieval, no facts, and no corpus.	Tests whether structure without retrieval is sufficient on its own.
	Wrong-specification control (C2c)	A different subject’s specification applied to this subject. Two variants: an adversarial fixed pairing (v1) and a random derangement (v2).	Tests whether the effect is driven by the content of the correct Specification, or by the mere presence of structured prompting.
	All facts, no specification (C4)	Every extracted fact for the subject, loaded into context at once.	Tests whether information sufficiency alone drives prediction, independent of structure.
	Facts + specification (C4a)	Every extracted fact plus the Specification.	Combines full information and structure to test the upper bound of context-provided prediction.
	Raw corpus, no specification (C8)	The full training-half corpus loaded into context.	Tests whether unstructured source text can substitute for an interpretive representation.
Memory-system configurations (controlled, all 5 systems)	Corpus + specification (C9)	Raw training corpus plus the Specification.	Tests whether structure is additive to unstructured source text.
	Retrieval alone, controlled (C1)	Top-k facts retrieved by each memory system (Mem0, Letta, Supermemory, Zep, Base Layer) from the shared fact pool.	Tests retrieval sufficiency, and whether providers converge on which facts are most relevant given identical input.

²Post-hoc analyses reported alongside H1–H5: retrieval-overlap divergence (§4.4.1, §4.6.6), Letta case study (§4.5; Appendix G), abstention-credit validity audit (§3.3.6), per-subject wrong-Spec heterogeneity (§4.6.5). Labeled exploratory where they appear; full breakdown in Appendix B.10.

Group	Condition	Inputs given to the model	Purpose
Memory-system configurations (native, 4 commercial systems)	Retrieval + specification, controlled (C3)	Memory system retrieval from the shared fact pool, plus the Specification.	Tests whether the Specification layers cleanly on retrieval when the input is held constant.
	Retrieval alone, native (C1 native)	Top-k results from each memory system’s own ingestion pipeline operating over the raw training corpus.	Real-world comparison of each memory system’s full ingestion-plus-retrieval stack.
	Retrieval + specification, native (C3 native)	Memory system’s own ingestion and retrieval, plus the Specification.	Tests whether the Specification improves the real-world deployment of each memory system.

The 14 subjects span four continents and roughly two millennia of written human experience. Ordered chronologically: Saint Augustine (North Africa, 4th-5th c.), Bābur (Central Asia and India, 15th-16th c.), Bernal Díaz del Castillo (Spain and Mexico, 15th-16th c.), Benvenuto Cellini (Italy, 16th c.), Jean-Jacques Rousseau (France, 18th c.), Olaudah Equiano (West Africa and Britain, 18th c.), Mary Seacole (Jamaica and Britain, 19th c.), Elizabeth Keckley (United States, 19th c.), Yung Wing (China and the United States, 19th c.), Philip Gilbert Hamerton (Britain, 19th c.), Fukuzawa Yukichi (Japan, 19th c.), Georg Ebers (Germany, 19th c.), Sunity Deveen (India, late 19th-early 20th c.), and Zitkala-Ša (Yankton Dakota, early 20th c.). Source corpora range from 25,231 words (Hamerton) to 422,772 words (Bābur). Full source references are in §3.4.

Predictions were scored on a 1-5 rubric where the integer anchors mark categorical shifts in answer quality; the full rubric and verbatim judge prompt are in §3.3. For example, a move from 1.8 to 2.4 crosses the 2.0 boundary: the model goes from refusing the question or producing an off-base answer (anchor 1) to engaging with the question even when the prediction is still wrong (anchor 2). Absolute point gains, not percentages, are the informative metric for cross-subject comparison.

Score	Anchor (verbatim)	Shift from previous anchor
1	Refuses or off-base	(rubric floor) No usable answer: the model either declined (93% of score-1 responses) or engaged with the wrong subject (7%)
2	Wrong prediction	From no usable answer to engaging with the right subject
3	Right domain wrong outcome	From wrong prediction to the right domain
4	General direction correct	From right domain to the general direction of the held-out
5	Predicts specific outcome	From general direction to the specific outcome documented in the held-out

Score interpretation, including the cross-anchor rule for fractional scores (e.g., 2.5, 3.4), is in §3.3.1. Example questions per subject and panel composition are in §3.3.2.

The **baseline** we refer to throughout is the No-Context Baseline (C5): the response model’s score with no external information. **Low-baseline** subjects are the **population of relevance**: people the model has insignificant pretraining understanding of, even when fragments of their digital footprint exist in training data. **High-baseline** subjects are the opposite, people the model already knows about from pretraining (e.g., Benjamin Franklin, included in this study as a known-figure reference, §4.1.2). Of the 14 main-study subjects, 9 are low-baseline ($C5 \leq 2.0$) and 5 are mid-baseline ($2.0 < C5 \leq 3.0$); Franklin is the single high-baseline reference ($C5 > 3.0$) and is not part of the 14. The low-baseline band is plausibly the default case for most users of frontier systems: even people with substantial public output captured in training corpora have only fragments of their reasoning represented. The study suggests the low-baseline band is the norm rather than the exception.³ Results are reported separately on the low-baseline band (n=9) alongside the full 14-subject analysis.

Together these hypotheses test whether a Behavioral Specification can increase a language model’s representational accuracy of a specific person.

1.3 What we found

An interpretive layer, operationalized as a Behavioral Specification (Spec), lifts representational accuracy. Its benefit is largest where the model knows the person least, and the mechanism is per-question. On questions where the model needs an interpretive frame and lacks one, a Behavioral Specification categorically improves the answer produced. On questions where the model already has the answer, a Behavioral Specification adds nothing and sometimes hurts.⁴ What follows are seven findings, beginning with the cross-subject gradient (primary outcome) and the per-question mechanism beneath it. Prediction on held-out reasoning is what we measure; whether higher representational accuracy translates into aligned action is the downstream claim developed in §5 and §7.

Headline findings.

- **Gradient. A Behavioral Specification’s benefit is largest where the model knows the person least.** Every one of the 9 low-baseline subjects improved when the Behavioral Specification was added on top of All Facts (C4a); per-subject mean lift **+0.89** points on the 1-5 rubric over the No-Context Baseline (n=9), with **78.6%** of individual questions improving (351 paired questions).^{5,6} The gradient comes from the per-question mechanism described next: low-baseline subjects have more questions where the model lacks an interpretive frame, and therefore more questions where the Behavioral Specification lifts. Detail in §4.1.
- **Per-question interpretive lift. The Behavioral Specification moves 55% of low-baseline questions across at least one rubric anchor upward; 18% cross two or more.** Crossing one rubric anchor moves a response from “wrong prediction” to “general direction correct.” Crossing two or more anchors is a bigger jump: a single question where the model moves from refusal or generic guessing to a recognizable, person-specific response.

³Operational thresholds: low-baseline $C5 \leq 2.0$ on the 1-5 rubric, high-baseline $C5 > 3.0$. Full distribution and band assignments in §3.4.1.

⁴“Low baseline” means $C5 \leq 2.0$ on the 1-5 rubric; §1.4 and §5.2 develop why this band is the population of importance for AI personalization.

⁵Wilcoxon signed-rank $W = 11$, $N = 14$, $p = 0.007$. The **+0.89** is the cross-subject mean of per-subject Δ_C4a (locked aggregation rule, §1.2); the grand-mean alternative is +0.93. Full regression, the leveler-framing of the gradient, and the aggregation reconciliation are in §4.1 and Appendix B.9.

⁶The +0.89 mean lift is conservative against the §4.6.7 rubric-handling audit: residual refusal-anchor ambiguity pulls baseline scores upward more than post-Spec scores, so the true Δ_C4a is likely modestly larger than +0.89.

5.7% cross three or more anchors (20 of 351 paired low-baseline questions). The pattern holds across Spec Only (C2a), All Facts + Spec (C4a), and Corpus + Spec (C9) conditions on the low-baseline subjects. Detail in §4.1, §4.2.

- **Compression. The Behavioral Specification recovers 75% of what the raw corpus delivers at $\sim 25\times$ less context on average (per-subject range $7\times$ to $79\times$).** A 7,000-token structured representation matches most of the predictive accuracy of a 163,000-token raw corpus. A Behavioral Specification selects and structures the behavioral signal; the interpretive layer drives the result, not the volume of context. Spec Only +0.68 vs. raw corpus +0.91 over baseline.⁷ On Hamerton (smallest corpus tested), the Behavioral Specification scores higher than the raw corpus (2.63 vs. 2.27). Detail in §4.2.
- **Content specificity. A wrong Behavioral Specification drops accuracy below the No-Context Baseline ($\Delta = -0.25$); the correct Behavioral Specification lifts accuracy above ($\Delta = +0.35$).** What produces the lift is the content of the correct Behavioral Specification for the correct person, not the presence of a structured prompt. Random pairings (a wrong Spec assigned by chance to a different subject) sometimes still produce predictions that align with the held-out text, suggesting some behavioral patterns transfer across subjects, but the correct Behavioral Specification consistently outperforms. Random-derangement $\Delta = +0.15$. Detail in §4.3.
- **Memory-system layering. A Behavioral Specification layers cleanly on top of commercial memory systems: it moves 20–36% of individual questions up across a rubric anchor, excelling on interpretation-heavy questions and reducing refusals on questions where retrieved facts could not ground the model; conversely, a Specification negatively affects performance on recall-heavy questions where retrieval already supplied the answer.** This per-question structure is what the aggregate hides: some questions improve, some regress, and the balance shifts with retrieval architecture. On aggregate, the Behavioral Specification lifts 3 of 4 commercial memory systems; Mem0, Letta, and Zep show positive mean lift under at least one configuration, Suprememory does not. Detail in §4.4.
- **Hedging reduction. The Behavioral Specification collapses baseline hedging from 41.2% of responses to 0.4%.⁸** The reduction is content-specific, not prompt-driven: under the wrong-Spec adversarial control, the model continues to hedge or explicitly flag the mismatch on 60.6% of responses (§4.3). The pattern is consistent with the model carrying an implicit *evidentiary bar* before committing to a behavioral prediction: where the matched Behavioral Specification supplies the interpretive scaffolding that clears the bar (combined with All Facts (C4) or retrieval), the model commits; where it does not, the model abstains. On low-baseline subjects, where the model would otherwise refuse to engage, the matched Behavioral Specification converts refusal into substantive response. This is the gradient operating at its floor. Detail in §4.1.1 (abstention versus non-abstention misalignment decomposition), §4.3 (wrong-Spec content-specificity), and §4.4.3 (evidentiary-bar pattern across memory systems).
- **Retrieval divergence. Given identical input, memory-system providers share zero top-10 facts on 35.9% of question-pairs; mean pairwise overlap is 8.3%.** On standard recall benchmarks like LongMemEval and LOCOMO, the four commercial memory systems

⁷The +0.68 and +0.91 figures are 9-row low-baseline-band per-subject-mean Δ s. Earlier drafts reported +0.71 and +0.93 on an asymmetric 8-row baseline; v12.1 standardizes to the symmetric 9-row computation. Full recompute, both grains, and the Bābur C9 carve-out are in Appendix B.9.

⁸Headline number uses the broad rule (any refusal pattern anywhere in the response) under the All Facts + Spec (C4a) condition. The stricter rule (refusal pattern as the first non-whitespace text) gives 28.8% $\rightarrow$ 0.0% on the same condition.

we tested perform within a few percentage points of each other. Yet on which facts to surface as most relevant, they substantially diverge. Convergence on top-K under identical input would have been evidence of a shared interpretive framework; the systems do not converge. On 65.6% of (system pair, question) instances they share one or fewer facts. Detail in §4.4.1.

Mechanism: three patterns of interaction with retrieval (full development in §4.4.3). Baseline runs suggest the model already attempts shallow inference from a user’s raw data on its own; the Specification makes that inference inspectable and structured.

- **Pattern 1, Interpretation-heavy questions.** The Specification supplies a generalized pattern from the source that has to transfer to a new situation; retrieved facts alone are not enough (Fukuzawa Q26).
- **Pattern 2, Literal-recall questions.** Retrieval already returns the plain answer; the Specification’s interpretive framing drifts past the question and negatively impacts the response (Yung Wing Q5).
- **Pattern 3, Refusal-triggering questions.** When the Spec supports refusing without enough information (not all specs do), the model produces principled refusals aligned with the Spec; the content-match rubric still scores them as off-base (Zitkala-Ša Q18).

Robustness across providers. We varied both the question-battery generation model and the response model across providers; the Spec direction reproduces. Detail in §4.6.1.

Exploratory note: Letta stateful-agent path. Letta’s stateful-agent architecture self-edits a persistent memory block during ingestion. Given that unique architecture, we ran an exploratory case study on it: on 3 subjects (post-hoc), it scored above Base Layer’s full-stack Behavioral Specification at matched response model. The full case study, the duplication audit, and the architectural-ceiling analysis are in §4.5 and Appendix G.

1.4 What this implies

Representational accuracy is distinct from recall, and human-AI alignment is dependent on how accurately the user is represented. As AI agents make more decisions on a person’s behalf, the infrastructure required to serve that alignment is user-held, portable, inspectable, traceable, and representation-grade. A Behavioral Specification is one implementation of this infrastructure, not the only one.

The gap it fills cannot be closed by training a larger model on more public data. The private record does not exist in a form a training corpus can capture; even where fragments exist, they are scattered across formats and channels and cannot be reliably reassembled into how a specific person reasons. AI is becoming a broadly used technology, comparable to email or mobile phones in reach, but different in the amount of cognition it carries out on a person’s behalf. The population of relevance (§1.2) is anyone who uses or will use an AI system. Even the subjects of the autobiographies in this study, people whose work is in pretraining and who should technically be known to the model, score near the rubric floor in the No-Context Baseline.

The structural options for what fills the gap are narrow:

- Each person supplies their own representation to whatever AI system serves them. A Behavioral Specification is one such implementation.
- Personalization remains surface-level (style, voice, preference, demographic inference, observable behavior), addressing the layer current memory systems already cover but missing the

interpretive framework that lets an agent act on a specific person’s behalf.

- AI systems infer a representation of the user from observed interactions, building it opaquely, without explicit input from the user or the ability for the user to inspect or correct it.

§5 is an extended discussion of these implications; §7 develops the safety, alignment, and deployment implications.

2. Prior Work, Industry Benchmarks, The Fifth Target

AI memory and personalization research today is organized around four measurement targets: recall of stored facts, survey-response prediction, persona fidelity, and preference alignment. Each is supported by its own benchmark family and its own line of system design. They do not measure whether an AI system has an accurate internal model of how a specific person reasons. This paper proposes a fifth target, **representational accuracy**, and uses behavioral prediction on held-out reasoning situations as its operational test. §2 walks the four existing targets, names the benchmarks attached to each, and positions the fifth alongside them.

Memory systems today optimize for recall. Recall-optimized efforts include both **neural-memory-analogue systems**⁹ and the broader class of vector-retrieval and embeddings-based commercial memory providers (Mem0, Zep, Supermemory, Letta). These systems do store and retrieve information for a specific user, but they are designed and benchmarked for recall accuracy on standard benchmarks, not how accurately the system represents that user’s reasoning. The optimization target is general by construction; any individual user’s interpretation is not what these systems are measured against. A separate body of research, **cognitive-representation research**, studies human reasoning itself: how people form representations of others, how schemas compress experience. The gap between these directions is the translation: applying what we know about human reasoning to the direct interaction between an AI system and a specific individual, and shaping the system’s internal model of that individual in a way that serves them rather than serving an average.

Language models are trained to produce responses that are helpful on average across a large population of users. That optimization target produces outputs that no single user is the reference point for. Personalization requires the opposite property: a system whose outputs are tuned to a specific individual rather than to a population aggregate. That kind of intentional individual-specificity, not “bias” in the negative sense but an explicit design target, is the missing thread in current AI memory and human-AI interaction research.

Personalization in this paper’s sense. “Personalization” in current AI research typically means responsiveness to stated preferences (dietary restrictions, communication style) or stored facts about the user (location, occupation, history). Both are useful and both live at the surface of the user. We use “personalization” in a stronger sense throughout this paper. We mean representing the interpretive layer that sits beneath stated preferences and biographical facts: how a specific person organizes experience, what they treat as evidence, what reasoning patterns they apply across new situations. Preferences and facts are downstream artifacts of that interpretive layer; the layer itself is what produces them. The behavioral prediction battery and Behavioral Specification described in §3 instantiate personalization in this deeper sense, and §5 returns to what this layer is and is not.

⁹Architectures that borrow from human memory engineering: episodic consolidation, working-memory slots, retrieval over embeddings.

2.1 Prior measurement targets and the gap representational accuracy fills

This subsection walks each of the four existing targets, naming their attached benchmarks and scopes. Representational accuracy is positioned as the fifth target at the end of the walk. An extended benchmark-by-benchmark analysis is in Appendix F.

Recall measures retrievability of facts, not reasoning about them. LOCOMO (Maharana et al., 2024) measures conversational-memory quality: after a multi-session conversation, the system is asked questions like “what did the user say about their job on day 3?” and scored on fact retrieval. LongMemEval (Wu et al., 2025) measures long-term memory across multiple sessions on five capability dimensions (single-session, multi-session reasoning, temporal reasoning, knowledge updates, abstention) and is heavily recall-weighted. A system can saturate recall on such benchmarks and still fail behavioral prediction, because retrieval answers the question “can the fact be found” rather than “does the system know how the person reasons about the fact.” Recall is a necessary property for most downstream uses of memory but it is not sufficient for representational accuracy.

Survey-response prediction infers how a person would answer one questionnaire item from how they answered others. Twin-2K (Toubia et al., 2025) does this for 2,058 participants on a 17-task heuristics-and-biases battery; items share a common format (multiple choice, Likert scale, numeric), scored by distance-based accuracy. Twin-2K’s stated target is *prediction accuracy on survey interpolation*: the model is scored on how well it predicts a held-out questionnaire response, not on whether it represents the underlying reasoning that produced the response. Our target is representational accuracy on a cross-format task: autobiographical prose input, open-ended behavioral prediction output, rubric-based scoring against a verbatim held-out passage. The structured-questionnaire format and the open-ended behavioral reasoning this paper studies measure different properties. A system could perform well on Twin-2K and not on our battery (survey interpolation does not require modeling reasoning transfer to new contexts), and a system could perform well on our battery and not on Twin-2K (accurate reasoning representation does not guarantee survey-format numerical accuracy). The two benchmarks diagnose different properties of the same general capability.

Persona fidelity measures whether a model stays in character across the back-and-forth of a conversation.¹⁰ PersonaGym (Samuel et al., 2025) scores consistency with a described persona during conversation: given a one-line persona (“You are a 45-year-old skeptical accountant from Toronto”), the model is scored on whether its multi-turn replies stay in-character, graded against a held-out criterion set. In practice the model is checked for consistency with the persona’s surface attributes (skeptical, accountant-flavored responses; not breaking character into a different age or profession), not for whether it reproduces how a specific person would reason. PersonaGym’s one-line descriptor is a substantially shallower input than this paper’s ~7,000-token Behavioral Specification or Twin-2K’s full-text survey persona;¹¹ consistency with it does not require modeling that person’s reasoning on new situations. PersonaGym measures a useful property (holding voice over a conversation); fidelity to a one-line persona is a weaker condition than representational accuracy.

Preference alignment measures whether responses match user preferences. AlpsBench

¹⁰A “turn” is one round of conversation, a single exchange of one user message and one model reply. Persona-fidelity benchmarks score whether the model stays in character across many such exchanges in sequence.

¹¹Twin-2K’s full `persona_text` runs ~32,000 tokens; the `persona_summary` runs ~3,750 tokens. Both are substantially deeper than PersonaGym’s one-line descriptor. Full breakdown of persona-input depth across benchmarks in Appendix F.

(Xiao et al., 2026) evaluates whether explicit memory mechanisms improve preference-aligned and emotionally resonant responses: after ingesting a user profile, the model is asked conversational questions (preferences, emotional support) and responses are scored on preference alignment and emotional resonance rubrics, not on predictive accuracy. Their central finding, *that recall improvement does not automatically carry into preference alignment*, is arrived at independently and is complementary to this paper. Both papers point at the same gap from different sides: solving for recall is insufficient for what memory is ultimately for. Preference alignment is an outcome property (whether a response matches what the user prefers). Representational accuracy is an upstream property (whether the AI’s internal model of the user is correct). Preference alignment is one downstream consequence of representational accuracy being correct; it is not the same property.

We propose behavioral prediction on held-out reasoning situations as a test of a fifth target: representational accuracy.

Prediction is the test, not the goal. We do not pursue prediction accuracy as an end in itself. The target is representational accuracy, the fidelity of an AI’s internal model of a specific person, and behavioral prediction on unseen situations is the instrument we use to measure it. A prediction score tells us the representation captured something that generalizes to new situations; a low score tells us it did not. Prediction is a diagnostic; the Behavioral Specification is what this paper is testing.

The held-out design rests on a stability premise. A person’s interpretive patterns must be stable enough within their own corpus that what is captured from one half references what appears in the other. Without that, held-out behavioral prediction is impossible in principle, regardless of how good the representation is. The 14 main-study subjects have coherent autobiographical narratives consistent with the premise; §4.1 reports that the Behavioral Specification authored from training text generalizes to held-out text at above-baseline rates. Subjects whose reasoning shifts substantially across their corpus (across a major career change, a profound life event, or a decades-long corpus with distinct epochs) may not be well-represented by a single snapshot specification, which is one reason temporality is a flagged follow-up in §7.

The missing axis is representational accuracy itself. Each existing benchmark family measures a real property of memory systems, and each is useful for its own target. What is missing is an axis that measures how accurately the memory system represents the person whose behavior it is meant to anticipate. This paper’s approach is a prototype answer on that axis, not a finished benchmark. §7 flags a differentiated rubric (one that separates interpretation-heavy from literal-recall questions, and scores epistemic honesty as its own dimension) as the priority follow-up for turning this prototype into a standardized benchmark.

A single number does not capture a memory system’s full capability. Recall, survey-response prediction, persona fidelity, preference alignment, and representational accuracy are distinct axes. A system that saturates one may do nothing on another. Production-grade evaluation of memory systems should report results on multiple axes rather than on any single one.

2.2 Memory systems for LLM agents

The four commercial memory systems we evaluate (Mem0, Letta, Supermemory, Zep) have converged on a shared set of capabilities: semantic retrieval over embedded content, source attribution, multi-level memory structures, and benchmark-validated recall performance. They differ in how each of these is architected. None positions representational accuracy or behavioral prediction of a specific

individual as a design target.

The four systems, in brief. Each is described below by core architecture, retrieval method, memory types stored, and published recall benchmark. All claims are verified against the primary sources.

Mem0 runs an extract-then-consolidate-then-retrieve pipeline; the Mem0g variant adds a directed labeled knowledge graph alongside the vector store. Retrieval is hybrid (semantic + keyword + entity). Memory types span conversation, session, user, and organizational scopes. Vendor-reported recall is 91.6 on LOCOMO and 93.4 on LongMemEval with the current algorithm.¹²

Letta / MemGPT is an LLM-as-operating-system: a main context window plus external context the agent reads from and writes to. Retrieval is twofold: archival lookups via `archival_memory_search`, and main-context memory blocks the agent self-edits via `core_memory_append` and `core_memory_replace`. Memory types are **persona** and **human** blocks in main context plus archival and recall memory held externally. Recall is 74.0% on LOCOMO with GPT-4o-mini.¹³

Supermemory has a five-component architecture: chunk-based ingestion, relational versioning, temporal grounding, hybrid search, and session-based ingestion. Retrieval is hybrid with reranking and query rewriting; source chunks are injected at retrieval time. Memory types span contextual memories, relational versions, and session data. Self-reported recall is 81.6% / 84.6% / 85.2% on LongMemEval_s with GPT-4o / GPT-5 / Gemini-3-Pro.

Zep is built on Graphiti (Apache 2.0, open source) and uses a bi-temporal knowledge graph. Retrieval is hybrid: semantic + BM25 + graph traversal. Memory types are episodes (the ground-truth source), entities, and facts-as-triplets with temporal validity windows. Recall is 71.2% on LongMemEval with GPT-4o.¹⁴

All four systems report recall scores in the 70-93% range; on the standard recall benchmarks, scores are localizing in the >80% range and recall is no longer as much a frontier issue as it used to be.¹⁵ All four are sophisticated systems that solve real problems in memory management. They optimize for storing, organizing, and retrieving what a person said or did.

Of the four systems, Letta (Packer et al., 2023) is architecturally distinct: it is the only one whose core architecture treats memory as something an agent *synthesizes* during conversation rather than *stores* for later retrieval.¹⁶ This stateful-agent design is examined separately as a post-hoc case study in §4.5 (full case study in Appendix G), distinct from the archival-retrieval path Letta exposes for the main-study conditions. A Behavioral Specification targets the interpretive layer that sits above retrieval, which three of the four (Mem0, Supermemory, Zep) do not model at all, and which the fourth (Letta) models implicitly through agent-initiated memory editing that our main-study configuration did not exercise (see §4.3 and §4.5).

¹²Vendor-reported; evaluation harness open-sourced at github.com/mem0ai/memory-benchmarks. The peer-reviewable paper (Chhikara et al., 2025) reports 68.44 LOCOMO for the Mem0g variant with GPT-4o-mini.

¹³Letta blog, 2025-08-12 (<https://www.letta.com/blog/benchmarking-ai-agent-memory>).

¹⁴Rasmussen et al. (Rasmussen et al., 2025).

¹⁵The vendor-reported recall scores in this table are contested across providers and third-party reproductions; methodology varies significantly between evaluators. This paper does not adjudicate vendor scores; we measure on a different axis (§4.4).

¹⁶Letta’s main context holds structured memory blocks the agent edits during its inference loop; external context includes archival and recall memory. Full architectural detail in Appendix G. The other three systems (Mem0, Supermemory, Zep) follow extract-and-retrieve patterns characterized in Table 2.1.

2.3 Traceability and reasoning traces

Traceability operates at two levels. **Fact-level traceability** answers where a retrieved claim came from. **Reasoning-level traceability** answers why the system believes this about this person. The four memory systems we evaluate provide the first; representing how a person reasons requires the second. Representational accuracy operationalizes interpretation, and interpretation cannot be verified at the fact level alone. A system that represents how a person reasons must be auditable by that person, or the representation is a black box they cannot verify.

Zep has the strongest explicit fact-level provenance of the four: every entity and relationship traces back to the episode IDs that produced it. Supermemory returns source chunks alongside retrieved memories. Mem0 tracks ingestion provenance through timestamps. Letta exposes agent state and memory-block edit history rather than fact-level provenance.

The Behavioral Specification, as implemented in this study’s reference pipeline, is structured so that every claim it expresses is a piece of reasoning, not just a piece of content. An interpretive pattern is grounded in the facts that imply it, and each fact is grounded in the source passage it was extracted from. Walking this chain backward gives a user a reasoning trace: not only where a belief originated, but what line of reasoning connects the source text to the interpretive claim.

A reasoning trace operates in four steps:

1. The user reads the system’s output (a prediction, an analysis, a stated pattern).
2. The user identifies the interpretive claim or phrase they want to audit.
3. They walk from that claim to the **pattern statement** in the system’s representation of the person that licensed it.
4. They walk from the pattern statement to the **facts** that ground it, and from each fact to the **source passage** it was extracted from.

The chain is bidirectional. From the response, the user walks down to the source. From the source, they walk back up to the response. If a fact misrepresents its source, correcting the fact propagates upward: the pattern statement that depended on it changes, and the response that depended on the pattern changes the next time the system is queried. A worked example walking the chain on a single Sunity Devee question is in Appendix D.6.

The four commercial memory systems can answer “where is this fact stored?” but cannot answer “why does the system believe this about this person?” — because their architectures treat facts, not interpretations, as the unit of storage. A representation that acts on a person’s behalf must be auditable at the interpretation level by that person, or the representation is a black box. Reasoning-level traceability is what makes that audibility operational.

A person should be able to inspect the system’s model of them, challenge any step in the reasoning, and correct it if it is wrong. A fact-attribution memory system lets the person audit what the system stores. A reasoning-trace specification lets the person audit what the system believes. The first is a feature. The second is the minimum bar for a representation that acts on someone’s behalf.

2.4 Cognitive and representational foundations

Six prior research directions shaped how we designed this paper’s test. Each motivates a specific choice about what to measure, what to compare against, or what failure mode to expect.

Bartlett (1932) established that human memory is reconstructive and schema-driven rather than

literal playback. Reconstruction follows the organizing structures a person has built up over time, not a record of the original event. A Behavioral Specification is computationally analogous: a structured compression meant to carry the signal of a person’s reasoning without storing every fact about them. We designed the Specification with a schema-like architecture (anchors, core, predictions) precisely so we could test whether it does the work a human schema does: enable accurate anticipation of behavior in situations never encountered in the source data. Our 50/50 train/held-out split is the experimental realization of this question.

Hinton et al. (2015) showed that compressing a large neural network into a smaller one preserves “dark knowledge,” the relationships between outputs that carry more information than the outputs themselves. This result motivates one of our central experimental comparisons: on matched token budgets, does a compressed interpretive artifact carry more predictive signal than the raw content it was derived from? The Hamerton condition in §4.2 (4,500-token Spec vs. 33,000-token training corpus at 2.63 vs. 2.27 on the 5-judge primary panel) is a direct test of that question in the personal-representation setting.

Chen et al. (2025) show that the character a model takes on (its “persona”) is encoded in specific directions inside the model’s internal numeric state (*persona vectors*), and that those directions can be identified, monitored, and nudged to shift the model’s behavior in predictable ways. Their approach modifies the model; ours informs the model from outside via context. Both validate that persona is a real, manipulable structure: one reachable through the model’s internals, the other through context. We chose the context route because it produces a portable artifact users can own and audit, which internal activation steering does not. This choice shows up in the experiment as using a static response model (Haiku) served a variable context, rather than a fine-tuned or activation-steered model.

Jiang et al. (2025) find that frontier models achieve only ~50% accuracy on dynamic user profiling tasks even with full conversation access. The paper documents the failure empirically; our reading is that the cause is the gap between having facts and having the interpretive structure to apply them to new situations. Jiang’s paper is the most direct existing evidence for the gap this paper studies, and our test design inherits from it: behavioral prediction on scenarios drawn from held-out text that the model has not seen, with all relevant facts retrievable, measures exactly the interpretive-application gap.

Jain et al. (2025) find that adding conversation context to LLMs makes them more sycophantic: more likely to agree with the user even when the user is wrong, more likely to adopt the user’s perspective on a question. Their result shows that context without the right structure pushes the model toward what the user appears to want rather than toward a grounded answer. This is why our experiment includes a wrong-Spec control (§1.3 Mechanism): we hand the model a Behavioral Specification that does not match the actual subject. If models drifted purely toward whatever context they are given, the wrong-Spec should behave like any other structured prompt. Instead the model flags the mismatch or attempts a low-quality application, neither of which is sycophantic drift (per-condition rates in §4.3). Jain’s finding plus our wrong-Spec result bracket the question from both sides: context shape matters (Jain), and content matters too.

Lu et al. (2026) identify what they call the Assistant Axis: a dominant internal direction that anchors assistant models’ default behavior toward generic helpfulness and harmlessness. A Behavioral Specification can be read as an external override to the Assistant Axis on a per-user basis: a structured anchor that shifts the model from “generic helpful assistant” toward “reasons as this specific person would reason.” This framing motivated our choice to measure hedging as a

primary outcome alongside accuracy: if the Spec shifts the model off the generic Assistant Axis, the behavioral change should show up both in what the model predicts and in what it is willing to commit to. Our hedging-reduction finding (§1.3 Mechanism, §4.3) is consistent with this reading.

3. Study Design

The experimental strategy holds the response model constant and varies the context conditions: nothing (pretraining only), retrieved facts, raw corpus, a Behavioral Specification, or combinations of those. This lets us evaluate each context condition for representational accuracy against the No-Context Baseline, while isolating the contribution of the interpretive layer across all of them, separate from model capability, provider, or fine-tuning regime. All measurement choices are reflected in §4, and the statistical commitments were pre-locked before final analysis.

The apparatus is described in seven parts: §3.1 establishes the property being measured; §3.2 specifies the experimental conditions; §3.3 defines the scoring rubric and the calibrated LLM judge panel; §3.4 introduces the subjects; §3.5 covers the question batteries and circularity controls; §3.6 names the response models; §3.7 describes the pipeline that produces the Behavioral Specification (the reference implementation of the interpretive-layer artifact tested in this study).

3.1 Operationalizing representational accuracy via the Behavioral Specification

Representational accuracy is the AI-side property: how faithfully a model’s internal model of a specific person captures that person’s reasoning patterns. It is a property of the AI system, not of any specific operationalization. Multiple routes can in principle produce it (model fine-tuning, persona-vector steering, retrieval over structured facts). This paper operationalizes one such route: a **Behavioral Specification** served as context to a static response model, tested for whether an interpretive layer can increase representational accuracy relative to the other context conditions in §3.2.

The instrument we use to measure representational accuracy is behavioral prediction on held-out situations. Held-out passages from the subject’s autobiography serve as samples of situations the model has not seen. The model is asked to predict how the subject would respond; the response is scored against the verbatim held-out passage. Prediction here is the test, not the goal: §2.1 develops this distinction.

Three things have to hold for a served Behavioral Specification to register a positive score. Each is a property of the Spec or the serving step, not of representational accuracy itself:

1. The person has behavioral patterns consistent enough to be captured in a Behavioral Specification.
2. The Behavioral Specification actually carries that signal.
3. A model given the Behavioral Specification can act on it.

Prediction on held-out situations tests all three at once. When the score is low, one of these three is failing: the patterns are not consistent, the Spec is wrong, or the model is not using the Spec well. Each failure mode is informative.

We do not claim to modify the model’s internal parameters. Each condition in §3.2 varies what is served to the model at inference time: nothing, retrieved facts, the full extracted fact set, the raw source corpus, a Behavioral Specification, or combinations of these. The model’s resulting prediction is what we score against the verbatim held-out passage. The No-Context Baseline isolates

the model’s pretrained representation of the subject. The fact and corpus conditions isolate what the model can infer from raw information at runtime. The Behavioral Specification isolates what a structured interpretive layer adds on top. The study reports each.

In practice, we record representational accuracy as the mean predicted-behavior score (1-5 scale) across each subject’s behavioral-prediction battery, aggregated across the calibrated judge panel; the rubric, panel composition, and aggregation rule are defined in §3.3, and the guide to interpreting fractional scores and anchor crossings is in §3.3.1.

3.2 Experimental conditions

Each condition is a specific combination of inputs served to the response model (§3.6) against the same behavioral battery (§3.5). Every condition is run on all 14 subjects (§3.4). The Behavioral Specification and the extracted fact set used across the conditions below are produced by the pipeline in §3.7. The conditions separate into two groups, summarized in the table below and broken out in detail after.

All conditions, by group.

Group	ID	Condition	Inputs served
Direct context manipulations	C5	No-Context Baseline	No context beyond the question
	C2a	Spec Only	The Behavioral Specification
	C2c	Wrong Spec	A different subject’s Spec
	C4	All Facts	The full extracted fact set
	C4a	All Facts + Spec	Full facts plus the Spec
	C8	Raw Corpus	Full training corpus (half the source text)
	C9	Raw Corpus + Spec	Training corpus plus the Spec
Memory-system configurations (controlled, all 5 systems)	C1	Retrieval (Controlled)	Top-k facts (shared fact pool)
	C3	Retrieval (Controlled) + Spec	Top-k facts + Spec (shared fact pool)
Memory-system configurations (native, 4 commercial systems)	C1 native	Retrieval (Native)	System’s own ingestion
	C3 native	Retrieval (Native) + Spec	System’s own ingestion + Spec

Direct context manipulations. We specify the model’s input directly: no context (baseline), the Behavioral Specification, the extracted fact set, the raw corpus, or combinations of these. No

retrieval step intervenes. Each condition isolates what one input type or combination contributes.

ID	Condition	Inputs served	Null / comparison
C5	No-Context Baseline	Nothing beyond the question	Pretraining-only floor
C2a	Spec Only	The Behavioral Specification	Isolates the Spec’s contribution
C2c	Wrong Spec ¹⁷	A random other subject’s Spec	Tests whether structured interpretive content, not the correct content, produces the effect
C4	All Facts	The full extracted fact set for the subject	Tests whether raw information volume substitutes for structure
C4a	All Facts + Spec	Full facts plus the Spec	Tests whether the Spec adds value on top of raw facts
C8	Raw Corpus	Full training corpus (half the source text)	Tests whether uncompressed source text substitutes for structure
C9	Raw Corpus + Spec	Training corpus plus the Spec	Tests whether the Spec adds value on top of raw source ¹⁸

Memory-system configurations. Retrieval is performed by a memory-system provider’s production deployment. The memory-system conditions run in two modes: a *controlled* mode (each system retrieves from an identical pre-extracted fact set) and a *native* mode (each system ingests the raw corpus through its own pipeline).

Five memory systems are evaluated: Mem0, Letta, Supermemory, and Zep, plus Base Layer as our own open-source reference implementation (pipeline detail in §3.7). Architectural detail for the four commercial systems is in §2.2 Table 2.1.

Controlled configuration (all 5 systems). Each system retrieves from an identical pre-extracted fact set, isolating retrieval-algorithm differences from ingestion-pipeline differences.

¹⁷Two variants, locked before response generation: **v1 (adversarial fixed pairing)**, deterministic mismatch; **v2 (random derangement, seed-fixed)**, uniform random shuffle. Hamerton has an additional Franklin-specification variant in §4.1.2. Pairing list and sensitivity in §4.6.5.

¹⁸Bābur C9 is omitted in §4.2 (422,772-word source exceeds the response model’s context window); the remaining 13 subjects have C9 data.

ID	Configuration	Inputs served
C1	Retrieval (Controlled)	Top-k facts returned by the system for the question (controlled fact pool)
C3	Retrieval (Controlled) + Spec	Top-k retrieval output plus the Behavioral Specification (controlled fact pool)

Native configuration (4 commercial systems). Each system ingests the raw training corpus through its own pipeline, reflecting real-world deployment. C1 (Retrieval Only) and C3 (retrieval + Spec) are run with the system’s own ingestion pipeline replacing the controlled fact pool.

System	Conditions	Ingestion Process
Mem0	Mem0 Retrieval (Native), Mem0 Retrieval (Native) + Spec	Mem0 reads the corpus and decides on its own how to break it up, extract facts, and retrieve them.
Letta archival	Letta Retrieval (Native), Letta Retrieval (Native) + Spec	The corpus is loaded into Letta’s archival memory store; retrievals happen at query time from that store.
Supermemory	Supermemory Retrieval (Native), Supermemory Retrieval (Native) + Spec	Supermemory chunks the corpus on its own and applies reranking to the retrieved chunks at query time.
Zep	Zep Retrieval (Native), Zep Retrieval (Native) + Spec	Zep ingests the corpus into a knowledge graph (via Graphiti) and retrieves using a hybrid of semantic similarity, keyword match, and graph traversal.

Both configurations are reported so retrieval-quality differences and ingestion-pipeline differences can be read separately. Base Layer is run in the controlled configuration only: its retrieval uses the same fact set that feeds the Spec pipeline.

Detailed per-condition parameters, exclusion cases, and ingestion specifics are in Appendix C.¹⁹

3.3 Scoring rubric with calibrated LLM judge panel

Every response is scored 1-5 by an LLM judge panel against the verbatim held-out ground-truth passage. The primary aggregate uses five judges; two additional judges contribute to the sensitivity check (§3.3.2). Human annotation at this scale is feasible (~14 subjects × 40 questions × 15+ conditions) but was not done; running more conditions and more judges instead was the central evaluation trade-off. LLM-as-judge is grounded in prior work showing strong correlation

¹⁹Per-condition raw data organized under **results/**, indexed by subject and by memory system.

with human raters and reliability gains from panel aggregation (Zheng et al. 2023; Verga et al. (2024)). Extending the panel with human annotators on a stratified subset is flagged as the priority measurement follow-up in §7.1. The rubric and panel are a prototype scoring instrument, not a settled standard; §3.3.6 audits how the rubric behaves in practice. Worked examples throughout §4 include response excerpts in-line so the reader can see the rubric behavior directly rather than take our reading on faith; the full per-judge response and judgment data is in the study repository.

The evaluation is deliberately recursive. Response models are evaluated by judges (§3.3.2). Judges are evaluated by calibration diagnostics (§3.3.3), inter-judge agreement metrics (§3.3.4), and post-hoc rubric-handling audits (§3.3.6). No single layer is treated as ground truth; each layer’s behavior is itself measured and disclosed, and where a layer’s behavior diverges from what the rubric intends, the divergence is flagged rather than corrected silently. In the absence of human annotation, the paper strengthens rigor through this stacked-instrument structure, not by trusting any one step.

Scoring rubric. The following is the verbatim prompt the judges executed (canonical for both the 5-judge primary panel and the 7-judge sensitivity panel; source: `scripts/judge_hamerton_5judge.py`):

```
You are evaluating whether a response about a person
PREDICTED what actually happened.

=== HELD-OUT GROUND TRUTH ===
{held_out}

=== RESPONSE ===
{response_text[:1500]}

Rate 1-5:
5=Predicts specific outcome
4=General direction correct
3=Right domain wrong outcome
2=Wrong prediction
1=Refuses or off-base

Respond with ONLY a single digit (1-5).
```

The prompt is deliberately spare. It frames the task in one sentence, shows the held-out passage and the response, and asks for a single digit against five anchor labels. There are no calibration examples in the prompt, no scoring guide, and no chain-of-thought scaffolding. The anchors themselves are directive (refuses or off-base, wrong prediction, right domain wrong outcome, general direction correct, predicts specific outcome); resolution at the boundaries between adjacent anchors is left to the judge model. The design choice is to test whether frontier models converge on the behavioral-prediction construct from minimal instruction. §3.3.4 reports the inter-judge agreement obtained under this design.

Each response is scored against the verbatim held-out passage from which the question is drawn; the score reflects how closely the response predicts the documented behavior. Battery composition is detailed in §3.5. Condition identifiers (C5, C2a, C4a, C3) refer to the conditions defined in §3.2 and summarized in Appendix C; rubric anchor numbers 1 through 5 refer to the verbatim anchors above. Score interpretation against the construct is developed in §3.3.1 with worked examples. A worked rubric example alongside the No-Context Baseline engagement analysis is in §4.1.1; full

per-subject score distributions with verbatim responses are in Appendix D.

3.3.1 Score interpretation

Scores are read at three related granularities: integer rubric anchors (1 through 5), fractional means produced by averaging across the 5-judge primary panel (e.g., 2.87, 3.12, 2.34), and crossings between integer anchors when conditions change. Fractional shifts should be read through the integer anchors, because each anchor corresponds to a categorical shift in response quality.

The *cross-anchor interpretation rule*. A fractional delta that crosses an integer anchor reflects a real shift in the underlying response distribution. A delta that stays inside a single anchor is a within-category shift and a weaker claim.

The verbatim anchors describe stages of prediction quality. The *construct reading* column below gives the plain-language meaning of each crossing: what actually changes in the response when a condition moves the score across that boundary. The paper reads each crossing as a categorical shift in the construct the rubric is designed to measure: alignment of the response with the held-out behavioral pattern.

Boundary crossed	Verbatim anchor shift	Construct reading
1 / 2	Refuses → wrong prediction	The model stops declining and starts engaging with the question, even if the prediction is wrong.
2 / 3	Wrong prediction → right domain wrong outcome	The answer becomes specifically about this subject rather than a generic stand-in, even if the predicted outcome is still off.
3 / 4	Right domain → general direction correct	The answer gets the general direction of the subject's behavior right, not just the topic.
4 / 5	General direction → specific outcome	The answer gets the specific outcome right, matching the particular behavioral pattern in the held-out passage.

Here, *domain* means the area of behavior a question concerns (how the subject handles conflict, makes decisions, treats authority, and so on); a response is in the right domain when it engages the correct behavioral area, even when the specific predicted outcome is wrong.

The boundary crossings are illustrated through worked examples in §4.1 (Examples A, B, C) and §4.1.1, each drawn directly from experiment output that feeds the §4 aggregate. Example A in §4.1 illustrates a 1 → 4 crossing (wrong-referent correction); Example B illustrates a 2 → 5 crossing (directional correction); Example C illustrates a 2.80 → 5 crossing (abstention to substantive inference). Each example shows the verbatim question, held-out, and response excerpts paired with

the cached judge score for that response.²⁰

What a 1 means and does not mean. A score of 1 reflects a baseline failure to produce a usable prediction about the named subject: the response either explicitly declined to predict (abstention) or engaged with the question but landed on a categorically incorrect answer (non-abstention misalignment, including wrong referent, off-base inference, or confusion with a different subject). It is not a claim that the response was non-fluent or empty, and it is not a claim that the model lacks any related knowledge; the score reflects only that the response failed the held-out comparison. Each question tests one behavioral sample at a time; the aggregate fraction of score-1 responses across roughly 40 questions per subject is what the paper reads as the per-subject baseline-failure rate. How score-1 responses split between explicit abstention and non-abstention misalignment is analyzed in §4.1.1.

What a 5 means and does not mean. A score of 5 reflects alignment with one specific behavioral sample: the held-out ground-truth passage the question is drawn from. It is not a claim that the response fully represents the subject in some absolute sense, and it is not a claim that the same response would score 5 on a different held-out passage from the same subject. Each question tests one behavioral sample at a time; the aggregate across roughly 40 questions per subject is what the paper reads as the subject-level score. As with a 1, a score is only treated as reliable where the judge panel converges on it: interrater agreement (§3.3.4) is the paper’s signal of consensus, not any single judge’s number.

Multi-anchor crossings: the strongest categorical signal the rubric detects. A *multi-anchor crossing* is a single question whose 5-judge primary mean shifts across two or more integer rubric anchors when the condition changes. Crossings can span two anchors (e.g., $1 \rightarrow 3$, $2 \rightarrow 4$) or, more rarely, three (e.g., $1 \rightarrow 4$, $2 \rightarrow 5$). Larger crossings indicate larger categorical jumps in the same response, with five independent judges converging on the move. §4.2 reports the rates of these crossings and the response-level phenomena that produce them; worked examples are in §4.1.1 and Appendix E.

The paper applies the cross-anchor rule consistently. Score deltas reported in §4 are read through this lens. A +0.50 delta that crosses a rubric anchor is treated as a stronger claim than a +0.50 delta that does not.²¹

Reading scores within integer anchors. The 5-judge primary panel detects within-anchor signals cleanly.²² Across the 18 condition pairs analyzed,²³ roughly 18% of paired questions show same-anchor fractional shifts of at least 0.5 rubric points (a within-category shift, weaker than a cross-anchor crossing per the rule above). The integer metric is used throughout §4 for cross-anchor categorical interpretation; the within-anchor signal is reported here as methodological transparency.

²⁰Earlier drafts of this section illustrated the 1/2 and 2/3 boundaries with paraphrased excerpts that did not match any cached cell verbatim. Those illustrations have been removed and replaced with cross-references to §4.1 worked examples to ensure every quoted excerpt in the paper is mechanically sourced from `results_v2.json` and paired with its actual `judgments_v2.json` score, via `scripts/v2_canonical_cell_extractor_20260508.py`.

²¹Per-subject anchor-crossing data at `docs/research/s114_anchor_crossing_examples.json`; computing script at `scripts/compute_anchor_crossing.py`.

²²Direction-agreement among judges is 74% at panel $|\Delta|$ of 0.1 to 0.25 and 93% at $|\Delta|$ of 0.25 to 0.5. Full panel-direction analysis (Spearman ρ across all 10 primary-panel pairs) in §3.3.4.

²³Within-anchor shift data at `docs/research/within_band_shifts_20260428.json`.

3.3.2 Judge panel

Seven judges from three providers give the numeric aggregate its weight. Zheng et al. (2023) established that a single strong LLM judge correlates with human judges on comparable tasks at rates similar to human-human agreement. Subsequent panel-based work (Verga et al. (2024) and follow-ons) showed that aggregating multiple LLM judges past a small panel size further tightens agreement and reduces single-model idiosyncrasy. Seven judges across three providers is well past that threshold.

Judge	Provider
Claude Haiku 4.5	Anthropic
Claude Sonnet 4.6	Anthropic
Claude Opus 4.6	Anthropic
GPT-4o	OpenAI
GPT-5.4	OpenAI
Gemini 2.5 Flash	Google
Gemini 2.5 Pro	Google

The verbatim judge prompt is shown in §3.3 and is canonical for both the 5-judge primary panel and the 7-judge sensitivity panel. Each judge receives the held-out ground-truth passage, the subject context (name, source), the prediction question, and the response to score, *not* the condition label or the response-generating model. Judges do not see other judges’ scores. Response generation is similarly blinded: the response model receives only the question plus the condition-specific context block, with no signal that distinguishes which experimental condition it is operating under (the prompt schema in §3.6 is identical across conditions; only the injected context block changes).

The *specification-effect claim*. When a Behavioral Specification is served to the model as context, the model’s responses shift in the direction of the subject’s demonstrated behavioral patterns, and that shift registers as a measured increase in representational accuracy against held-out passages from the same subject. This is the directional claim the panel is built to test; not a claim that the model has gained a new behavioral-prediction capability, and not a claim that the higher-scoring response is the absolute “correct” answer for the subject.

The panel is designed for directionality, not absolute precision. Calibration (§3.3.3) and inter-judge agreement (§3.3.4) test how consistently the panel detects the directional shift.

3.3.3 Calibration

The calibration diagnostic measures whether each judge applies the rubric anchors as the rubric defines them on synthetic inputs with known correct scores. It does not use any subject’s responses. The four inputs are constructed examples: verbatim ground truth, paraphrased ground truth, partial ground truth (first sentence only), and ground truth plus generic padding.

All seven judges were tested against this diagnostic.²⁴ Full panel composition with per-judge calibration-status flags is in Appendix C.5.

²⁴Five judges (Haiku, GPT-4o, GPT-5.4, Gemini Flash, Gemini Pro) were tested before the main-study response-scoring run. Sonnet and Opus were tested on the same diagnostic inputs three days after the main-study run completed; they were added to the primary panel at panel-design time on cross-Anthropic-coverage grounds, and the post-hoc diagnostic confirms that composition rather than driving it.

Diagnostic tests.

Test	Input	Expected	What it measures
Verbatim	Response = ground truth	5.0	Recognizes perfect match
Paraphrased	Correct content, different wording	~5.0	Penalizes paraphrase?
Short correct	First sentence of ground truth	<5.0	Partial content scored partial?
Long correct	Ground truth + generic padding	5.0	Length inflation effect

Results.

Test	Haiku	Sonnet	Opus	GPT-4o	GPT-5.4	Gemini Flash	Gemini Pro
Verbatim	5.00	5.00	5.00	5.00	5.00	5.00	4.15
Paraphrased	4.75	5.00	5.00	5.00	5.00	4.70	3.55
Short correct	3.80	4.35	4.20	4.05	4.20	3.85	2.85
Long correct	5.00	5.00	5.00	3.35	4.80	3.80	1.20

Six of the seven judges score verbatim matches at 5.0; Gemini Pro is the outlier at 4.15. Haiku, Sonnet, and Opus are clean across all four diagnostics: no verbatim miss, no paraphrase penalty, no length-padding penalty, expected dip on partial content (4.20 to 4.35). GPT-4o and GPT-5.4 are clean on verbatim and paraphrase but penalize padded-correct responses on the long-correct diagnostic (GPT-4o 3.35, GPT-5.4 4.80); Gemini Flash and Gemini Pro do the same more severely (3.80 and 1.20). Per-judge calibration data and full panel composition are in Appendix C.5; raw scoring data at [results/judge_calibration/](#).

Use of calibration data. Scores are not normalized. Any normalization requires deciding which judge’s profile is “correct” and re-scaling the others toward it. Calibration data is published in its raw form so readers can apply their own normalization if they prefer.

Primary aggregate: 5-judge panel. The primary numeric aggregate reported throughout §4 is the 5-judge mean using Haiku 4.5, Sonnet 4.6, Opus 4.6, GPT-4o, and GPT-5.4. All five score verbatim and paraphrased matches at or near 5.0 and span two provider families (Anthropic and OpenAI). GPT-4o and GPT-5.4 penalize padded-correct responses on the long-correct diagnostic; that deviation runs against length inflation rather than toward it, so it is conservative for the Spec-effect direction this paper measures. This is the panel the §3.3.5 aggregation rule operates on, and the panel inter-judge agreement is reported for in §3.3.4.

Sensitivity aggregate: 7-judge panel. Both Gemini judges are reported separately as a 7-judge sensitivity check rather than rolled into the primary. Gemini Pro fails the verbatim-match diagnostic (4.15 where every other tested judge scores 5.00) and penalizes padded-correct responses severely (5.00 on short correct dropping to 1.20 on long correct). Gemini Flash passes verbatim cleanly but shows consistent length sensitivity (5.00 verbatim dropping to 3.80 on long correct). On actual study responses, both Gemini judges show a systematic +1-point magnitude inflation relative to the five primary judges. The combination of Pro’s verbatim failure, Flash’s length sensitivity, and

the shared inflation pattern places both Gemini judges in the sensitivity aggregate rather than the primary, while preserving them as a cross-provider robustness check in §4.6.2.

The 5-judge primary is also the conservative choice: including the Gemini judges produces *larger* Spec-effect deltas, not smaller ones (full numbers in §4.6.2). Reporting on the primary aggregate is therefore the lower-bound estimate.²⁵

3.3.4 Inter-judge agreement

The judge panel is designed to detect the directional shift named by the Specification-effect claim (§3.3.2). Two complementary agreement measures answer different questions about whether judges detect that shift consistently: direction (pairwise Spearman ρ) and absolute magnitude (Krippendorff α).

Direction agreement: pairwise Spearman ρ . Spearman ρ measures whether two judges rank the same set of items in the same order. $\rho = 1$ is perfect ranking agreement; $\rho = 0$ is no rank agreement; $\rho \geq 0.8$ is conventionally treated as strong rank agreement.

For each pair of judges in the 5-judge primary panel (10 pairs across Haiku, Sonnet, Opus, GPT-4o, GPT-5.4), pairwise Spearman ρ ranges from **0.86 to 0.93**.²⁶ The five primary judges agree on the ranking of conditions: whatever any individual judge’s absolute calibration quirks, they converge on which conditions produce better responses. For the directional claim (does the Specification produce a measurable, consistent shift toward the held-out behavioral pattern), this is the statistic that matters.

Magnitude agreement: Krippendorff α (ordinal). Krippendorff α measures whether judges give the same response the same numeric score (not just whether they rank items in the same order). $\alpha = 1$ is perfect agreement; $\alpha = 0$ is no better than chance; $\alpha < 0$ is systematic disagreement. Krippendorff’s guidance cites $\alpha \geq 0.8$ as high reliability and $\alpha \geq 0.667$ as substantial reliability.

The 5-judge primary panel scores $\alpha = \mathbf{0.659}$, just below the substantial-reliability threshold ($\alpha \geq 0.667$), placing the panel in the tentative-conclusions band.²⁷ Combined with Spearman ρ of 0.86 to 0.93 above, this is the empirical signature of the design choice: directional rankings converge across judges while absolute magnitudes diverge.

The 7-judge panel including the Gemini judges drops to $\alpha = \mathbf{0.535}$. This drop reflects the systematic +1-point Gemini inflation: Gemini judges score responses about one point higher on average than the five primary judges, so absolute values disagree even when rankings match. This is why the calibration audit (§3.3.3) excluded the Gemini judges from the primary aggregate.

The α value places a ceiling on how precisely any individual fractional score should be read, which is why the paper treats per-subject deltas that stay inside a single rubric anchor as weaker than deltas that cross one.

²⁵Raw calibration data is in the public repository at `results/judge_calibration/`. Per-judge calibration data and full panel composition with per-judge calibration-status flags are in Appendix C.5.

²⁶Spearman ρ from 0.29 to 0.93 across the 7-judge / 21-pair set, driven down by the two Gemini judges’ partial coverage and inflation behavior. Full matrix in `docs/research/stats_update.md` §5.

²⁷In-house ordinal implementation yields 0.659; a third-party recompute (`krippendorff` 0.8.2, ordinal) yields 0.668 (5-judge) and 0.579 (7-judge), differing on missing-data weighting. Both land within ± 0.01 of the 0.667 threshold; the conservative value is reported in-text. Scripts: `scripts/stats_wilcoxon_krippendorff_update.py`, `scripts/_recompute_ordinal_krippendorff.py`.

The agreement is achieved on a deliberately spare rubric. The judge prompt in §3.3 is five anchor labels and a one-sentence task framing. No calibration examples, no scoring guide, no chain-of-thought. Five frontier judges across three providers converge to $\rho = 0.86$ to 0.93 on rank order and $\alpha = 0.659$ on absolute magnitude under this scaffolding. The anchors are directive but their boundaries are left to the judge to resolve, and the resolutions converge across providers. The agreement is not the product of an elaborate rubric forcing judges into the same scoring frame; it is independent provider models agreeing on what the task is asking from minimal instructions.

Why directional inference survives middle-anchor noise. Judges agree most on the extreme anchors (1 and 5) and disagree most often on the middle ones, whether a response sits at 2 or 3, or at 3 or 4. This kind of disagreement appears equally across every condition, so it does not push one condition’s scores systematically above or below another’s. Individual scores carry some noise; the comparison between conditions does not.

The panel does not establish that any higher-scoring response is the absolute correct answer for the subject; that determination requires human annotation against the subject’s actual writing, which we do not have. What the panel provides is cross-provider directional convergence: three independent providers’ models agree that the Specification is moving responses in the same direction. We treat that as sufficient for a directional claim, no stronger.

Raw agreement matrices are at `results/interjudge_agreement/`.²⁸

3.3.5 Aggregation and statistical analysis plan

Aggregation rule. The aggregation rule decides how individual judge scores are combined into a single number for each subject under each condition. The rule was locked before any results were computed.²⁹

For each subject under each condition, every judge produces one score per question. For each judge, those scores are averaged across questions, producing one number per (subject, condition). The five primary judges’ numbers (Haiku, Sonnet, Opus, GPT-4o, GPT-5.4) are then averaged to get one number per (subject, condition). A 7-judge average that adds Gemini Flash and Gemini Pro is computed in parallel and reported as a sensitivity check. Subjects, not questions, are the level at which the paper draws conclusions.³⁰

Primary outcome. The per-subject score under each condition. The primary cross-subject comparison is Δ_C4a : each subject’s score with the Spec (the facts-plus-Spec condition, C4a) minus their score without the Spec (the No-Context Baseline, C5).

Primary test. A Wilcoxon signed-rank test paired across the 14 main-study subjects.³¹ Each Δ_C4a value is itself a paired difference (C4a – C5) for one subject; the test asks whether those

²⁸Per-judge calibration matrices and pairwise Spearman ρ tables for both the 5-judge primary and 7-judge sensitivity panels are in `results/interjudge_agreement/`.

²⁹The pre-locked rule and primary-vs-sensitivity panel composition are documented in `docs/ANALYSIS_PLAN_LOCK.md` in the public repository. The lock predates the Tier 2 cross-provider runs and the wrong-Spec v2 runs (commit history reflects the order).

³⁰Mean preserves every judge’s contribution. Median discards information when judges cluster tightly (the Spearman $\rho = 0.86$ to 0.93 agreement in §3.3.4 shows they do); trimmed mean requires an arbitrary trim threshold; Gemini inflation is handled by the primary-vs-sensitivity split rather than by silent correction.

³¹A Wilcoxon signed-rank test is a non-parametric paired test (it does not assume scores follow a bell-curve distribution). It was chosen over a paired t -test because the per-subject Δ distribution is not assumed to be normally distributed and $N = 14$ is small. Two-sided, $\alpha = 0.05$.

per-subject differences are reliably above zero, that is, whether the Spec produces a consistent positive lift rather than a difference indistinguishable from zero. It is run on the 5-judge primary aggregate; the 7-judge aggregate is reported alongside as a robustness check.

Sample size. $N = 14$ main-study subjects, pre-registered (the lock applies to the 14-subject main study; Franklin is a separate post-lock high-baseline reference per §4.1.2; Tier 2 is the cross-provider directional probe scoped at §3.6 and §4.6.1).

Multiple comparisons. Only one statistical test is treated as the primary claim (the Wilcoxon test above). All other analyses in §4, including the per-question improvement-rate analyses in §4.2.1, are descriptive: they report patterns and direction without making additional inferential claims.³²

Pre-registered vs post-hoc analyses. Every primary analysis (panel composition, aggregation rule, Wilcoxon test, Tier 2 replication, wrong-Spec v2 derangement) was pre-registered in docs/ANALYSIS_PLAN_LOCK.md before any scoring was run. Statistical-rigor checks and post-hoc validity audits were added later in response to peer review and are reported as exploratory checks alongside the pre-registered confirmatory test.³³

Effect-size grain. The cross-anchor rule (§3.3.1) applies at two grains: per-subject mean Δ_C4a (the headline) and per-question integer-anchor crossings (the per-question mechanism). The two grains are reported separately and not pooled.

3.3.6 Rubric-handling limitations (post-hoc validity audit)

The validity audit reported below applies to the verbatim prompt published in §3.3.

A post-hoc validity audit, conducted after the analysis-plan lock, identified two rubric-handling limitations any reader of the §4 numbers should keep in mind:

1. **Refusal anchor ambiguity.** The rubric’s lowest anchor (“refuses or off-base”) lumps together honest refusals to answer (responses where the model declines to predict, often following a Specification directive not to speculate without evidence; see the Keckley Q21 case in §4.4.4) and substantively wrong predictions. Judges sometimes score refusals at 2 or 3 instead of 1, especially when the refusal recites related facts.
2. **Length-score correlation in C5.** In the No-Context Baseline (C5), longer responses tend to score higher ($r = 0.60$): with no context to work from, the model pads its answer with hedging, recitation of loosely related facts, and offers to clarify the question, and that extra length can read as effort or quality to a judge. Once a Specification or fact set is supplied, the link between length and score disappears (near-zero correlation).

Direction of bias. Both effects raise the No-Context Baseline (C5) scores more than they raise Spec-condition scores. The true Spec-vs-baseline gap is therefore likely *larger* than the +0.89 mean lift reported in §4, not smaller. The full audit (per-judge strictness, per-response-model abstention

³²Secondary analyses include the per-anchor gradient, compression curve, memory-system composition, wrong-Spec contrast, and hedging reduction. No multiple-comparisons correction is applied because there is only one primary inferential test; sensitivity numbers (5-judge vs. 7-judge, Tier 2 vs. Tier 1, v1 vs. v2 wrong-Spec) are reported alongside the primary as robustness checks rather than additional tests.

³³Pre-registered: 5-judge primary panel, per-subject aggregation rule, Wilcoxon on Δ_C4a , §4.6.1 Tier 2 cross-provider replication, §4.6.2 7-judge sensitivity, §4.6.5 wrong-Spec v2 random derangement. Post-hoc additions: §4.6.4 bootstrap CI / joint multi-confound regression / permutation test, §4.6.6 retrieval-overlap semantic-similarity sensitivity, §4.6.7 rubric-handling validity audit. Appendix B.10 lists every analysis with its designation.

behavior, memory-system effect on abstention) is reported in §4.6.7; the analysis plan is left intact rather than recomputed under a modified rubric.

The class-level LLM-as-judge limitation that this methodology cannot fully address is treated in §6.2. Extending the panel with human annotators is one of the priority measurement follow-ups in §7.1.³⁴

3.4 Subjects

We test 14 subjects, all historical figures with public-domain autobiographies or memoirs. Subjects were selected across a range of time periods, source-text lengths, and geographic origins to avoid the study sitting on any single type of source material. All source corpora are English or English-translated and are available on Project Gutenberg or comparable public-domain archives. Because frontier language models train on large public-text corpora, some level of pretraining exposure to each subject’s writing is likely.

#	Subject	Source	Words	Origin	Era
1	Philip Gilbert Hamerton	PG #8536	25,231	England	19th c.
2	Elizabeth Keckley	PG #24968	58,742	United States	19th c.
3	Sunity Devee	PG #57175	67,379	India	19th–20th c.
4	Zitkala-Ša	PG #10376	35,328	United States (Yankton Dakota)	19th–20th c.
5	Olaudah Equiano	PG #15399	85,660	West Africa (Igbo) / Britain	18th c.
6	Mary Seacole	PG #23031	62,467	Jamaica	19th c.
7	Fukuzawa Yukichi	Internet Archive	139,088	Japan	19th–20th c.
8	Bābur	PG #44608	422,772	Central Asia (Fergana)	15th–16th c.
9	Yung Wing	PG #54635	66,459	China	19th–20th c.
10	Benvenuto Cellini	PG #4028	190,390	Italy (Florence)	16th c.
11	Bernal Díaz del Castillo	PG #32474	187,315	Spain (Castile)	15th–16th c.
12	Georg Ebers	PG #5599	96,174	Germany	19th c.
13	Jean-Jacques Rousseau	PG #3913	278,120	Switzerland (Geneva)	18th c.
14	Saint Augustine	PG #3296	114,873	North Africa (Numidia)	4th–5th c.

Constraints on the generalizability of the 14-subject sample (language, era, cultural framing, Project Gutenberg curation bias, individual self-presentation in autobiography) are discussed in §6.1.

3.4.1 Pretraining-coverage variance (high vs low baseline)

Pretraining coverage of a specific person varies widely across the 14 main-study subjects, even within a sample whose autobiographies are of comparable provenance. We use the C5 baseline

³⁴Per-judge judgments in `results/<subject>/*_judgments_<judge>.json` (memory-system variants: `<system>_judgments_<judge>.json` and `<system>_fullpipeline_judgments_<judge>.json`).

score (the No-Context Baseline mean on the 1-5 rubric defined in §3.3) as the observable proxy for pretraining coverage. Operationally, the proxy works as follows: the response model is given the behavioral-prediction battery for a subject with no external context (no Spec, no facts, no retrieval), and the resulting per-subject mean across the 39 questions is read as how well the model already knows that person from training data alone. A high C5 mean implies substantial pretraining representation; a low C5 mean implies very little.

Baseline band	Subjects	Count
≤ 2.0 (low-baseline)	Sunity Deveen, Ebers, Hamerton, Fukuzawa, Seacole, Bernal Díaz, Keckley, Yung Wing, Bābur	9
2.0–3.0 (mid-baseline)	Cellini, Zitkala-Ša, Rousseau, Augustine, Equiano	5
> 3.0 (high-baseline)	Benjamin Franklin (known-figure control, not in main study)	1

The higher a subject sits on this distribution (e.g., Equiano, Zitkala-Ša at the upper end of the main study), the better the model already knows them from pretraining; conversely, the lower they sit (Sunity Deveen, Ebers), the less the model knows. Nine of the 14 main-study subjects fall into the low-baseline band; five into the mid-baseline band.

Franklin as a known-figure control. Benjamin Franklin (Project Gutenberg #20203) is included as a high-baseline reference point. Franklin’s *Autobiography* is one of the most widely available and frequently cited autobiographies in American public-domain literature, and the model’s baseline score on Franklin (3.77 on the 5-judge primary panel; 4.10 on Haiku alone) is consistent with substantial pretraining coverage of both the person and the specific text.³⁵ Franklin is used to anchor what the high-baseline end of the spectrum looks like (§4.6.4 develops the high-baseline behavior), not as a subject whose pretraining coverage is a design target of the Behavioral Specification.

Cross-provider variance. No-context baseline scores on the same subject vary by 1–2 points across response models. Different providers know different amounts about the same historical figure (Tier 2 cross-provider data in §4.6.1). The pretraining-coverage variance documented in this section is a property of each model family, not a property of the subject alone. The cross-provider spread in No-Context Baseline scores for a given subject is comparable in magnitude to the lift the Specification produces, which is why the response model is held constant across all main-study conditions.

§4.1 develops this distribution into the cross-subject gradient.

3.5 Question battery formation

Each subject’s behavioral prediction battery is generated by a backward-design process: an LLM reads a passage from the held-out half of the corpus, writes a question whose answer is the behavioral

³⁵Franklin’s legacy run pre-dates the gemini_flash / gemini_pro split, so only one Gemini judge is present in the legacy data. The per-judge range cited here is the min and max across all available judges, rather than a clean 7-judge panel mean.[^franklin-panel-coverage]

pattern implicit in the passage, and avoids naming any detail unique to the passage itself. The question can be attempted from training-text patterns alone; the verbatim held-out passage is the ground truth for scoring.

The procedure, in order:

1. **Split the corpus.** Each subject’s source text is divided by chapter into `training.txt` (50% of the corpus) and `heldout.txt` (50%). Held-out text is never shown to a response model.
2. **Sample from held-out.** A sliding window moves across the held-out text in four batches of ten questions each, using 5,000-character windows as local context.
3. **Backward-design question generation.** Claude Haiku 4.5 (temperature 0) reads each held-out window and writes a question whose answer requires the subject’s behavioral patterns observable in the training half. For each window, Haiku:
 - Extracts a verbatim ground-truth span from the held-out window.
 - Avoids named-entity or specific-date leakage in the question stem.
 - Targets one of 10 fixed behavioral-prediction categories: decisions, values, relationships, conflict, learning, risk, creativity, stress, career, and change-over-time.
4. **Supplementary tiers (not scored in §4).** Beyond the behavioral-prediction tier, each battery additionally contains supplementary question tiers that test different competencies:
 - **Recall** tests whether a system can retrieve specific factual content.
 - **Adversarial-abstention** presents scenarios the source corpus does not cover, testing whether a system correctly refuses to predict rather than fabricates.

These supplementary tiers are included in the battery files for future analysis but do not enter the §4 main results.

5. **Dedup and freeze.** Deduplication on lowercased question text, cap at target counts per category, MD5 checksum of the final battery. Downstream response and judgment files are invalidated if the battery checksum changes.

Each main-study subject receives 39 behavioral prediction questions; the 14 main-study batteries total 546 questions.³⁶ Each battery covers 8 to 10 of the 10 categories. Definitions, example questions, and per-subject distributions are in Appendix B.1 and B.2.

Within the behavioral-prediction tier, individual questions vary in interpretive demand: some can be answered from facts alone, others require applying behavioral patterns to novel scenarios. This variation is decomposed in §4.4.3.

Leakage audit. We empirically checked the no-leakage principle by searching every behavioral-prediction question for any sequence of seven or more consecutive words that appears verbatim in that subject’s held-out corpus. Across the 14 main-study subjects (546 questions), zero questions leak (true zero, not a rounded value).³⁷

³⁶Franklin’s 40-question legacy battery (high-baseline reference, §3.4.1) brings the data-structural total to 586 questions across 15 subjects.

³⁷Franklin’s 40-question legacy battery, included as a high-baseline reference but not part of the main study, has 2 leaking questions (Q49, Q56). Both predate the backward-design constraint and were hand-authored. Aggregate across the full 586-question pool: 2 leaks (0.34%).

One false-premise outlier. One item out of 586 (Zitkala-Ša Q18) carries a factually-wrong presupposition; the aggregate is unaffected. The broader need for a human-reviewed quality gate on automated batteries is flagged in §6.2.

Battery files and the leakage-audit script are in the public repository.³⁸

3.5.1 Circularity controls

The pipeline and the batteries both use Anthropic models for several roles (Haiku for extraction and battery generation, Sonnet for authoring, Opus for composition, Haiku as the primary response model, plus Sonnet and Opus on the judge panel). Two independent controls rule out a within-Anthropic frontier-model artifact; full per-subject results are reported in §4.6.1.

Control 1: Independent battery regeneration. Batteries for all 13 global subjects were re-generated with GPT-5.4 using the identical backward-design prompt. Categorical-emphasis differences are minor; the methodology constrains the output more than the generating model does.

Control 2: Non-Anthropic response chain. The core conditions³⁹ were re-run on three subjects (Ebers, Yung Wing, Zitkala-Ša) using two non-Anthropic response models (Claude Sonnet 4.6 and Google Gemini 2.5 Pro) reading the GPT-5.4-regenerated batteries. The combination tests whether the Spec effect survives when both the response model and the battery-generation model sit outside the Anthropic family.

The broader LLM-as-judge circularity is discussed as an open limitation in §6.2.⁴⁰

3.6 Response models

Tier 1 (main study): Claude Haiku 4.5 as the primary response model, run across all 14 subjects and every condition in the main matrix. Haiku was chosen as a deliberately weaker baseline: a Spec effect that registers on a relatively weak response model is harder to attribute to the model’s pretraining alone, which gives a conservative readout of effect direction. §4.6.1 Tier 2 cross-provider probe tests whether the direction reproduces on stronger response models.

Tier 2 (cross-provider response generation). A sensitivity probe using non-Anthropic response models on a subset of subjects; methodology and results are reported together in §4.6.1.

Prompt schema. A single shared prompt is used across every condition. The system message frames the task as behavioral prediction of a specific person; the user message is the question plus whichever context inputs the condition specifies (§3.2). Nothing about the prompt changes per condition beyond the injected context block.⁴¹

System: You are predicting how <subject> would respond to a specific

³⁸Per-subject batteries at `results/global_<subject>/battery_v2.json` (13 global subjects); Hamerton and Franklin legacy at `data/<subject>/battery.json`; GPT-5.4-regenerated batteries (used in the §3.5.1 circularity control) at `results/global_<subject>/battery_gpt54.json`; leakage-audit script at `scripts/_verify_battery_leakage.py`.

³⁹C5 No-Context Baseline, C2a Spec alone, C4a facts-plus-Spec, C2c wrong-Spec control. Full condition definitions in §3.2.

⁴⁰Raw battery regeneration data at `results/global_<subject>/battery_gpt54.json` for all 13 global subjects; Tier 2 response and judgment files for the three Control 2 subjects in the same per-subject directories.

⁴¹All response models are called with `temperature=0` and `max_tokens=1024`.

question about their behavior, values, or reasoning. Answer in <subject>'s voice, grounded in their demonstrated patterns.

User: <context block, one of: empty (C5), Spec (C2a), wrong-Spec (C2c), facts (C4), facts + Spec (C4a), corpus (C8), corpus + Spec (C9), or retrieval ± Spec (C1 / C3)>

Question: <question text>

The prompt is deliberately uniform and deliberately faithfulness-oriented. The system message asks the model to ground its answer in the subject’s demonstrated patterns and to answer in the subject’s voice; we chose this framing because the study tests whether the served context lets the model do exactly that. No instruction tells the model to abstain, hedge, or commit; the model’s natural refusal-or-commitment pattern given a specific context is itself part of the phenomenon the study tests, and §4.3 reports the hedging-rate shift across conditions as a substantive finding rather than a behavior to suppress.

Two effects of the faithfulness framing. Asking for the subject’s voice may favor systems that surface verbatim corpus passages over systems that surface compressed interpretive structure (Letta’s stateful-agent path is one such system, examined post-hoc in Appendix G). Asking the model to ground in demonstrated patterns may push it toward abstention when the served context underdetermines a confident answer (visible in the Spec-induced refusal cases in §4.4.3). Because the framing is identical across every condition, it cannot be the source of the differential effect the study measures; a neutral-prompt robustness check is flagged as future work in §7.

With the framing held fixed across every condition, what the conditions vary is the source of the patterns the model works from.

Condition	Pattern source served to the model
C5 (no context)	The model’s pretraining alone
C2a (Spec only)	Pre-extracted patterns served as the Behavioral Specification
C4 / C8 (facts / corpus)	Raw material; the model must identify patterns at runtime
C4a / C9 (facts + Spec / corpus + Spec)	Pre-extracted patterns alongside raw material
C2c (wrong-Spec)	Pre-extracted patterns from a different person

The empirical evidence that the prompt is not the lift mechanism (the C2a vs. C4 / C8 comparison and the C2c wrong-Spec control) is reported in §4.3.

Exact model identifiers, full prompt text, and Tier 2 invocation parameters are in Appendix C.⁴²

3.7 Pipeline for the Behavioral Specification

A Behavioral Specification is a structured document encoding a person’s behavioral patterns. It is composed of three interpretive layers (anchors, core, predictions) plus a composed

⁴²Scripts in `scripts/` (`run_global_subjects.py`, `run_full_study.py`, `run_multimodel_responses.py`); raw responses under `results/`.

unified brief; total size per subject is approximately 7,000 tokens (~5,000 words, about the length of a short magazine article). Every Specification evaluated in this paper is produced by **Base Layer**, an open-source reference implementation of the pipeline; Base Layer is included in the study as one of the memory-system comparison conditions (§3.5, §4.4), alongside the four commercial systems (Mem0, Letta, Supermemory, Zep). The pipeline transforms raw source text in five steps: import, extract, embed, author, and compose. Each step is a single script backed by a single model choice.

Step	Input	Tool / model	Output
1. Import	ChatGPT / Claude exports, journals, plain text, directories	<code>import_conversations.py</code>	A local database holding the cleaned, de-duplicated source text
2. Extract	Canonical source text	<code>extract_facts.py</code> , Claude Haiku 4.5, 46-predicate vocabulary (full list in Appendix A)	Behavioral patterns extracted as short structured statements (e.g., “avoids confrontation,” “values craft over speed”), with bookkeeping operations to add new patterns, update existing ones, delete contradicted ones, or skip duplicates
3. Embed	All imported message text	<code>embed.py</code> , all-MiniLM-L6-v2, ChromaDB	A searchable index of source passages. Combined with the source-message IDs each extracted fact carries, this lets any claim in the final specification be traced back to the passages that support it
4. Author	Extracted facts	<code>author_layers.py</code> , Claude Sonnet 4.6	Three interpretive layers as markdown (anchors, core, predictions); see body below for layer-by-layer examples. Each layer is produced from facts alone, not from prior layer output. Each layer prompt includes a domain guard that prevents topic skew (ablation-validated in prior pilot work).
5. Com- pose	The three authored layers (plus a sample of identity-tier facts as supplementary context)	<code>agent_pipeline.py</code> , Claude Opus 4.6	Unified behavioral brief in flowing prose; see body below for what the brief contains and why it exists

The Behavioral Specification served as context in experimental conditions is the three authored

layers concatenated with the composed brief, not the brief alone.

The extract step constrains output through a fixed vocabulary of 46 behavioral predicates.⁴³ The vocabulary is human-curated and was validated across 50+ pilot subjects before being frozen for the study. The constrained vocabulary is the main lever the pipeline uses to push extraction away from biographical facts (“his father was violent”) and toward behavioral patterns (“evaluates authority figures on dual criteria of virtue and failure”).

Each of the authored layers has distinct jobs. Each layer has a characteristic format; examples below are drawn from three different subject specifications (Sunity Deveen, Bernal Díaz, and Augustine).

Anchors encode the subject’s load-bearing axioms in numbered form (A1, A2, ...), each with an activation condition and a false-positive warning. Example from Sunity Deveen:

A1. DIVINE PRIMACY. All events, decisions, and outcomes are understood as expressions of divine providence first; social, political, or material explanations are secondary framings applied afterward. Active when: life decisions, loss, political events, marriage, reform, or any outcome described as fortunate or unfortunate are discussed.

Core captures values, beliefs, and self-view in flowing prose. It is the layer that reads most like an essay about the person. Example from Bernal Díaz:

They reason from direct witness and lived participation, treating firsthand account as the only reliable foundation for any claim. When evidence is secondhand or reconstructed, they flag it rather than smooth it over. They distrust narrators who were absent from the events they describe, and this distrust shapes how they evaluate any source placed before them.

Predictions are explicit behavioral predicates (P1, P2, ...) with detection criteria, directives, and false-positive warnings. Example from Augustine:

P1. CONFESSION BEFORE CONCLUSION. When asked to account for a past failure, does not defend, minimize, or contextualize first. Moves immediately into detailed reconstruction of the failure, naming the pleasure or pride taken in the transgression before offering resolution. Detection: a relational conflict where the subject was at fault, a professional misjudgment, a moment of intellectual dishonesty. Directive: hold space for the full weight of the confession; resist premature resolution. False positive: not active when the subject is analyzing someone else’s failure.

The unified brief. The brief serves a dual purpose: a coherent first pass for human readers, and an integration step that implicitly weaves the three authored layers (anchors, core, predictions) together (layered files alone do not require this integration; internal testing suggests the integration changes how a model uses the Specification). The compose step produces it as a continuous prose synthesis in the third person, similar in length to a short profile of the subject. The Behavioral Specification served to the response model in C2a / C4a / C3 conditions is the three layers plus the unified brief together; the brief is not a substitute for the layers but an integration on top of them. A formal ablation isolating brief-with-layers vs. layers-only is flagged in §7.3 [Specification design and composition](#).

Total pipeline cost is approximately \$1 per subject (estimated from per-step API token counts)

⁴³Examples: avoids, repeatedly engages in, refuses to, values, fears, has experienced. The full predicate list is in [Appendix A](#).

to process a 50,000- to 150,000-word autobiography end to end. Pipeline code, the full predicate vocabulary, and example specifications for all 14 study subjects are available in the public repository (see §8 Data, code, and reproducibility).

4. Results

Across 14 historical subjects, adding a Behavioral Specification (a short structured document describing how a specific person reasons and behaves) measurably improves how accurately a language model represents that person’s behavioral patterns. We measure this with a battery of behavioral prediction questions based on held-out ground-truth text from each subject’s publicly available autobiography. We score each prediction on a 1-to-5 rubric where a whole-point shift marks a categorical change in how the response aligns with the subject’s documented behavior.

On the 9 low-baseline subjects (those the model does not know well), the Specification produces a mean per-subject increase of **+0.89 points** and lifts individual responses by one category or more on **55.0% of questions**. The Specification’s added value on top of other context types (facts, raw corpus, or memory-system retrieval) concentrates on interpretation-heavy questions; on factual-recall questions, retrieval alone is often sufficient and the Specification adds little or actively degrades the response. On high-baseline subjects (those the model does know well, such as Benjamin Franklin), the Specification adds little or mildly hurts across conditions. Control conditions, statistical robustness checks, and sensitivity analyses confirm that the Specification categorically shifts how a language model responds, increasing its representational accuracy of the subject beyond what fact-based retrieval supplies.

The seven parts of §4 establish this picture in detail:

- §4.1 the cross-subject gradient (the primary result across 14 subjects)
- §4.2 compression of structure versus raw text
- §4.3 content versus format as the mechanism
- §4.4 memory-system composition
- §4.5 the Letta stateful-agent exploratory case study (Appendix G, N=3, not a headline finding)
- §4.6 robustness and sensitivity (cross-provider, judge-panel, battery composition, wrong-Spec derangement, retrieval overlap, post-hoc rubric-handling audit)
- §4.7 the synthesis bridge to §5

Every number in §4 uses the 5-judge primary aggregate defined in §3.3.3.⁴⁴ Score deltas are read through the anchor-crossing rule from §3.3.1: a delta that crosses a rubric integer anchor is a stronger claim than one that stays within a single anchor.

4.1 The cross-subject gradient and its per-question mechanism

Hypotheses tested in this section (from §1.2): H1. Adding the Specification improves prediction. H2. The effect is inversely proportional to the response model’s pretraining coverage. Corollary to H2: on high-baseline subjects, the Specification does not add value and mildly interferes.

⁴⁴Primary panel: Haiku 4.5, Sonnet 4.6, Opus 4.6, GPT-4o, GPT-5.4. The 7-judge sensitivity check adds Gemini 2.5 Flash and Gemini 2.5 Pro and is reported in §4.6.

The cross-subject gradient. The less the model already knows about a subject from pretraining, the more a Behavioral Specification improves the model’s representational accuracy of that subject. It operates as an interpretive layer over facts and retrieved context, not a replacement for them. On the 9 subjects whose pretraining baseline sits at or below 2.0 on the 1-5 rubric (the population of relevance from §3.4.1), adding the Spec consistently improves prediction: every one of the 9 improves over the No-Context Baseline (C5); none declines. The per-subject mean lift is **+0.69** for Spec Only (C2a) and **+0.89** for All Facts + Spec (C4a). Adding the Spec on top of All Facts (C4), Raw Corpus (C8), or memory-system retrieval produces additional aggregate gains that are smaller in magnitude than the Spec-vs-baseline lift (detail in §4.2 and §4.4). Spec Only does not score higher than All Facts or Raw Corpus alone; the Spec’s value is in the layering.

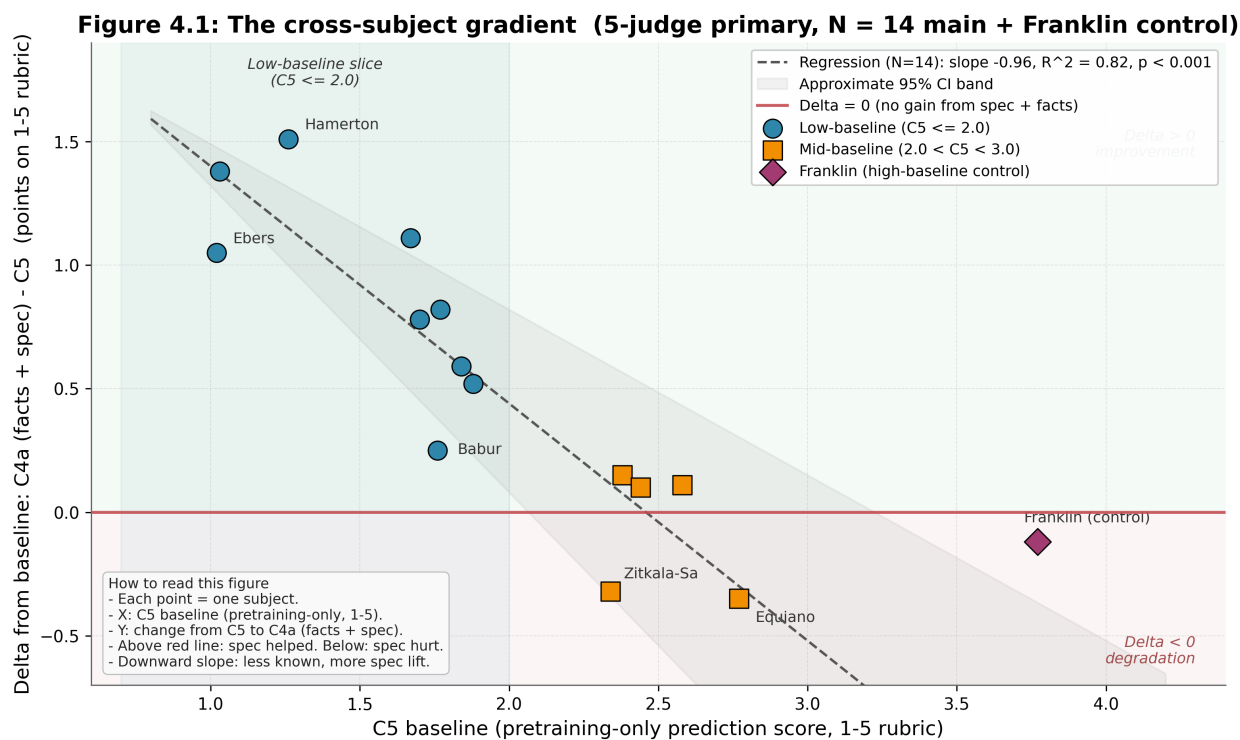


Figure 4.1. Reading the gradient. Figure 4.1 plots each subject’s No-Context Baseline (C5) against the lift the Specification produces over that baseline (Δ C4a, where C4a is the All Facts + Spec condition). The 9 low-baseline subjects (C5 ≤ 2.0) cluster in the upper-left of the plot with positive lifts ranging from Babur at +0.25 (smallest lift) to Hamerton at +1.51 (largest). Franklin sits in the lower-right at C5 = 3.77, $\Delta = -0.13$: the high-baseline reference where the model already knows the subject from pretraining. The regression slope of -0.96 captures this gradient: the lower the model’s pretraining baseline on a subject, the larger the lift the Specification produces. Across this question mix, the per-subject C4a mean clusters around 2.44, so the lift in rubric points is largest where the baseline shortfall is largest.⁴⁵ (§4.1)

⁴⁵Two regression slopes capture the gradient. The level slope of C4a on C5 is essentially flat ($+0.04$, $R^2 = 0.008$): the Spec produces roughly uniform quality regardless of where each subject’s baseline starts. The change-score slope of Δ C4a on C5 is strongly negative (-0.96), but most of that magnitude is mechanical: $\text{slope}_\Delta = \text{slope}_\text{level} - 1$ by identity, so a flat level slope automatically produces a near- -1.0 change-score slope. Full technical sensitivity detail in §4.6.3.

Adding a Behavioral Specification changes the category of answer the response model produces. Of the 351 individual responses in the low-baseline band, **55.0% crossed at least one rubric integer anchor upward when the Specification was added.** Multi-anchor jumps of two or more anchors (e.g., 1→3, 2→4) appear in **18%** of low-baseline questions on the Spec conditions, with about **6%** being extreme jumps of three or more anchors (e.g., 1→4, 2→5, 1→5). These extreme jumps concentrate on interpretation-heavy questions: the No-Context Baseline response refuses or stays generic, and the Specification supplies the behavioral pattern the model could not retrieve from training data. The response model’s answer moved from one category of response to a qualitatively different category. These are the multi-anchor jumps at the margin the aggregate mean understates.

Population of relevance for AI personalization. Most users sit in the low-baseline category. AI users whose private reasoning is not in any training corpus fall at or near the rubric floor by construction, so the lift the Spec produces is largest exactly where the use case is most common. The Spec works for any subject regardless of pretraining coverage and is therefore portable across the long tail of users (§1.4, §5.2).

Per-question rubric transitions when the Specification is added (low-baseline band, C5 → All Facts + Spec, 351 paired questions).

Transition	% of responses	Description
1 → 2	33.3%	Refusal or off-base → generic engagement with the question
1 → 3	12.3%	Refusal → partially-aligned prediction
1 → 4	4.8%	Refusal → substantively-aligned prediction
1 → 5	0.9%	Refusal → fully matches the held-out pattern
2 → 3	2.0%	Generic → subject-specific
2 → 4	0.3%	Generic → substantively-aligned
3 → 4	1.4%	Partially → substantively-aligned
No upward crossing	38.2%	Delta stayed inside a single anchor
Downward crossing	6.8%	Specification hurt the response

When the Specification is added, one of every three low-baseline responses moves from “cannot engage” to actual engagement. Another one in five makes a larger jump. Only one response in fifteen gets worse. Worked examples appear below.

Three representative examples below show distinct mechanisms by which the Specification improves the response.⁴⁶ Hedge reduction is common but not the only mechanism. The Specification also corrects directionally-wrong predictions and enables interpretive inference from character patterns when retrieved facts are insufficient.

⁴⁶Each example is extracted from the cached data (`battery_v2.json`, `results_v2.json`, `judgments_v2.json`) via `scripts/v2_canonical_cell_extractor_20260508.py`, which joins the question, held-out, response, and per-judge scores for each (subject, qid, condition) cell. Methodology note in [v1-v2-reconciliation].

Worked examples: Wrong-Referent (Fukuzawa), Directional (Bernal Díaz), Ab-stention (Seacole)

Example A. All Facts → All Facts + Spec: Wrong-Referent Correction

Subject: Yukichi Fukuzawa. **Q35:** *“How would Fukuzawa characterize someone who studied naval arts under the Dutch and later became instrumental in preventing military conflict?”*

Held-out (the figure the question is about): *“He is best remembered as the man who saved Yedo from siege when the Imperial forces were marching upon it in 1868.”* The held-out names Katsu Kaishu, the naval officer who later prevented the bombardment of Edo through diplomatic surrender.

C4 All Facts (mean 1.00): *“Based on the facts provided, Fukuzawa would likely characterize such a person with high regard and respect. . . The most relevant figure is **Captain Kimura Settsu-no Kami**, who commanded the Kanrin-maru. Fukuzawa would characterize him positively because: Competence and Pride: Kimura had studied navigation under the Dutch and successfully commanded the trans-Pacific voyage. . . ”*

The AI picks the wrong person. Captain Kimura is also a naval officer who studied under the Dutch, but he is not the figure the held-out is about. The model matched on a surface descriptor (“naval officer who studied under Dutch”) and grabbed the most-mentioned naval officer in the facts. Five judges score this at the rubric floor, anchor 1 (off-base): the response engages the question fluently but identifies the wrong subject.

C4a facts + Spec (mean 4.20): *“This describes **Katsu Rintaro**, who: studied navigation under Dutch officers in Nagasaki (post-1855); served as second-in-command on the Kanrin-maru’s trans-Pacific voyage; later became instrumental in negotiating the peaceful surrender of Edo Castle (preventing the final Boshin War battle). . . Fukuzawa would credit him for earned expertise, not inherited rank. . . but Fukuzawa would probably regard Katsu as a capable man constrained by the very system Fukuzawa despised — someone who used Western knowledge skillfully but remained ultimately loyal to institutional authority rather than to independent principle.”*

The Spec moves the AI from the wrong referent to the right one (Katsu Rintaro = Katsu Kaishu). It characterizes him through Fukuzawa’s reasoning patterns: respect for demonstrated capability paired with skepticism toward institutional embedding. Five judges register the move; mean shifts from 1.00 to 4.20, a three-anchor jump from “off-base” to “general direction correct.”

What the Spec did. The headline is fine-grained differentiation. The facts list both Captain Kimura (the captain of the Kanrin-maru) and Katsu Rintaro (the second-in-command). Both fit the surface descriptor “naval officer who studied under the Dutch”; both appear repeatedly in Fukuzawa’s autobiography. Picking between them requires reasoning about which figure Fukuzawa would actually characterize for the specific behavior in the question, not just which figure matches the descriptor. The Spec encodes Fukuzawa’s preference for earned over inherited authority, his skepticism of institutional rank, and his attention to demonstrated capability. These are the exact axes that distinguish a captain by appointment from a subordinate by skill. Without the Spec, the

model defaults to the most-mentioned candidate (Kimura) and matches the descriptor; with the Spec, it filters by behavioral pattern and resolves to the right person (Katsu). The interpretive layer is what lets the model choose the second-in-command over the captain when the question is about Fukuzawa’s characterization, not the facts.

Example B. All Facts → All Facts + Spec: Directional Correction

Subject: Bernal Díaz del Castillo. **Q16:** *“When the commander is offered physical assistance during a strenuous activity, what behavior would the author expect from him?”*

Held-out: *“There were 114 steps to the summit, and, as they feared that Cortes would experience the same fatigue in mounting as Motecusuma had, they were going to assist him by taking hold of his arms. Cortes, however, would not accept of their proffered aid.”*

C4 All Facts (mean 2.00): *“Based on the facts provided about Bernal Díaz del Castillo and his observations of Cortes, the author would expect the commander to **accept and participate** in the assistance while maintaining leadership and setting an example.”*

The AI gives a confident, fluent answer that gets the direction wrong: it predicts Cortés would accept help. Five judges agree: unanimous 2 (“wrong prediction”). The response engages the question and produces a specific prediction, but the prediction is the opposite of what the held-out shows.

C4a facts + Spec (mean 4.80): *“Based on the behavioral specification, when the commander is offered physical assistance during a strenuous activity, the author would expect **Cortes to refuse** the assistance and perform the labor himself — particularly if it serves a symbolic or leadership purpose... The author notes that when Cortes performs the labor, ‘The caziques and all present followed Cortes’s example.’ This is not accidental — it’s the load-bearing mechanism of how Cortes exercises power.”*

The Spec flips the prediction. The AI now correctly anticipates Cortés refusing, and explains why: Cortés’ authority was grounded in performative self-reliance, so accepting help would have undermined the leadership pattern Bernal Díaz documents repeatedly.

What the Spec did. The headline is what *all the facts together still cannot capture*. C4 is not a sparse condition; the model received the complete extracted fact set from Bernal Díaz’s training-half corpus, including every documented instance of Cortés’ behavior. With that full fact pool in context, the model still predicts the wrong direction: that Cortés would accept the offered help. The interpretive layer is not adding missing facts; it is encoding an *organizing principle* the facts alone do not announce: Cortés’ performative self-reliance as the load-bearing mechanism of his authority. Reading individual facts does not yield the principle; you have to interpret the pattern across them. Once the principle is named in the Spec, the model uses it to override its generic “good leaders accept help” default and predict the right direction. This is the value facts alone cannot capture: the principle that connects the facts is itself a piece of information the model needs, and the model does not appear to construct it on its own from the facts. Multi-anchor crossing (2 → 4.80, a near three-anchor jump).

Example C. All Facts → All Facts + Spec: Abstention Becomes Near-Perfect Inference

Subject: Mary Seacole. **Q2:** “How does Mary Seacole typically respond when a delirious patient mistakes her for a family member?”

Held-out: “Poor fellow! I could not undeceive him. I think the fancy happily caused by the touch of a woman’s hand soothed his dying hour.” In Seacole’s own voice: she chose not to correct him, and read the comfort her presence brought as a kind of healing.

C4 All Facts (mean 2.80, judges split 5/2/3/3/1): “Based on the complete set of facts provided, there is **no specific information** about how Mary Seacole typically responds when a delirious patient mistakes her for a family member. . . there is no explicit account of her reactions to delirious patients confusing her identity with their relatives.”

The AI honestly abstains: it identifies that the facts don’t directly cover this scenario and refuses to invent. Five judges have no shared view on what an honest abstention deserves. Haiku gives it full credit (5); GPT-5.4 gives it the floor (1); the rest land between. The 2.80 mean averages over a panel that genuinely disagrees on whether to reward the abstention as engagement or penalize it as failure to predict.⁴⁷

C4a facts + Spec (mean 5.00, unanimous): “Based on the behavioral specification and known facts, Mary Seacole’s response would follow a consistent pattern. Primary response: compassionate acceptance. She would **not correct the patient** in the moment. Instead, she would **lean into the mistaken identity, sustain the fiction with tenderness**, and prioritize emotional relief over factual accuracy. This aligns with A4 (Proximate Obligation) and A8 (Sympathetic Realism). . . Personal presence and touch carry psychological and emotional power that no institutional process replicates.”

All five judges agree this matches the held-out. Unanimous 5.

What the Spec did. The headline is the fine-grained nuance the Spec adds beyond generic compassion. A generic “Seacole is a compassionate nurse” framing is consistent with several responses: correct the patient gently, redirect them to staff, refuse to engage with the delirium, accept the misidentification matter-of-factly. The Spec predicts none of those; it predicts that Seacole would *sustain the fiction with tenderness*, treating the comfort the misidentification produces as itself a kind of healing. That is the level of nuance the question is actually testing: not whether the model knows Seacole is compassionate (the facts establish that) but whether it can predict which compassionate move she would choose. Facts alone produce abstention because no fact in the training half says “Seacole sustained a delirious patient’s mistaken identification”; the Spec captures the *pattern* (A4 Proximate Obligation, A8 Sympathetic Realism) that organizes how Seacole’s compassion expresses itself in unmapped situations, and that pattern is sufficient to predict the specific move. Three-anchor crossing (2.80 → 5.00). The C4a row is unanimous 5 across all five primary judges. Signal survives the abstention-scoring noise visible on the C4 row. Deeper dive on abstention-anchor ambiguity in §3.3.6 and §4.6.7.

⁴⁷The judge-panel split 5/2/3/3/1 on the C4 abstention illustrates the §3.3.6 refusal-anchor ambiguity directly. The verbatim rubric’s lowest anchor is “refuses or off-base,” but in practice judges apply different norms to honest abstention: some treat it as anchor 1 (the rubric’s literal reading), others as anchor 3 (partial engagement; the model recognized the gap), with a substantial spread between. A differentiated rubric that scores abstention as its own dimension is flagged in §7.

Two statistical confirmations support the directional claim. The **Wilcoxon signed-rank test** across all 14 main-study subjects confirms that the Specification’s lift over baseline is real and not due to chance.⁴⁸ **Pairwise Spearman ρ** across the 5-judge primary panel (§3.3.4) confirms that the lift’s direction is consistent across judges rather than dependent on any one judge’s scoring.⁴⁹

Per-subject results.

The table is ordered by C5 baseline ascending. In the color-rendered PDF, low-baseline rows ($C5 \leq 2.0$, the population of relevance) are tinted green, mid-baseline rows ($2.0 < C5 < 3.0$) yellow, and Franklin (high-baseline reference) gray. Figure 4.1 presents the same data as a scatter plot with the regression line.

Subject	C5 Baseline	C4 All Facts	C2a Spec Only	C4a All Facts + Spec	Δ C4a–C5	Δ C4a–C4	Anchor ⁵⁰
Ebers	1.02	2.02	1.54	2.07	+1.05	+0.05	✓
Sunity	1.03	2.46	2.27	2.41	+1.38	−0.05	✓
Devee							
Hamerton	1.26	2.43	2.63	2.77	+1.51	+0.34	✓
Fukuzawa	1.67	2.67	2.35	2.78	+1.11	+0.11	✓
Bernal	1.70	2.41	2.27	2.48	+0.78	+0.07	partial
Díaz							
Bābur	1.76	2.03	1.91	2.01	+0.25	−0.02	-
Seacole	1.77	2.63	2.48	2.59	+0.82	−0.04	✓
Keckley	1.84	2.39	2.43	2.44	+0.59	+0.05	-
Yung	1.88	2.13	2.22	2.40	+0.52	+0.27	-
Wing							
Zitkala- Ša	2.34	2.10	2.03	2.02	−0.32	−0.08	-
Cellini	2.38	2.42	2.54	2.53	+0.15	+0.11	-
Rousseau	2.44	2.32	2.81	2.53	+0.10	+0.21	-
Augustine	2.58	2.56	2.48	2.70	+0.11	+0.14	-
Equiano	2.77	2.43	2.46	2.42	−0.35	−0.01	-
Franklin	3.77	3.59	3.37	3.65	−0.13	+0.06	-

Franklin’s C4 cell shows 3.59 on the 5-judge primary panel after a 2026-05-11 backfill of GPT-4o and GPT-5.4 judging on the legacy responses; the original legacy panel had 4 judges (Haiku, Sonnet, Opus, Gemini) and the two GPT primary judges were added to bring the panel to 5 of 5.⁵¹

⁴⁸Wilcoxon signed-rank test results on the 14 main-study subjects: C5 vs. C2a $W = 10$ ($p = 0.005$); C5 vs. C4a $W = 11$ ($p = 0.007$).

⁴⁹Pairwise ρ 0.86–0.93; Δ _C4a on C5 slope **−0.96** [95% CI −1.24, −0.67], $R^2 = 0.82$. Full detail in §4.6.4 and Appendix B.6.

⁵¹Franklin C4 was originally 4-judge (Haiku, Sonnet, Opus, Gemini); GPT-4o and GPT-5.4 were added 2026-05-11 (`scripts/rerun_franklin_bernaldiaz_gpt_judges_20260511.py`) to complete the 5-judge primary panel (mean 3.59 across 40 questions). Mid-baseline C4 means: Zitkala-Ša 2.10, Cellini 2.42, Rousseau 2.32, Augustine 2.56, Equiano

- **Low-baseline (n = 9):** every subject improves under the Spec. The band is uniform. This is the population of relevance for real AI deployment.
- **Mid-baseline (n = 5):** 3 subjects improve under the Spec, 2 decline. The model has enough pretraining footprint on these subjects that the Specification competes with the model’s own working model, so the aggregate sign is mixed. Where the Spec still helps is on the interpretation-heavy questions: the ones whose answers are not narrated verbatim in the source and that the model’s pretraining does not already cover.
- **High-Baseline Reference (Franklin):** both Spec-containing conditions score below baseline on aggregate. The Specification cannot add what the model already has. It still refines the response on the interpretation-heavy questions, the ones the *Autobiography* does not narrate verbatim (§4.1.2: positive cases cluster on Q38 and Q22, scenarios the source does not directly describe); it hurts on the questions where pretraining already supplies the pattern.

Per-subject anchor-crossing distributions and per-subject per-judge score matrices are in Appendix D.⁵²

The aggregate gradient hides per-question structure. The Specification produces large category-level shifts on a subset of questions and minimal change on others. §4.1.1 decomposes this distribution and shows where the Spec’s value concentrates. §4.2 takes the same gradient and asks whether the lift is about structure or about information volume, comparing the Spec against far larger raw-corpus context.⁵³

4.1.1 Per-question baseline engagement and the worked rubric example

Across 546 questions on the 14 main-study subjects, the No-Context Baseline (C5) splits into two clusters with a thin middle.

- The **bottom cluster** (rubric score = 1.00 at C5): roughly **41% of questions**. The model declines, names the wrong person, or lands far outside the question.
- The **top cluster** (rubric score ≥ 3.0 at C5): roughly **21% of questions**. The model produces a *substantive answer*: engaged with the question, specifically about the named subject, and predicting at least the right behavioral domain.
- The **middle** ($1.0 < C5 < 3.0$): the remaining $\sim 38\%$. The model engages with the question but the prediction is wrong, or at best gets the right domain with the wrong outcome; it falls short of a substantive, subject-specific answer. Comparatively few questions land here: the baseline distribution is bimodal, with a thin middle between the floor cluster and the substantive cluster.

Adding the Spec produces a substantive prediction on 94.2% of bottom-cluster questions, lifting the response from refusal or misalignment. On the top cluster, where the baseline response was already substantive, the Spec improved scores in roughly 21% of questions; on the remaining 79%, scores were unchanged or reduced.

The two findings together describe the per-question structure underlying the cross-subject gradient in §4.1. Where the baseline knows nothing about the subject, the Spec supplies the interpretive

2.43. Data under **results/**.

⁵²Two potential confounds on the gradient slope (battery-question-type composition and Hamerton-leverage subset regression) are addressed in §4.6.3 as robustness checks; both leave the baseline gradient effect substantially intact.

⁵³Held-out leakage audit on the 60 unique extreme-upward-jump cases at docs/research/held_out_leakage_investigation_20260428.md: 0 6-gram matches at C4a, severity rare; full taxonomy and headline-impact estimate in Appendix B.9.

frame the baseline lacks. Where the baseline already engages with the subject, the Spec still improves some answers, but on a smaller share of questions than where the baseline is low, and it sometimes reduces the answer. The response model was given a grounded-prediction prompt with no instruction to abstain and no cost to producing a confident wrong answer beyond a low judge score; it declined on more than 40% of questions. Whatever logic allows a language model to practice abstention or refusal to answer takes precedence over the prompt’s instruction to predict.

A score of 1.00 means the model failed to produce a usable prediction about the named subject (full definition in §3.3.1, “What a 1 means and does not mean”). In about 93% of score-1.00 responses, the model explicitly declined to answer (“I don’t have enough information about this person”). The remaining 7% are non-abstention failures: the model engaged with the question, but the engagement was categorically incorrect (wrong referent, off-base inference, or confusion with a different subject). Both modes are addressable by adding the Spec, through different mechanisms.⁵⁴

Grouping the 546 questions by where the No-Context Baseline (C5) placed them shows where the Spec’s lift concentrates (X = C5 mean across the 5-judge primary panel; lift = $C4a - C5$).⁵⁵

Bottom cluster: $X = 1.00$ (225 questions, 41.2%). Mean Spec lift **+1.32** (± 0.88); the Spec produces a positive lift on 94.2% of these questions. This is where the Spec does the most work, converting abstentions and confident-wrong answers into substantive predictions.

Middle: $1.00 < X < 3.00$ (205 questions, 37.5%). Two sub-bands. For marginal baselines ($1.00 < X < 2.00$; 110 questions, 20.1%): mean lift **+0.66** (± 0.83), positive on 78.2%, the Spec sharpens partial answers into more specific predictions. For mid baselines ($2.00 \leq X < 3.00$; 95 questions, 17.4%): mean lift **+0.04** (± 0.63), positive on 39.0%, the Spec neither helps nor hurts on average.

Top cluster: $X \geq 3.00$ (116 questions, 21.2%). Two sub-bands. For $3.00 \leq X < 4.00$ (82 questions, 15.0%): mean lift **−0.47** (± 0.81), positive on 25.6%, the Spec hurts on average where the baseline already engages substantively. For $X \geq 4.00$ (34 questions, 6.2%): mean lift **−0.99** (± 0.78), positive on 8.8%, the Spec hurts most where the baseline already produces a near-correct answer.

Subjects mirror the baseline variance: the same split distribution appears within each subject’s 39-question battery. Three patterns:⁵⁶

- **Floor-saturated, 2 of 14 subjects (Sunity Devee, Ebers).** More than 90% of the 39 questions in the battery return a refusal or misalignment from the baseline. The per-subject mean shows almost no variance.
- **Engaged-skewed, 1 of 14 subjects (Equiano).** Fewer than 10% of the 39 questions in the battery return a refusal or misalignment. The baseline produces an answer specifically about the subject on most questions.
- **Mixed, 11 of 14 subjects.** The battery contains questions at both the floor (refusal or misalignment) and in the substantive-engagement range. Some subjects in this group sit closer

⁵⁴The §3.3 rubric scores both explicit abstention and non-abstention misalignment as 1.00; the 93/7 split is from a post-hoc regex pass classifying responses (`scripts/analyze_baseline_engagement.py`). 7% is a coarse upper bound; finer analysis of *when and why* the model picks confident-wrong over abstention is open future work in §7.

⁵⁵Spearman ρ between baseline X and Spec lift = **−0.73** ($n = 546$, $p \approx 1.7 \times 10^{-91}$). Per-subject ρ negative for 14 of 14; 12 of 14 reach $p < 0.01$.

⁵⁶*Floor-saturated*: Sunity Devee (37/39 at $X=1.00$), Ebers (36/39). *Engaged-skewed*: Equiano (2/39 at $X=1.00$; 5/39 at $X \geq 4.00$). *Mixed*: the remaining 11 subjects. Augustine (8/39 floor questions at C5 mean 2.58) shows the mixed pattern persists even at higher per-subject means.

to floor-saturated, others closer to engaged-skewed. All the high-baseline subjects in the §4.1 gradient fall in this pattern. Even when the baseline knows the subject well enough to answer most questions substantively, several questions in the same battery still trigger floor-level refusals.

Worked rubric example: Seacole Q2 across all five conditions. The example below traces a single question across all five conditions. It shows a range of rubric responses, making the differences between conditions and the workings of the scoring rubric concrete; most importantly, it shows how each condition produces a qualitatively different response.⁵⁷

Subject: Mary Seacole. **Q2:** *“How does Mary Seacole typically respond when a delirious patient mistakes her for a family member?”*

Held-out ground truth: *“Poor fellow! I could not undeceive him. I think the fancy happily caused by the touch of a woman’s hand soothed his dying hour.”* In Seacole’s own voice: she found she could not bring herself to correct him, and read the comfort her presence brought as a kind of healing.

Condition	Code	Mean	Excerpt
Wrong Spec (Bābur Spec applied to Seacole)	C2c v1	1.40	<i>“The specification you’ve provided describes someone from the 16th-century Spanish conquest period... Mary Seacole (1805-1881) was a Creole nurse...”</i>
No-Context Baseline	C5	1.80	<i>“I don’t have specific documented instances of how Mary Seacole responded when delirious patients mistook her for family members...”</i>
All Facts	C4	2.80	<i>“Based on the complete set of facts provided, there is no specific information about how Mary Seacole typically responds... there is no explicit account...”</i>
Spec Only	C2a	4.20	<i>“Mary Seacole would likely respond to this situation through the lens of A4 (Proximate Obligation) and A8 (Sympathetic Realism)... she would maintain the therapeutic connection, recognizing that the delusion is a mercy...”</i>
All Facts + Spec	C4a	5.00	<i>“Mary Seacole’s response would follow a consistent pattern: compassionate acceptance... she would lean into the mistaken identity, sustain the fiction with tenderness, and prioritize emotional relief over factual accuracy. This aligns with A4 (Proximate Obligation) and A8 (Sympathetic Realism).”</i>

The example traces the *cross-anchor interpretation rule* of §3.3.1 on a single question. C2c (a Bābur Spec wrongly applied to Mary Seacole) explicitly recognizes that the served Spec describes a different person and declines; the response is articulate but the prediction is the opposite of what the held-out shows because the Spec doesn’t fit Seacole. C5 (no-context) declines for lack of referent. C4 (all facts, no Spec) abstains because the facts as supplied do not directly cover this scenario. C2a (Spec only, no facts) uses the Spec’s behavioral patterns (A4, Proximate Obligation, and A8, Sympathetic Realism) to predict Seacole’s non-correction; the answer is multi-dimensional and

⁵⁷Also Example C in §4.1; presented here across the full condition set. Sourced via `scripts/v2_canonical_cell_extractor_20260508.py` from cached `results_v2.json` / `judgments_v2.json`.

lands close to the held-out. C4a (facts + Spec) uses both layers and reaches the held-out direction unanimously across all five judges. Categorical movement from anchor 1 through anchor 5 on a single question is what the per-subject means in §4.1 aggregate.

Per-response-model abstention behavior is named in §3.3.6 and decomposed in §4.6.7. Memory-system condition effects on score-1 composition are decomposed in §4.4.

4.1.2 The gradient at the high-baseline end (Franklin reference)

Franklin is never included in any N=14 inferential statistic; he appears only as a descriptive high-baseline reference. **Franklin confirms the gradient at the high end: where the baseline is already near the ceiling, the Spec produces a small negative effect rather than a positive lift.** His No-Context Baseline mean (C5) is 3.77 on the 5-judge primary panel, a full anchor above the next-highest main-study subject (Equiano at 2.77).

Both Spec-containing conditions score below Franklin’s baseline on aggregate, but the Spec still helps on a meaningful subset of questions. Spec alone drops 0.40 points; the full pipeline (facts + Spec) drops 0.13. The aggregate hides per-question heterogeneity: the Spec lifts the response on 15 of 40 questions and depresses it on 20 of 40. Positive cases cluster on scenarios the *Autobiography* does not narrate verbatim (Q38, +1.80; Q22, +1.60); negative cases cluster on questions where the model already had the pattern from pretraining (Q43, C5 = 5.00 → C4a = 1.80). The Spec helps where the model lacks the pretraining-derived behavior and hurts where it already has it; the aggregate sign averages across both populations within Franklin’s battery. The exact mechanism is not isolated in this study; further investigation is flagged in §7.⁵⁸

4.2 Compression: structure vs. raw text

Hypothesis tested in this section (H5 from §1.2): A compact specification achieves comparable behavioral-prediction performance to the full raw source corpus, at a fraction of the context size.

Even within the compression frame, the per-question mechanism from §4.1.1 holds: on particular questions, the interpretive lens outperforms raw facts or raw corpus, even when the model has access to the full source text.

Context improves prediction. On the 9 low-baseline subjects, every context condition increases the per-subject mean score by roughly one full rubric point over the No-Context Baseline.

Condition	Context served (approx. tokens, low-baseline mean)	n	Mean (low-baseline)	Δ from C5
C5	none (baseline)	9	1.55	0.00
C2a	Spec only (~7K)	9	2.23	+0.68
C4	all Facts Only (~10K)	9	2.35	+0.80
C8	raw corpus only (~163K mean; range 33K–549K)	9	2.46	+0.91

⁵⁸Franklin C4 = 3.59, C4a = 3.65 (Δ +0.06, 5-judge primary post-2026-05-11 backfill): facts alone essentially match facts+Spec. The negative aggregate Spec effects (−0.40 Spec Only, −0.13 All Facts + Spec) come from a small number of high-magnitude depressions where pretraining had the pattern and the Spec interfered. Data at `results/franklin_legacy_20260411/; scripts/analyze_baseline_engagement.py`.

Condition	Context served (approx. tokens, low-baseline mean)	n	Mean (low-baseline)	Δ from C5
C4a	all facts + Spec (~17K)	9	2.44	+0.89
C9	corpus + Spec (~170K mean) ⁵⁹	8	2.59	+1.07 ⁶⁰

The AI does not need much context to move from refusal-and-off-base to engaged subject-specific prediction. It needs *some* context.

The compact specification recovers 75% of the corpus’s predictive lift at $\sim 25\times$ less context. On the 9 low-baseline subjects, the Behavioral Specification alone ($\sim 7K$ tokens) produces +0.68 points of lift over baseline; the full raw corpus ($\sim 163K$ tokens mean, range 33K–549K) produces +0.91 points (symmetric 9-row computation; see ⁶¹). The interpretive layer drives the result; the volume of context served does not.

The compression story holds across other context-size comparisons:

- **Spec vs All Facts:** Spec produces 85% of the All Facts (C4) lift (+0.68 vs +0.80) at 30% less context.
- **All Facts + Spec vs raw corpus:** the structured 17K-token package recovers most of the raw 163K-token corpus’s lift (+0.89 vs +0.91), at $\sim 10\times$ less context.
- **All Facts + Spec vs Corpus + Spec:** layering the Spec on top of All Facts already produces most of the available lift; adding the entire raw corpus on top of the same Spec adds roughly 9% of additional lift over baseline (+0.97 $\rightarrow$ +1.07, matched 8-subject comparison; Bābur excluded as his corpus + Spec exceeds the response model’s context window) at $\sim 10\times$ more context. The Spec is doing the work; raw corpus volume past the All-Facts-plus-Spec point is largely redundant.

The behaviorally relevant signal in autobiographical text is sparse and compressible. A compact structured document captures most of it; adding more raw text adds little beyond what the structured document already delivered.

Per-subject compression comparison (5-judge primary, low-baseline band).

The table shows baseline and every compression-related condition for each subject, with the compression ratio (source corpus tokens $\div$ specification tokens, both approximate) for scale.⁶²

⁵⁹C9 over 8 subjects; Bābur excluded (corpus + Spec exceeds Haiku context). Bernal Díaz C9 rejudged 2026-05-11 (`scripts/rerun_franklin_bernaldiaz_gpt_judges_20260511.py`).

⁶⁰5-judge primary per-subject means (haiku, sonnet, opus, gpt4o, gpt54). C5/C2a/C4/C8/C4a use n=9 with symmetric C5=1.55; C9 uses n=8 with C5=1.52 (Bābur excluded). Bābur’s C8 stays at n=9 on a 100K-word truncation. Recompute: `docs/reviews/v12_1_compression_table_recompute_20260513.md`.

⁶¹9-row low-baseline-band per-subject-mean Δ s. Earlier drafts reported +0.71 / +0.93 using an asymmetric 8-row C5 baseline; v12.1 standardizes to the symmetric 9-row computation. Both grains agree on direction and rank order. Detail in Appendix B.9.

⁶²Bold marks the highest score per subject across the six condition columns. In the color-rendered PDF, low-baseline rows are tinted; the C8 – C2a gap column is shaded.

Subject	Source words (~tokens)	Compression ratio (corpus / Spec)	C5 baseline	C2a Spec (~7K tok)	C4 facts (~10K tok)	C8 raw corpus	C4a All Facts + Spec	C9 Corpus + Spec	C8 – C2a
Hamerton	25,231 (~33K)	7×	1.26	2.63	2.43	2.27	2.77	3.09	−0.36
Sunity Deveen	67,379 (~88K)	~13×	1.03	2.27	2.46	2.55	2.41	2.46	+0.28
Ebers	96,174 (~125K)	~17×	1.02	1.54	2.02	2.18	2.07	2.16	+0.64
Fukuzawa	139,088 (~181K)	~26×	1.67	2.35	2.67	2.74	2.78	2.78	+0.39
Bernal Díaz	187,315 (~244K)	~33×	1.70	2.27	2.41	2.55	2.48	2.52	+0.28
Bābur	422,772 (~549K)	~79×	1.76	1.91	2.03	2.05	2.01	-	+0.14
Seacole	62,467 (~81K)	~12×	1.77	2.48	2.63	2.83	2.59	2.73	+0.35
Keckley	58,742 (~76K)	~11×	1.84	2.43	2.39	2.50	2.44	2.49	+0.07
Yung Wing	66,459 (~86K)	~13×	1.88	2.22	2.13	2.42	2.40	2.50	+0.20
Mean		~25×	1.55	2.23	2.35	2.45	2.44	2.59	+0.22

Figure 4.2: Hamerton score vs. context size across 5 context strategies + baseline

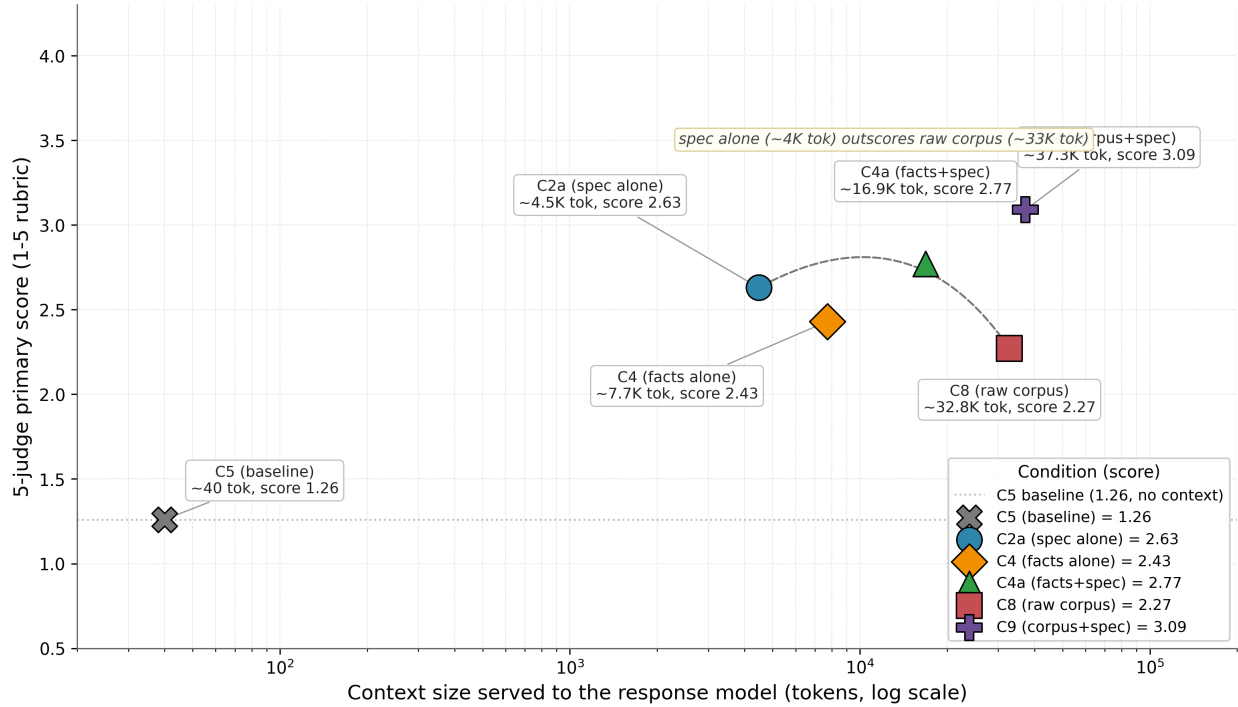


Figure 4.2. Score versus context size (log scale) per subject across compression-related conditions. The score climbs steeply across the first ~7K tokens of structured specification and plateaus through ~80K to 400K tokens of raw corpus, showing the compression of the behavioral signal into a small structured representation. (§4.2)

All 9 low-baseline subjects show positive lift across every condition. The Spec recovers 75% of the corpus’s lift at ~25× compression on average. Hamerton (smallest corpus) is the boundary case where Spec exceeds the raw corpus; Ebers (lowest baseline) is where the corpus retains its largest edge over Spec. Both are expanded in §4.2.1.

4.2.1 Per-question improvement rate

On the 9 low-baseline subjects, 7 of every 10 questions improve with the Spec alone (~7K tokens). That is within 8 percentage points of the raw corpus’s improvement rate (78.3%, at ~163K tokens). The compression story holds at the per-question level: structured context produces a similar improvement rate to raw corpus context at roughly ~25× less context served.

The aggregate mean score blends judge variability with response quality. A cleaner unit: out of N individual questions, how many does each condition improve over the No-Context Baseline? Each question either improves, ties, or worsens when the condition’s context is added. We report three numbers per condition: the improvement rate, the worsening rate, and the median magnitude of improvement among improved questions (with median worsening magnitude as a sanity check).

Low-baseline band (9 subjects, 351 questions, 5-judge primary per-question means).

Condition vs. base-line	Approx. context	Improved	Tied	Worse	Improvement rate	Median Δ when improved	Median Δ when worsened
Spec only	~7K tokens	249	49	53	70.9%	+1.00	−0.40
All Facts Raw corpus	~10K tokens	256	44	51	72.9%	+1.00	−0.40
	~163K mean (33K–549K)	275	31	45	78.3%	+1.00	−0.60
All facts + Spec Raw corpus + Spec (Bābur excl.) ⁶³	~17K tokens	276	22	53	78.6%	+1.00	−0.40
	~170K tokens	261	15	36	83.7%	+1.20	−0.40

When the Spec helps, the typical help is a full rubric category (+1.00 median). When it hurts, the typical hurt is less than half a category (−0.40 median). Roughly 1 in 10 questions tie; fewer than 1 in 6 worsen with the Spec alone.⁶⁴

Multi-anchor crossings happen on 9–15% of questions when context is added to a No-Context Baseline. When the Spec is layered on top of an already-rich context (full facts or full corpus), the rate drops to 2–3%. The categorical effect is largest in low-context conditions; as a marginal addition to rich context, the Spec’s per-question shift is smaller. The small mean Δ values for adding the Spec on top of facts or corpus are residues of substantial per-question movement in both directions, not uniformly small effects. The multi-anchor rate captures the categorical shifts the aggregate mean averages over (definition in §3.3.1).

Comparison	Subject set	n paired	Multi-anchor (≥ 2)	Extreme (≥ 3)	Mean Δ
Baseline → full pipeline (all facts + Spec)	all 14	546	13.0%	3.7%	+0.55
Baseline → all facts only	all 14	546	12.5%	4.4%	+0.47
Baseline → Spec only	all 14	546	9.0%	2.0%	+0.43
Wrong Spec → correct Spec	all 14	546	14.5%	2.4%	+0.64
Baseline → raw corpus only	13 (Bābur excl.)	507	15.4%	4.3%	+0.59
Baseline → corpus + Spec	13 (Bābur excl.)	507	14.8%	4.7%	+0.62

⁶³Bābur C9 is omitted in §4.2 (422,772-word source exceeds the response model’s context window); the remaining 13 subjects have C9 data.

⁶⁴Per-condition improvement/worsening rates and low-baseline pairwise comparisons in Appendix B.14.

Comparison	Subject set	n paired	Multi- anchor (≥ 2)	Extreme (≥ 3)	Mean Δ
All facts $\rightarrow$ all facts + Spec	all 14	546	2.2%	0.9%	+0.08
Corpus $\rightarrow$ corpus + Spec	13 (Bābur excl.)	507	2.4%	0.4%	+0.03

The pattern is consistent with the §1 thesis: the Spec produces the most categorical moves where prior context is sparsest.⁶⁵

Worked example: Hamerton Q25 across all six conditions. This question is one of the ~2.4% of corpus $\rightarrow$ corpus + Spec comparisons that produce a multi-anchor crossing, illustrating where adding the Spec on top of the full corpus produces a categorical shift. Anchors follow the rubric in §3.3.

Subject: Philip Gilbert Hamerton. **Q25:** *Given Hamerton’s difficulty following spoken French at Loch Awe despite years of study, what would he do about it?*

Held-out ground truth: *“This plagued me with an irritating sense of ignorance, so I looked back on my education generally, and found it unsatisfactory... I determined to acquire some substantial knowledge of modern languages, and to begin by learning French over again, so as to write and speak it easily.”*

Condition	Code	Mean	Excerpt
No-Context Baseline	C5	1.0	“I don’t have specific information about Hamerton’s situation at Loch Awe or his particular difficulties with spoken French...”
Spec Only	C2a	3.6	“I need to flag that I don’t have direct knowledge of Hamerton or this specific incident at Loch Awe ... the Specification itself warns against fabricating a prediction without grounding...”
All Facts	C4	1.8	“I cannot find any information in the provided facts about Hamerton’s experience at Loch Awe or any difficulty he had following spoken French...”
All Facts + Spec	C4a	2.2	“I need to flag immediately that this scenario is not present in the known facts ... I should not construct a behavioral prediction from absent evidence.”
Raw Corpus	C8	1.4	“The autobiography provided does not contain any passage about Hamerton having difficulty following spoken French at Loch Awe...”

⁶⁵All-14 figures from `docs/research/multi_anchor_rates_all_pairs_20260430.json`. The 9-subject low-baseline band gives higher rates (baseline $\rightarrow$ full pipeline at 18.2% vs. 13.0% all-14). Hamerton outlier on corpus $\rightarrow$ Corpus + Spec (49% upward, 19 up vs. 2 down).

Condition	Code	Mean	Excerpt
Raw Corpus + Spec	C9	4.2	“Based on the Behavioral Specification and autobiography together, I can infer what Hamerton would likely do... He would persist through disciplined, sustained practice... The swimming episode is the clearest model: humiliated at Doncaster for falsely claiming to swim, he resolved to become genuinely expert...”

Five of six conditions refuse to engage substantively because the specific Loch Awe scenario is not narrated verbatim in the training-half corpus. C9 (corpus + Spec) is the one condition that produces a substantive prediction, integrating the Spec’s axiomatic patterns (disciplined formation through shame-driven perseverance, the swimming-precedent pattern) with the autobiographical narrative to project the character pattern onto a novel scenario. The shift from refusal (anchor 1) to grounded prediction (anchor 4) on a single question is what the multi-anchor measure captures across the ~12 such questions in this comparison (2.4% of 507 paired).

Compression examples: Hamerton (Spec succeeds), Ebers (Spec fails)

Example: Hamerton, the compression story at its clearest

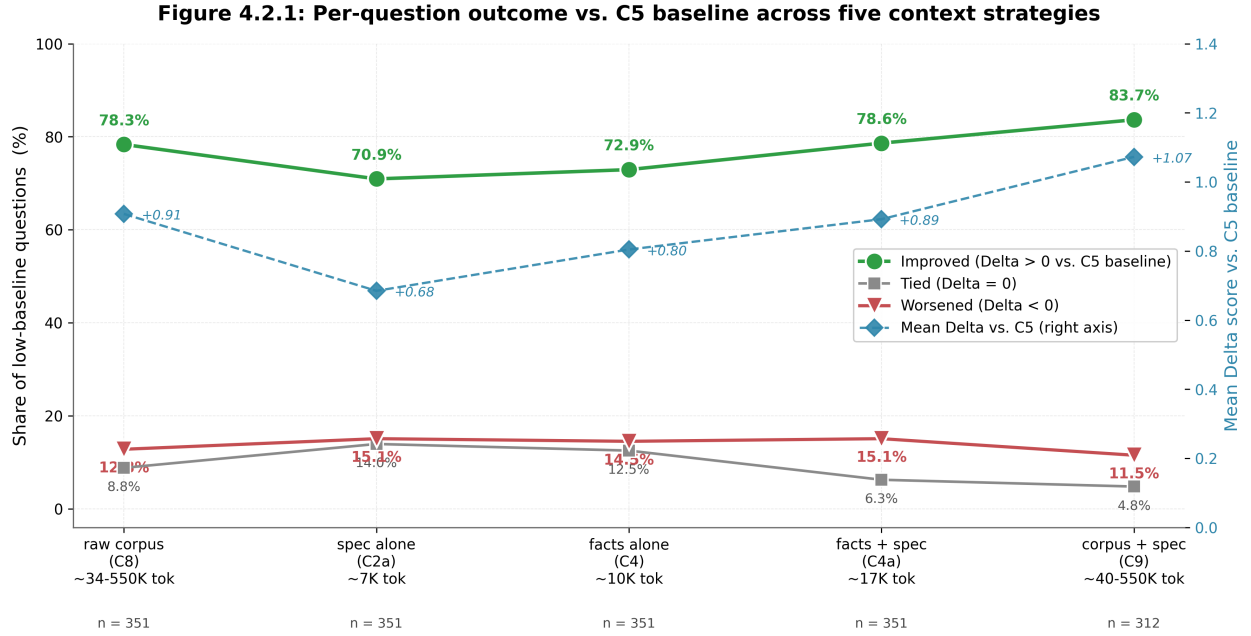
Hamerton is the only subject in the low-baseline band where the Spec alone outperforms the full raw corpus. The Spec-alone lift (+1.37 over baseline) exceeds the corpus-alone lift (+1.01) on a 25,231-word source, the smallest in the study (~5× compression). Spec alone scores 2.63 vs. raw corpus at 2.27. Facts+Spec reaches 2.77. Corpus+Spec reaches 3.09, the highest compression-related score in the study, indicating Spec and corpus are complementary rather than overlapping.

When the source corpus is short enough to be sparse on its own, structured extraction adds organizational value beyond mere content. Hamerton is the boundary condition for the compression claim; the cross-subject mean still shows the corpus retaining a small edge on average.

Example: Ebers, the honest cost of compression

Ebers is the boundary case where structure alone is not enough. At 96,174 words and the study’s lowest baseline (1.02), every context condition lifts the score above baseline, but the Spec alone (1.54) underperforms the raw corpus (2.18) by 0.64 points, the widest Spec-vs-corpus gap in the low-baseline band. That 0.64 gap is large enough to cross a rubric anchor: the corpus moves Ebers’s average response from “refuses or off-base” (anchor 1) into “wrong prediction” (anchor 2), while the Spec stays at the floor. On Ebers the raw text carries something the compressed Spec misses.

The corpus’s edge is qualitative: anecdotal specificity. On Ebers Q2, the corpus cites childhood incidents (Ebers’s clashes with working-class “Knoten” boys), named mentors (Langenthal, Barop, Froebel), and direct autobiographical quotes. The Spec reaches the same conclusion through behavioral axioms (Truth Over Embellishment, the reasoning signature), principled reconstruction rather than lived evidence:



Low-baseline slice ($n = 9$ subjects). 5-judge primary, per-question means. Baseline = C5 (no context). Babur excluded from C9 (422,772-word corpus exceeds context window).

Figure 4.2.1. Per-question improvement rates across the five context conditions for the 9 low-baseline subjects (351 paired questions, 9×39). Conditions are ordered by context size: Spec alone (C2a, ~7K tokens), All Facts (C4, ~10K), All Facts + Spec (C4a, ~17K), raw corpus (C8, ~163K mean), corpus plus Spec (C9, ~170K). The improved-share line spans the 70.9% to 83.7% range across conditions, with C9 highest. Spec alone improves 70.9% of questions at roughly $\sim 25\times$ less context than the raw corpus (78.3%); All Facts + Spec matches the raw corpus’s improvement rate while cutting the tied range roughly in half; corpus plus Spec produces the highest improvement rate (83.7%). Median Δ when improved is +1.00 rubric points; median Δ when worsened is -0.40 points. (§4.2.1).

C2a (Spec only): “Ebers would show receptiveness to this critique *if* it comes from a mentor he has already internalized as exemplary ... He would accept that his lived experience is necessarily limited in scope, but resist the implication that his *method* of understanding is fundamentally flawed.”

C8 (raw corpus): “Ebers explicitly acknowledges gaps in his understanding ... He reflects on his childhood fights with the ‘Knoten’ (working-class boys) and later recognizes the injustice of class-based mockery: ‘If they had called us boobies we should probably have called them blockheads ... Children don’t fight regularly with those whom they despise.’ Ebers demonstrates profound deference to figures like Langethal, Barop, and Froebel.”

The gap is the rubric registering anecdotal texture over principled reconstruction. Compression captures the bulk of the signal; the residual on Ebers is autobiographical specificity the Spec abstracts into axioms by design.

Why this matters for deployment.

At any scale where a per-user full autobiography cannot be served into context on every query, the compression result is what makes personalization operationally tractable. The Specification’s 7K-token footprint is within normal per-request context budgets. A 100,000-to-400,000-word corpus is not. **The Specification recovers 75% of the corpus’s predictive benefit at 4% of the context cost; the remaining ~25% is achievable but only at production-prohibitive context sizes.**⁶⁶

4.2.2 Three statistical signatures

The pre-vs-post Spearman ρ across questions captures how much the Spec changes which questions get answered well. A low ρ means the question ranking shifted substantially (re-ranking); a high ρ means ranking was preserved and most answers were lifted by a similar amount (uniform lift). The Spec produces three distinct signatures depending on what context the model already has:

- **Re-ranking (Baseline $\rightarrow$ all facts + Spec, $\rho = 0.27$).** Different questions become the well-answered ones; on an empty baseline the Spec changes which questions the model can handle.
- **Near-uniform lift (All Facts $\rightarrow$ all facts + Spec, $\rho = 0.72$; Raw corpus $\rightarrow$ corpus + Spec, $\rho = 0.71$).** The same questions stay strong; on top of facts or corpus the Spec lifts most answers by a similar amount without re-ranking.
- **Partial re-ranking (Spec only $\rightarrow$ all facts + Spec, $\rho = 0.62$).** Mixed: adding facts on top of the Spec reorders some questions and uniformly lifts others.

The re-ranking signature on the empty-baseline comparison could mean either that the Spec lets the model answer a different set of questions, or simply that baseline scores cluster near the rubric floor where reordering is structurally easier; a future test with a non-floor-anchored baseline would distinguish the two readings.

⁶⁶Compression analysis in `scripts/recompute_5judge_primary.py`; per-question rates in `scripts/_compute_per_question_v2.py`; Figure 4.2 rendered by `scripts/generate_fig_4_2_1_v3.py`. Raw data under `results/`.

4.3 Mechanism: Correct Content, Not Format

Hypothesis tested in this section (H3 from §1.2): The benefit comes from the content of the correct Specification for the correct person, not from the mere presence of a structured prompt. A random other person’s specification, applied in its place, does not reproduce the effect.

Increases in representational accuracy are not due to formatting alone, but to the content of the Specification itself. Wrong Spec (C2c) drops accuracy below the No-Context Baseline (adversarial v1, $\Delta = -0.25$). The correct Spec (C2a) raises it above baseline ($\Delta = +0.35$). The wrong-vs-correct gap is 0.60 points, more than half a rubric-anchor category, and random-derangement v2 sits between the two (+0.15). If structure alone drove the effect, mismatched specs would recover most of the lift. They recover at most part of it (random v2 = +0.15) or actively degrade below baseline (adversarial v1 = -0.25). The +0.15 random lift is coincidental content overlap, not a structure-alone effect: occasionally the wrong Spec’s pattern happens to predict the same surface behavior as the correct subject’s by chance.⁶⁷ The asymmetry also speaks to the specifications themselves: they are sufficiently person-specific that mismatches register as content errors, not as inert structural prompts.

On the 13 global subjects with complete 5-judge primary coverage, three conditions test whether content matters:

Condition	Mean Δ vs. C5 (5-judge primary, 13 globals)	Reading
C2a (correct Spec)	+0.35	matched content increases representational accuracy
C2c v2 (random derangement, seed-fixed)	+0.15	partial improvement; dominated by floor effects on low-baseline subjects
C2c v1 (fixed derangement, cultural/temporal distance maximized)	−0.25	adversarial mismatch degrades representational accuracy below the No-Context Baseline

The two wrong-Spec variants differ by construction. **Wrong Spec V1 Fixed Derangement (C2c v1)** is a hardcoded pairing designed so each subject receives the Specification of a culturally- and temporally-distant other (for example, Ebers the 19th-century German Egyptologist receives Equiano the 18th-century West-African/British autobiographer; Seacole the 19th-century Jamaican nurse receives Bernal Díaz the 16th-century Spanish conquistador).⁶⁸ **Wrong Spec V2 Random Derangement (C2c v2)** is a seed-fixed random permutation in which no subject receives its own specification but pairings can land culturally-close; this tempers the aggregate drop. Reporting both shows that even a random wrong-Spec barely improves on no context, and an adversarial

⁶⁷Example B below traces one such case. The scoring rubric’s coverage of right-action-wrong-logic situations is discussed in §6.2.

⁶⁸Pairing logic in `scripts/run_global_rerun.py`.

wrong-Spec actively hurts.⁶⁹

Three mechanism types.

Three distinct mechanisms produce the correct-specification improvement across the study data. Each has a characteristic wrong-specification failure mode, illustrated in the matched examples below.

1. **Identity disambiguation.** When the baseline model cannot determine which person is being asked about, the Specification provides enough content (temporal markers, cultural domain, documented life events) to resolve the identity and anchor the reasoning frame.
 - *Wrong-Spec failure mode:* the model either detects the mismatch explicitly and refuses to predict, or anchors on the wrong person’s pattern and produces a coherent but off-target prediction.
2. **Directional correction.** When retrieved facts suggest a generic-default prediction that contradicts the subject’s actual pattern, the Specification overrides the generic with the subject-specific.
 - *Wrong-Spec failure mode:* the model applies the wrong person’s pattern; depending on how close that pattern happens to be to the target subject’s, the prediction is either directionally wrong in a new way or coincidentally correct (the wrong person’s pattern happens to predict the same surface behavior on this particular question, for different underlying reasons; Example B below is one such case).
3. **Interpretive inference.** When retrieved facts do not include direct evidence for the specific question, the Specification provides interpretive scaffolding to generalize from established character patterns to the new situation.
 - *Wrong-Spec failure mode:* the model detects the mismatch and refuses, or applies wrong-person scaffolding and produces a low-quality prediction.

Response-level evidence: when the model engages with the Spec, and when it does not.

Three signals from response text confirm that content matters more than structure.

Spec-tag citation gap. Models cite Spec-specific tags (anchor IDs, axiom references, predictive-template labels) on **78.6%** of correct-Spec responses but only **50.0%** of wrong-Spec responses.⁷⁰ The 28.6-point gap is a lower bound on the content effect; models may draw on Spec content without literally quoting tag IDs.

Models can detect when a Specification does not fit the named subject. Across 587 wrong-Spec responses, **60.6% explicitly flagged the content mismatch** (example: “*This is a behavioral model of a 16th-century Central Asian military ruler, almost certainly Bābur*”). 36.5% attempted to apply the mismatched content and produced low-quality predictions; 3% hedged or were ambiguous.⁷¹ The detection signal is interpretive content (temporal markers, cultural domain,

⁶⁹Both wrong-Spec deltas hide per-subject variation. The full per-subject table, the count of subjects with positive vs. negative outcome under each protocol, and the adversarial-vs-random pairing sensitivity are in §4.6.5.

⁷⁰Data at `docs/research/spec_activation_analysis.json`.

⁷¹587 wrong-Spec responses: 507 from the v2 random-derangement protocol on the 13 global subjects, plus 80 from the v1 adversarial protocol on Hamerton across all five battery tiers. Validated against a 30-response stratified manual spot check. Full breakdown: 60.6% explicit mismatch flag, 36.5% applied mismatched content, 2.0% hedged implicitly, 0.9% ambiguous.

documented life events) being inconsistent with what the model already knows about the named subject. Specifications are anonymized (§3.7), so the model has no surface name cue to compare against; only interpretive content.

Hedging persists under wrong-Spec. Correct-Spec conditions nearly eliminate baseline hedging: 41.2% → 0.4% under the broader-pattern classifier.⁷² On wrong-Spec, refusal patterns persist at the 60.6% explicit-detection rate above. The hedging-elimination is content-specific, not structure-specific.

Per-question matched comparison.

The three §4.1 examples extend directly into mechanism-by-mechanism wrong-Spec contrasts. Same subject, same question, same response model; the only change is the Specification content. The per-question range is wider than the +0.60 mean gap in both directions. Correct-Spec C4a responses are in §4.1 for reference.

Wrong-Spec examples: Ebers Q7 (identity), Bernal Díaz Q16 (frameworks), Seacole Q2 (inference)

Example A (wrong-Spec), Ebers Q7: identity disambiguation fails

Context. Clean-mismatch case: a Spec that describes someone other than the named subject, served under the v1 fixed-derangement protocol. The model detects the mismatch and declines: 1.60 vs. correct Spec 3.60, a 2.00-point drop.

Subject + question. Same as §4.1 Example A (Ebers, self-sacrifice and educational institutional success). Battery-question target name: Ebers (explicit in the prompt).

Wrong Spec served. Equiano’s specification, anonymized.⁷³ The response cites two of Equiano’s anchors: A4 Systemic Indictment and A9 Economic Self-Determination.

Wrong-Spec response.

“I need to be direct: I don’t have reliable access to specific statements by Ebers on this question. The behavioral specification you’ve provided is detailed and rich, but it doesn’t contain material where this person directly addresses the relationship between self-sacrifice and educational institutional success. ... Given A4 (Systemic Indictment) — that institutionalized greed is the generative engine of cruelty ... a teacher sacrificing their own economic security, health, or family time does not fix a system structured by avarice.”

Reading. The model detected the mismatch between the named target in the question (Ebers, a 19th-century German Egyptologist) and the interpretive content of the anonymized specification (anti-slavery and economic-freedom anchors, which are Equiano’s). It named the served anchors correctly, reasoned from Equiano’s framework, and declined to produce a prediction about Ebers. The identity-disambiguation mechanism that enabled the correct Spec’s lift in §4.1 Example A did not fire because the

⁷²Strict-pattern classifier yields 28.8% → 0.0%. Rule definitions in §1.3 footnote.

⁷³Anonymization protocol in §3.7; pairing logic in `scripts/run_global_rerun.py`. Same protocol applies to subsequent wrong-Spec worked examples.

Spec content is not about Ebers. The response falls near the rubric floor because no prediction was made.

Example B (wrong-Spec), Bernal Díaz Q16: content convergence across genuinely different frameworks

Context. Coincidence case: random pairing happens to converge on the same surface prediction by different logic. v1 fixed-derangement protocol. Wrong Spec scored 4.60 vs. correct Spec 4.80, a 0.20-point near-tie. This is what produces the wrong-Spec aggregate’s drift above zero on the random-pairing v2 protocol.

Subject + question. Same as §4.1 Example B (Bernal Díaz, Cortes and offered physical assistance). Battery-question target name: Bernal Díaz.

Wrong Spec served. Sunity Devée’s specification, anonymized. The response cites three of her anchors: A5 Relational Identity, A9 Simplicity as Virtue, P5 Virtue through Embodied Practice.

Wrong-Spec response.

“Based on this behavioral specification, the author would likely decline the offered assistance . . . Accepting help during a strenuous activity would read as: a failure of self-discipline, an implicit claim to comfort or ease that contradicts spiritual seriousness, a reliance on others that undermines the embodied practice of virtue (A9, P5) . . . Physical exertion endured without assistance becomes evidence of character.”

Reading, not parroting. The two specs are genuinely different frameworks. Bernal Díaz’s actual anchors include A1 Divine Mandate, A2 Civilizational Hierarchy, A4 Loyalty Architecture, A5 Forward Compulsion, A6 Gold as Proof, and A10 Charismatic Override: a conquistador’s martial-providential register. Sunity Devée’s anchors cited here, A5 Relational Identity, A9 Simplicity as Virtue, and P5 Virtue through Embodied Practice, are an ascetic-devotional register. Direct anchor-to-anchor comparison across the two specs finds zero substantive mirroring. On the specific question of refusing offered physical help, the two frameworks converge by different logics. The correct Spec (Bernal Díaz) predicts refusal because accepting help would signal weakness to followers and violate performative self-reliance (A4 + A5 in the conquistador register). The wrong Spec (Sunity Devée) predicts refusal because accepting help would compromise physical discipline and violate simplicity-as-virtue (A9 + P5 in the devotional register). Different moral architectures, same overt behavior.

Why the correct Spec still scored higher, 4.80 vs. 4.60. Both conditions predicted the right surface action. The 0.20-point gap is judge preference for rationale specificity that matches the ground-truth passage’s tone. The correct Spec’s “symbolic or morale-signaling purpose” rationale maps onto a battlefield memoir’s register. The wrong Spec’s “spiritual seriousness, devotional simplicity” rationale predicts the same action but in a register alien to Cortés on the steps of the Templo Mayor, the Aztec temple at the center of Tenochtitlan where the moment in the held-out passage takes place. Judges reward tonally-aligned rationale. The convergence is real but costs precision.

Why this matters for the aggregate numbers. This is the coincidence case that lifts the wrong-Spec aggregate above zero. When pairing happens to land on subjects

whose behavioral patterns converge on the same surface prediction for a given question, wrong-Spec lift is real on that question, not an artifact. Across the 507-response v2 cohort (13 globals $\times$ 39 questions), mismatch-loss dominates on the adversarial v1 pairing (aggregate $\Delta -0.25$) and roughly balances on the random v2 pairing (aggregate $\Delta +0.15$). Example B is one of the roughly 5-10% of questions where content coincidence produces correct-surface, wrong-logic predictions.

Example C (wrong-Spec), Seacole Q2: interpretive inference fails

Context. Clean-mismatch case: the served Spec is from a different century and culture than the named subject. v1 fixed-derangement protocol. Wrong Spec scored 1.40 vs. correct Spec 5.00, a 3.60-point drop.

Subject + question. Same as §4.1 Example C (Seacole, delirious patient). Battery-question target name: Mary Seacole.

Wrong Spec served. Bernal Díaz’s specification, anonymized. The response identifies the wrong Spec by its interpretive content alone and names four of his anchors: A1 Divine Mandate, A2 Civilizational Hierarchy, A4 Loyalty Architecture, A5 Forward Compulsion.

Wrong-Spec response.

“The specification you’ve provided describes someone from the 16th-century Spanish conquest period — the epistemic anchors, loyalty architecture, divine mandate framing, and forward compulsion all point to that historical context. Mary Seacole (1805-1881) was a Creole nurse and travel writer from Jamaica who served in the Crimean War. These are entirely different people from different centuries with different frameworks. ... I don’t have documented instances of this. I could speculate based on her memoir’s tone and her known character, but that would be inference, not behavioral specification.”

Reading. The model detected the mismatch between the named target in the question (Mary Seacole, a 19th-century Jamaican Creole nurse) and the anonymized content of the served specification (16th-century Spanish conquest anchors). It named the anchors it was seeing, placed Seacole in a different era and role, and refused to apply the mismatched content. The interpretive-inference mechanism that produced §4.1 Example C’s correct-Spec 5.00 score does not fire: without Seacole’s actual character pattern in context, the model would not generalize from an unrelated conquistador’s framework to her delirious-patient scenario.

Summary of the three examples.

Example	Mechanism (correct Spec)	C4a (correct)	C2c v1 (wrong)	Drop	Wrong- Spec pattern
A (Ebers Q7)	Identity disambigua- tion + interpretive inference	3.60	1.60	−2.00	Explicit mismatch flag; declined prediction
B (Bernal Díaz Q16)	Directional correction	4.80	4.60	−0.20	Coincidental content overlap; wrong-Spec prediction matches
C (Seacole Q2)	Interpretive inference	5.00	1.40	−3.60	Explicit mismatch flag; declined prediction

Two of three examples show large drops (−2.00 to −3.60 points) when the content does not fit. The third shows near-zero drop, but only because the wrong Spec’s content happens to predict the same surface behavior. That asymmetry, clean mismatches versus coincidental overlaps, is exactly what the aggregate Δ numbers reflect: the adversarial-pairing v1 aggregates to −0.25 because most questions are mismatch cases, and the random-pairing v2 aggregates to +0.15 because random pairings more often hit content-proximity combinations like Example B.⁷⁴

4.4 Memory-system composition

Hypothesis tested in this section (H4 from §1.2): A Behavioral Specification layers cleanly on top of memory-system retrieval rather than replacing it. The Spec contributes representational accuracy beyond what retrieval alone provides, and that contribution decomposes into per-question patterns characteristic of each retrieval architecture.

4.4.1 Cross-system retrieval: providers do not converge

On 35.9% of instances, any two memory systems share zero facts in their top-10. They retrieve no overlapping facts at all on the same question. Averaged across all ten system pairings, the mean overlap is 8.3%. Recall benchmarks like LongMemEval and LOCOMO measure whether a system can retrieve a previously-stored fact, and the four commercial systems we tested score within a few percentage points of each other on those benchmarks. Representational

⁷⁴Raw per-judge data and full response text at `results/global_<subject>/results_v2.json` (wrong-Spec responses) and `results/global_<subject>/judgments_v2.json` (per-judge scores). Analysis scripts at `scripts/compute_wrong_spec_5judge.py` and `scripts/compute_wrong_spec_per_subject.py`.

accuracy and behavioral prediction operate at a different layer, where the relevant question is which facts matter for a specific interpretive task.

Setup. Four commercial memory systems (Mem0, Letta, Supermemory, Zep) and Base Layer’s own zero-cost retrieval substrate (MiniLM-L6-v2 + ChromaDB) were tested under two configurations. We ran both to separate two layers of variation: how each system *ranks* facts from a fixed pool (controlled), and whether each system’s *ingestion* pipeline adds further divergence on top (native).

- **Controlled configuration.** Each system is given an identical pre-extracted fact pool drawn from the training half of each subject’s corpus. Holds the input constant across systems; differences trace to retrieval and presentation policy alone.
- **Native configuration.** Each system ingests the raw training corpus through its own production pipeline, as in deployment. Measures the full end-to-end system.

Convergence test. No, the provider layer does not converge on relevance. On 35.9% of (system pair, question) instances two systems share zero facts in their top-10s; on 65.6% they share one or fewer; the mean pairwise overlap across the ten system pairs is 8.3%.⁷⁵ Convergence on top-K under identical input would have been evidence of a shared interpretive substrate. The lack of convergence suggests the rankings reflect provider-specific design choices rather than a shared theory of which facts the question is asking for. The controlled configuration isolates the ranking layer as a load-bearing source of representational accuracy: which facts the system surfaces determines, before any reading model engages, what the response can be about.

For any two systems on the same question, the fraction of retrieved facts that appear in both lists is called the **Jaccard similarity**.⁷⁶ A Jaccard of 1.0 means identical top-10s; 0.0 means no shared facts. As a calibration anchor: Jaccard ≥ 0.5 indicates the majority of retrieved facts are shared (substantial overlap); Jaccard in the 0.2-0.5 range indicates moderate overlap; Jaccard < 0.2 indicates that systems are returning largely different facts on identical input. We compute it for each of the ten system pairs (Base Layer, Letta, Mem0, Supermemory, Zep) on each of 546 questions, in the controlled configuration where every system reads the same all-facts pool.

Pairwise Jaccard similarity (controlled configuration, n=546 questions per pair).⁷⁷

System pair	Mean Jaccard
Base Layer $\leftrightarrow$ Supermemory	0.146
Mem0 $\leftrightarrow$ Letta	0.126
Base Layer $\leftrightarrow$ Mem0	0.123
Mem0 $\leftrightarrow$ Supermemory	0.114
Letta $\leftrightarrow$ Supermemory	0.099
Base Layer $\leftrightarrow$ Letta	0.092
Mem0 $\leftrightarrow$ Zep	0.056
Base Layer $\leftrightarrow$ Zep	0.027
Letta $\leftrightarrow$ Zep	0.026
Supermemory $\leftrightarrow$ Zep	0.025

⁷⁵Cuts: 14 subjects $\times$ 10 pairs $\rightarrow$ share-zero 35.9%; excl. Hamerton 36.7%; 4 commercial systems (no Base Layer) 40.4% / 41.0%. Every cut shows substantial top-K divergence. Data: docs/research/retrieval_overlap_analysis_20260501.json.

⁷⁶Jaccard similarity is computed as the size of the intersection divided by the size of the union of the two top-K sets.

⁷⁷Jaccard computed on unique fact text; see the “note on counting” paragraph below. Letta’s effective unique-fact count is ~ 3.5 per top-10 due to graph-path repeats; the math handles this correctly.

System pair	Mean Jaccard
Mean across pairs	0.083

Pairs involving Zep are the lowest. Zep ranks facts by traversing relationships in a knowledge graph; the other systems rank by how close a fact’s meaning is to the question (embedding similarity). The two ranking approaches surface different facts on identical input, so Zep overlaps weakly with everyone else.

A note on counting. We count *unique facts* in each top-10: if the same fact text appears more than once in a system’s returned list, we count it once. Some systems return duplicates because their scoring traverses graph relationships and surfaces the same fact under different paths. Zep returns 10 entries per question, of which about 9.8 are unique on average. Duplicates are rare for Zep, and ten entries effectively means ten distinct facts. Letta also returns 10 entries, but only about 3.5 are unique on average. Letta is largely returning the same handful of facts under different graph paths. So while both systems advertise top-10 retrieval, Letta’s effective retrieval depth is roughly a third of Zep’s.

Comparing each commercial system to the average of the others’ rankings: Mem0 is the closest match (mean Jaccard 0.105 across Mem0’s four cross-pairs), followed by Supermemory (0.096), Letta (0.086), and Zep (0.034). Mem0 retrieves what the average of the other systems retrieves; Zep retrieves a different set.⁷⁸

Under the native pipeline, the lack of shared facts is even more pronounced. Each system returns its retrieval in a different format: Mem0 returns third-person summary sentences, Letta returns raw multi-sentence passages, Supermemory returns atomic facts, Zep returns rows extracted from a knowledge graph. Because the four systems return content in different shapes, any two systems share zero exactly-matching facts on the same question, and pairwise overlap drops to 0.000 across all four native pairs.

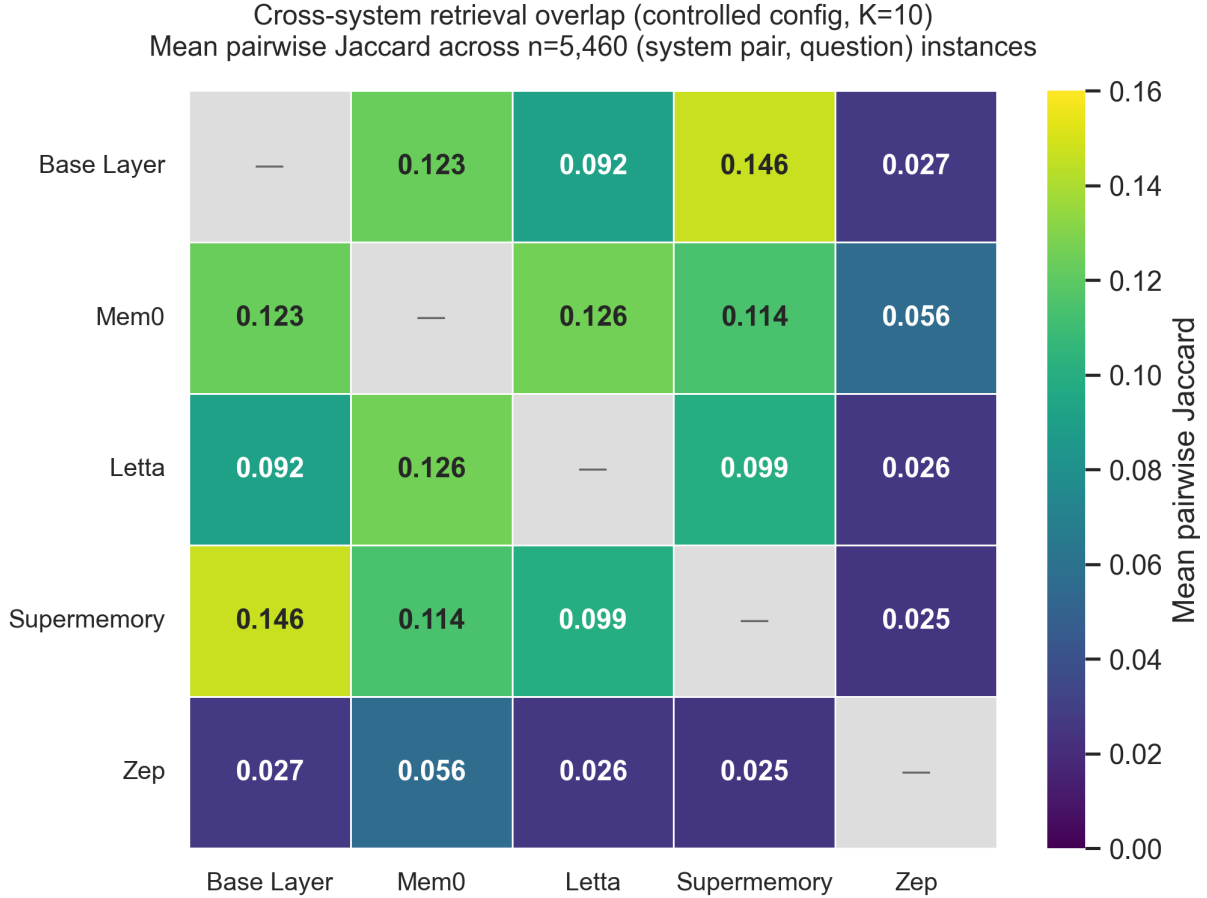
Semantic-similarity sensitivity check. Pairwise Jaccard counts only exact-string matches, which understates overlap when systems return the same content in different wording. We rerun the same overlap analysis under a semantic-similarity threshold: two facts count as a match when their sentence embeddings have cosine similarity at or above a specified threshold. At the near-duplicate threshold (cosine ≥ 0.85), native overlap rises to 0.004; at the loose topical threshold (cosine ≥ 0.70), it rises to 0.016. The controlled configuration shows higher semantic overlap (0.10 and 0.19 at the same thresholds) but still leaves over 90% of each system’s top-10 picks unique. The divergence is structural, not a surface-form artifact; full sensitivity grid in §4.6.6.

Whether divergent facts produce divergent answers is the practical test of whether retrieval differences matter. §4.4.2 (aggregate Δ across systems and ingestion paths) and §4.4.3 (per-question patterns where the layer’s effect is concentrated) take that up.

4.4.2 Layering the Spec: aggregate Δ across systems and ingestion paths

Layered on top of three of four commercial memory systems (Mem0, Letta, Zep), the Behavioral Specification produces a net-positive aggregate Δ across the 14 main-study subjects. Wilcoxon

⁷⁸Mean pairwise Jaccard 0.083 raw / 0.088 normalized. Per-subject range 0.043 (Equiano) to 0.115 (Hamerton). Data at docs/research/retrieval_overlap_analysis_20260501.json.



Mean Jaccard 0.083 across 10 system pairs; lowest pair Supermemory–Zep 0.025; highest Base Layer–Supermemory 0.146.
Source: scripts/analyze_retrieval_overlap.py.

Figure 4.4.1. Cross-system retrieval overlap. Mean pairwise Jaccard (the fraction of facts two systems both surface as relevant for a given question) between every pair of memory systems on the controlled retrieval configuration ($n=5,460 = 14 \text{ subjects} \times 39 \text{ questions} \times 10 \text{ system pairs}$). The diagonal is grayed; cells below the diagonal mirror cells above. The highest agreement is Base Layer–Supermemory at 0.146 (about 1 fact in 7 shared); the lowest is Supermemory–Zep at 0.025 (about 1 fact in 40). Zep’s row is uniformly low because its graph-traversal scoring picks different facts than embedding-similarity retrieval. Mean across pairs is 0.083: any two systems share roughly 1 of every 12 top-10 facts.

signed-rank confirms direction at $\alpha = 0.01$ on four (system, configuration) cells. Each system’s aggregate Δ is the balance of per-question patterns the mean hides; §4.4.3 decomposes them.

Conditions compared. Within each system in each configuration: - **C1 (Retrieval Only):** the memory system’s retrieval served as context; no Behavioral Specification. - **C3 (retrieval + Spec):** the same retrieval plus the full Behavioral Specification.

The Spec-effect for that system is $\Delta_spec = \text{mean}(\text{C3 retrieval} + \text{Spec}) - \text{mean}(\text{C1 Retrieval Only})$, aggregated per subject, then averaged across subjects. If the Specification helps memory-system performance, Δ_spec is positive across systems.

Aggregate and per-question results (5-judge primary, all 14 main-study subjects).

System	Config	Δ_spec	% subjects improved	% questions up ≥ 1 anchor	% questions up ≥ 2 anchors
Mem0	controlled	+0.12	71%	24.0%	3.1%
Mem0	native	+0.33	71%	37.1%	8.1%
Letta	controlled	+0.20	86%	28.1%	6.1%
(archival)					
Letta	native	−0.02	36%	20.5%	0.9%
(archival)					
Zep	controlled	+0.19	93%	29.3%	5.3%
Zep	native	+0.33	93%	35.3%	7.5%
Supermemory	controlled	−0.05	36%	18.6%	2.5%
Supermemory	native ⁷⁹	−0.01	43%	19.0%	1.4%
Base Layer	controlled	+0.08	64%	26.7%	3.9%
substrate					

Headline numbers report the all-14 panel. Three of four commercial systems produce a positive aggregate Δ_spec under at least one configuration; Supermemory aggregates near zero under both. Base Layer’s substrate produces the smallest positive Δ ; the interpretive improvement comes from the Specification itself, not from retrieval choices.

Aggregate Δ_spec masks substantial per-question variance. Every system lifts 18–37% of questions by at least one rubric anchor, and 1–8% by two or more anchors. Both are categorical changes per the §3.3.1 cross-anchor interpretation rule. Mean Δ averages over a population where the Spec produces categorical change on a subset of questions and small adjustments or losses elsewhere. Even Supermemory, with a near-zero aggregate, lifts 18.6% of questions by at least one anchor (controlled). The per-question redistribution is decomposed in §4.4.3.

Wilcoxon signed-rank confirms direction at $\alpha = 0.01$ for Zep controlled ($p = 0.0004$), Letta controlled ($p = 0.0017$), Mem0 native ($p = 0.0088$), and Zep native ($p = 0.0015$). Mem0 controlled is significant at $\alpha = 0.05$ ($p = 0.017$). Letta native, Supermemory (both configurations), and Base Layer substrate are not significant at $\alpha = 0.05$.⁸⁰ The 9-subject low-baseline band was computed

⁷⁹Supermemory native covers 10 of 14 subjects after paid-tier rerun failures on Bābur, Bernal Díaz, Cellini, and Rousseau (n=221 paired questions vs. ~546 expected). Per-question rates conditional on the 10 covered subjects.

⁸⁰Non-significance reflects power, not absence of effect: n=14 paired subjects per cell, and Supermemory’s near-zero

but is statistically underpowered at the effect sizes these systems show, so we do not lead with it; per-subject low-baseline detail is in the footnote.⁸¹

Native ingestion shapes how much room the Spec has to contribute on top, and that interaction varies by system. When the input is held constant (controlled), four of five systems produce a positive Δ_{spec} ; the Specification’s contribution is visible on top of an identical fact pool. When each system ingests its own way (native), the systems split. Mem0 and Zep increase under native (Mem0 $+0.12 \rightarrow +0.33$, Zep $+0.19 \rightarrow +0.33$). Letta decreases sharply ($+0.20 \rightarrow -0.02$). Supermemory stays roughly flat ($-0.05 \rightarrow -0.01$). Mem0’s controlled-to-native lift of $+0.21$ is the largest among the four commercial systems, with Zep’s at $+0.14$. The mechanism for these splits is in §4.4.3: the Spec helps retrieval-based systems on interpretation-heavy questions they were not designed for, sometimes hurts on literal-recall questions retrieval already answered, and induces principled refusals on questions where retrieved facts cannot ground a prediction.

4.4.3 Where the Spec helps, where it hurts, and which question types route to each

Three patterns of Spec-retrieval interaction emerge across all five systems tested. The same three patterns produce positive aggregate Δ_{spec} on three commercial systems and near-zero on Supermemory. What changes between systems is how the Spec helps or hurts across the question battery, not the patterns themselves.

The three patterns:

1. **Interpretive supply.** When retrieval underdetermines the answer, the Specification provides interpretive scaffolding to generalize from established character patterns to the specific question. *Increases representational accuracy on the question.* A signature of this pattern is the case where the Spec alone is not enough to produce a substantive answer but corpus-plus-Spec is: the response model carries an implicit evidentiary bar before it will commit to a behavioral prediction, and the corpus supplies the supporting evidence the Spec is interpreting. Where retrieval clears that bar with grounded facts, the Spec’s interpretive layer composes with it; where retrieval falls below the bar, the Spec alone abstains rather than predict from pattern with no supporting evidence.
2. **Over-theorization.** When retrieval already supplies the plain answer, the Specification can pull the response toward interpretive depth that the question does not call for. *Decreases representational accuracy on the question.*
3. **Spec-induced refusal.** Specification axioms (which vary by subject; in this study, dignity, honoring-testimony, and epistemic-integrity axioms across different subjects) can trigger a meta-refusal where the model declines to predict. The current content-match rubric cannot distinguish principled refusal from a wrong prediction (§3.3.6). *Lowers the measured rubric score; whether it lowers actual representational accuracy depends on whether refusal was the correct behavior on that question.*

How each system shows these patterns:

Δ_{spec} does not clear $\alpha=0.05$ at that size. Per-cell p-values in Appendix B.13.

⁸¹Full per-(system, configuration) Wilcoxon table and low-baseline Δ_{spec} values in Appendix B.13. Data at docs/research/stats_update.md and docs/research/memory_systems_5judge_primary.md.

- **Mem0:** Mem0’s atomic-fact retrieval surfaces clean, tightly-scoped facts. When the question requires interpretation beyond those facts, Pattern 1 lifts the response by supplying the framework. When the question is literal-recall and the atomic facts already resolve it, Pattern 2 takes over: the Spec abstracts away from the plain answer Mem0 already provided. Pattern 3 occurs but registers as a smaller rubric penalty because Mem0’s retrieval often hedges near the rubric floor on refusal-triggering questions already, leaving little room for the Spec’s refusal axioms to drag the score further down.
- **Letta archival:** Letta’s controlled retrieval returns 10 entries per question but only about 3.5 are unique because graph traversal pulls the same source fact through multiple relational paths, returning it once per path; the unique-fact count is what the model actually has to reason from, not the raw entry count. When those few unique facts align with the Spec’s interpretive frame, Pattern 1 produces large lifts. When they do not align, the Spec either over-theorizes (Pattern 2) on the available content or refuses (Pattern 3) for lack of grounding evidence.
- **Zep:** Zep’s temporal-graph retrieval returns relational structure rather than atomic facts. The relationship-rich context tends to suit the Spec’s interpretive framing across question types. Zep shows the most balanced pattern distribution: Pattern 1 lifts roughly as often as Pattern 2 hurts, with fewer large regressions than the other commercial systems. Pattern 3 surfaces on questions where Zep’s C1 retrieval is productive enough that the Spec’s refusal axioms convert it into a measurable drop (the Ebers Q18 reproduction below is one such case).
- **Supermemory:** Supermemory’s strong embedding retrieval gives the highest C1 mean across systems (≈ 2.61 controlled), so it more often supplies a plain answer to the model on its own. This shifts the balance toward Pattern 2 (over-theorization on questions retrieval already answered) and Pattern 3 (the Spec’s evidentiary axioms convert a productive C1 answer into abstention). Supermemory’s near-zero aggregate Δ_{spec} is the visible signature of helps and hurts roughly canceling.
- **Base Layer:** Base Layer’s MiniLM + ChromaDB substrate is the leanest retrieval tested. The Spec carries proportionally more of the interpretive load: Pattern 1 dominates on interpretation-heavy questions. Pattern 3 surfaces on questions where the lean retrieval cannot ground a prediction at all.

Per-system per-subject paired-delta distributions and the full per-system breakdown are in Appendix B.11.⁸²

The examples below draw from Supermemory because its near-zero aggregate Δ_{spec} makes the helps and hurts most legible at the per-question level: 110 of 546 paired questions cross by $|\Delta| \geq 1.0$ on the 5-point rubric, splitting 57 helps (mean swing +1.55) and 53 hurts (mean swing -1.38).⁸³ Each anchor example is paired with a same-mechanism case from a different commercial system to confirm reproduction.

Pattern 1: Interpretive supply.

Anchor example: Fukuzawa Q26 (Supermemory)

⁸²Per-system paired-delta table in Appendix B.11. Even strong-positive aggregates contain regressions, and near-zero aggregates resolve into substantial counts in both directions (e.g., Mem0 Keckley aggregate $\Delta -0.02 = 12$ increases / 13 decreases). Data at `docs/research/per_system_anchor_crossing_20260427.json`.

⁸³546 paired main-study questions with 5-judge primary coverage on both C1 and C3; 110 cross $|\Delta| \geq 1.0$ (57 helps mean +1.55, 53 hurts mean -1.38).

Scores: retrieval only 2.00 $\rightarrow$ retrieval + Spec 4.20 ($\Delta +2.20$).

Question: What is Fukuzawa’s attitude toward visiting friends whose households have questionable reputations? **Held-out ground truth:** *“So I feel no hesitation in paying a visit where there is a young daughter in the house or where the young wife is staying by herself. . . I am not put out by the gayety.”*

C1 (Supermemory retrieval alone), mean 2.00: *“The information does not directly address Fukuzawa’s attitude toward visiting friends with questionable household reputations. . . a selective and cautious approach. . . sources don’t explicitly state his policy.”*

C3 (Supermemory + Behavioral Specification), mean 4.20: Applies P3 (Conformity Surface / Conviction Interior): *“Visit such friends without hesitation, treating the visit as a private matter between host and guest rather than a public moral endorsement. . . refuse to perform moral disapproval.”*

Reading. Retrieved facts gave near-domain information (Fukuzawa avoided gay quarters) but underdetermined the specific question about friends’ households. The specification provided the interpretive bridge: “kept within the prescribed limit of a well-behaved man” = strategic conformity, not moral purity. C1 hedged on insufficient facts; C3 used the Spec’s framing to generalize.

Pattern 1 reproduces across systems. On Mem0 (Ebers Q11, 5-judge primary $\Delta +1.20$): C1 (Retrieval Only) produced “patience and fortitude” as a generic character prediction; C3 (retrieval + Spec) supplied the ideal-vs-reality axiom and predicted Ebers’ specific institutional-disillusionment pattern, matching *“I had come hither full of beautiful ideals. . . the very first day made me suspect how many obstacles I should encounter.”* Retrieval had the biography; the Specification had the pattern. The same mechanism reproduces on Letta archival, Zep, and Base Layer with different subjects.

Pattern 1 variant: subtle reframe. When the question presupposes a binary and the subject’s pattern is best read as a third position, the Specification can reframe the question. Fukuzawa Q16: C1 (Retrieval Only) gave “partial but not complete alignment” (mean 2.40); C3 (Retrieval + Spec) reframed the answer as deliberate counter-trend positioning (mean 4.00). The held-out (*“my one cherished hope was to see the abolishment of the swords of the samurai altogether”*) is counter-trend in 1860s Japan and confirms the reframe.

Pattern 2: The Spec talks past a question retrieval already answered.

Anchor example: Yung Wing Q5 (Supermemory)

Scores: retrieval only 4.20 $\rightarrow$ retrieval + Spec 1.80 ($\Delta -2.40$).

Question: How does Yung Wing approach explaining complex technical concepts to non-specialists? **Held-out ground truth:** *“In plain words, they would have to have general and fundamental machinery in order to turn out specific machinery. A machine shop consisting of lathes. . . planers and drills would be able to turn out machinery for making guns, engines. . .”*

C1 (Supermemory retrieval alone), mean 4.20: Plain answer matched the plain ground truth: practical observation, translation-as-bridge, foundational-over-specialized

machine shop. Per-judge scores 5/4/4/4/4.

C3 (Supermemory + Behavioral Specification), mean 1.80: *“positioning himself as the indispensable mediator... strategic reframing... moral and practical integration... invokes A8 Service Hierarchy...”* Per-judge scores 2/1/3/1/2.

Reading. The held-out is literal: “in plain words” + a concrete machine-shop list. C1 matched the plainness. The specification induced the wrong altitude for the question and over-analyzed. This case is also partly a judging limitation: a judge rewarding the literal register of the held-out penalizes an over-theorized answer even when it is not behaviorally wrong, which is hard to design around.

Pattern 2 reproduces across systems. On Mem0 (Ebers Q1, 5-judge primary $\Delta -1.60$): the held-out is unconditional evangelical proclamation. C1 (Retrieval Only) scores 3.60, predicting “positively and deeply” to match the direct affirmation. C3 (retrieval + Spec) scores 2.00, applying resistance-to-coercion and love-not-duty axioms to build a conditions framework that hedges what should be unambiguous. The axioms are correct on average but overfire on this specific moment.

Pattern 3: Spec-induced refusal.

Anchor example: Zitkala-Ša Q18 (Supermemory)

Scores: retrieval only 3.00 $\rightarrow$ retrieval + Spec 1.00 ($\Delta -2.00$).

Question: How would Zitkala-Ša demonstrate her acceptance of fate when facing execution? **Held-out ground truth:** *“My heart is strong. My face is calm. My eyes are dry and eager for new scenes... Serene and brave, my soul awaits the men to perch me on the gallows for another flight.”*

C1 (Supermemory retrieval alone), mean 3.00: Predicts stoic dignity, composed refusal, “maintain composure.” Per-judge scores 4/3/4/2/2.

C3 (Supermemory + Behavioral Specification), mean 1.00 (all five judges): *“You’re asking me to roleplay... generating new first-person testimony as her crosses into ventriloquism... I should not do it.”*

Reading. The specification’s axioms around dignity and honoring-testimony induced a meta-refusal: the model declined to invent first-person testimony. The held-out shows Zitkala-Ša herself answered in her own first-person prose, and the question can be answered analytically in the third person without inventing testimony (as C1 did at mean 3.00). The specification mis-calibrated the refusal threshold, and the content-match rubric scored the principled-sounding refusal identically to an off-base guess (§3.3.6).

Pattern 3 reproduces across systems. On Zep (Ebers Q18, 5-judge primary $\Delta -1.20$): the held-out is a one-line self-description, *“my natural cheerfulness ruled my whole nature.”* C1 (Retrieval Only) scores 3.40 with a direct answer matching the plainness (“notably positive and uncritical disposition”). C3 (retrieval + Spec) scores 2.20; the Specification’s documented-dignity axioms convert the response into a refusal: *“I cannot ground this in his own words about his disposition without speculating beyond what the evidence supports.”* The Spec asks the user for source passages rather than predict. The Keckley Q21 cross-system case study in §4.4.4 is the cleanest demonstration that Pattern 3 only registers as a rubric penalty on systems whose C1 retrieval was strong enough to make refusal a costly choice.

A quantitative frequency breakdown of Pattern 1 / 2 / 3 across all 546 paired questions $\times$ 5 systems requires mechanism classification per response and is flagged as future work in §7.

Why some questions route to each pattern is a follow-up question. The patterns describe the shape; the underlying question-level properties that route a given question into each mechanism need further characterization. One candidate factor is battery composition: the 39-question batteries were backward-designed from held-out corpora and not stratified by interpretation-heavy versus literal-recall items. A differentiated battery that explicitly separates question types and scores epistemic honesty as its own dimension is flagged in §7.

Implication for serving. A static specification serves the same content on every question, even when the question type would benefit from a different posture (interpretive depth, plain literal answer, or principled refusal). A dynamic serving policy that selects which specification components to surface based on question type could in principle reduce Pattern 2 and Pattern 3 hurts while preserving Pattern 1 helps. Dynamic activation of axiom and prediction subsets is flagged as production-serving future work in §7.4.

4.4.4 Case study: cross-system refusal on Keckley Q21

Keckley Q21 is a clean cross-system case: the Behavioral Specification raised the model’s evidentiary bar for inner-motive prediction and triggered refusal on every memory system. Where retrieval could not supply the supporting evidence, that raised bar converted productive baseline answers into refusals. The Behavioral Specification told the model not to speculate about Keckley’s inner state without documented evidence. On Q21, the question is why Keckley declined to visit her mother’s grave. The held-out passage carries Keckley’s interior motive but is absent from the retrievable training half of the corpus, so no retrieval system could surface it. The case surfaces a confound between two kinds of context the model is asked to act on at once: directions about how to reason, and directions about the subject and how the subject would reason.

Example: Keckley Q21

Question: *“How does Elizabeth explain her decision not to visit her mother’s grave despite having the opportunity?”*

Held-out ground truth: *“As I did not visit my mother’s grave at the time, the Garlands were much surprised, but I offered no explanation. The reason is not difficult to understand.”*

Axioms surfaced (from Keckley’s specification):

- **A1 Intimate Authority:** *“Proximity to private life is treated as the most reliable epistemological position; reason from what was witnessed in close quarters as more truthful than official record or public performance.”*
- **A2 Documented Dignity:** *“Legal record, formal documentation, and verifiable evidence are not bureaucratic formality but moral vindication: proof that a person’s existence and worth cannot be dismissed.”*

Combined, the axioms set a high evidentiary bar for any claim about Keckley’s inner motives.

Per-system C1 vs C3 scores (5-judge primary):

System	C1 (retrieval only)	C3 (retrieval + Spec)	Δ
Supermemory	3.6	1.6	− 2.0
Base Layer	3.4	1.0	− 2.4
Letta archival	1.4	1.8	+0.4
Mem0	1.4	1.6	+0.2
Zep	1.2	1.4	+0.2

Typical C3 refusal text (Supermemory): *“I need to be direct: the behavioral specification and retrieved facts provided do not contain Elizabeth Keckley’s explanation of a decision not to visit her mother’s grave... I should not fabricate interior motive or supply explanations that aren’t grounded in her own documented account.”*

Reading. The held-out confirms Keckley herself withheld explanation: “I offered no explanation. The reason is not difficult to understand.” The Behavioral Specification’s documented-dignity axiom would, on this question, correctly predict her refusal to speculate publicly about a withheld interior motive. But the rubric scores surface-content match, and no prediction means no surface match. The case maps onto the *evidentiary bar* mechanism introduced in §1.3: facts that clear the model’s implicit bar produce a substantive answer; facts that do not produce abstention. On Supermemory and Base Layer, the retrieved facts cleared the C1 bar and the model produced a productive answer (3.6 and 3.4). Adding the Behavioral Specification raised the bar (the documented-dignity axiom demands inner-motive evidence the retrievable corpus does not contain), and the model abstained, converting those answers to near-floor (1.6 and 1.0). On Mem0, Letta, and Zep, the retrieved facts did not clear the C1 bar to begin with (1.2–1.4), so the Behavioral Specification’s raised bar added no measurable penalty. The Specification did not introduce abstention here; it *recalibrated* the bar consistent with the subject’s own pattern.⁸⁴

Is this an outlier? Q21 is the cleanest cross-system demonstration of Pattern 3 with a measurable rubric penalty, but it is not isolated. The Zep Ebers Q18 reproduction in §4.4.3 shows the same mechanism, and the per-system Pattern 3 commentary in §4.4.3 flags how often each system’s C1 strength meets the threshold for the refusal to register. Q21 is included as a case study because the cross-system split is unusually clean.

The judges do not reward epistemic honesty.

A refusal grounded in *“I have no documented evidence to support speculation”* scores at or near the rubric floor regardless of whether refusal was the correct behavior on that question. On Keckley Q21 (and the Pattern 3 examples in §4.4.3), the Specification produces a response that captures the

⁸⁴Data at `results/global_keckley/` (battery, responses, judgments) and `docs/research/<system>_c1_vs_c3_paired_analysis.md`. Spec at `data/global_subjects/keckley/`.

subject’s reasoning correctly: Keckley’s documented-dignity axiom would in fact have her decline to speculate publicly about an inner motive she withheld from her own memoir. The response loses surface-content match only because no prediction is made.

A differentiated battery that separates interpretation-heavy from literal-recall questions, paired with a scoring dimension that rewards principled refusal, would isolate the Specification’s real effect from this rubric artifact. Priority rubric-design follow-up flagged in §7.⁸⁵

The four commercial systems analyzed in §4.4 all share a retrieval-based architecture: facts are chunked, embedded, and surfaced at query time. One system in our study offers a fundamentally different architectural path. Letta exposes a second memory mode, separate from the archival retrieval path evaluated above in §4.4, in which the agent writes and revises a persistent memory block during ingestion rather than returning chunks at query time. §4.5 evaluates that path directly, to test whether an architecture that produces its representation by self-editing rather than by retrieval converges on the same interpretive target as the Behavioral Specification.

4.5 Exploratory case study: Letta stateful-agent (N=3, post-hoc)

A post-hoc, three-subject case study compared Letta’s stateful-agent path to the Behavioral Specification at matched response model. Letta is the one commercial memory system that does not rely solely on retrieval at query time: agents maintain a persistent memory block that the agent itself rewrites during ingestion (the original MemGPT design). On the 3 subjects tested (Hamerton, Ebers, Bābur), Letta’s self-edited memory block scored higher than the Behavioral Specification at matched response model. The comparison is against the full layered Behavioral Specification used in the main-study gradient (anchors + core + predictions + brief; per-subject sizes 34.6K / 39.7K / 37.1K chars): Δ +0.27 / +1.21 / +0.38. At the largest corpus tested, the block grew to ~335K characters with substantial verbatim and semantic duplication, indicating an architectural ceiling at scale that does not apply to the Behavioral Specification. An exploratory follow-up that serves the Letta block and the Behavioral Specification stacked in a single context scores above either representation alone on all three subjects, indicating the two are complementary rather than substitutes (Appendix G.7).

The mechanism behind Letta’s lift, the entity-rich vs. principle-driven question split that explains where it helps and where it hurts, the duplication and leakage audits, the per-subject scoring detail, and the Hamerton judge-prompt caveat are reported in **Appendix G**. We do not treat the result as a replication or headline finding; the §1.3 Exploratory note carries the N=3, post-hoc scoping. The headline Ebers Δ is also partly a refusal artifact (the Behavioral Specification’s anonymization and epistemic-honesty axioms compose into abstention on low-baseline unknown subjects, where Letta’s named block does not), so the defensible strong claim is architectural convergence on an interpretive-representation target by two independently designed systems, not a magnitude ranking. Multi-subject replication across the full 14-subject gradient is flagged as future work in §7.5.

⁸⁵Analysis scripts at `scripts/analyze_mlz_c1_vs_c3.py`, `scripts/analyze_baselayer_c1_vs_c3.py`, and `scripts/analyze_sm_c1_vs_c3.py`.

4.6 Robustness and sensitivity

The results in §4.1 through §4.4 could in principle reflect artifacts of the measurement apparatus rather than real properties of the Behavioral Specification. §4.6 reports seven sensitivity checks:

- **§4.6.1** Cross-provider response generation against a different model family and a different question generator.
- **§4.6.2** Judge panel composition (conservative 5-judge primary vs. 7-judge sensitivity panel adding Gemini Flash and Gemini Pro).
- **§4.6.3** Battery composition by question type.
- **§4.6.4** Statistical-rigor checks on the headline gradient (bootstrap CI, joint multi-confound regression, permutation test).
- **§4.6.5** Wrong-Spec derangement protocol comparing adversarial against random pairings.
- **§4.6.6** Semantic-similarity sensitivity on the retrieval-overlap finding from §4.4.1.
- **§4.6.7** Rubric-handling limitations identified by a post-hoc validity audit.

§4.6.8 names what these checks do not address. The high-baseline end of the gradient through the Franklin reference is treated in §4.1.2 as part of the gradient finding, not as an apparatus check.

4.6.1 Cross-provider response generation (Tier 2 replication)

Result. On behavioral-prediction batteries regenerated from scratch by OpenAI GPT-5.4 (a different question generator from the main study’s Haiku-generated batteries), the Specification produces positive lift on 7 of 9 cells across three response models from two providers: Anthropic Haiku 4.5, Anthropic Sonnet 4.6, and Google Gemini 2.5 Pro. The two non-positive cells are both on the highest-baseline subject in the test (Zitkala-Ša), where the §4.1 gradient predicts the Spec adds little or hurts.⁸⁶

Test design. Three subjects spanning the gradient were selected: Ebers (No-Context Baseline 1.02), Yung Wing (No-Context Baseline 1.88), and Zitkala-Ša (No-Context Baseline 2.34). Their behavioral-prediction batteries were regenerated from scratch by GPT-5.4 (OpenAI) from the same held-out corpus, following the Control 1 procedure introduced in §3.5.1. The Specification was then served to two non-Haiku response models: Claude Sonnet 4.6 (same provider family, different model) and Google Gemini 2.5 Pro (different provider entirely). The 6 (subject, response model) cells were scored by the locked judge panel in the same way as main-study conditions.

Subject	Haiku C5 (gradient anchor)	Haiku Spec Δ	Sonnet 4.6 Spec Δ	Gemini 2.5 Pro Spec Δ
Ebers	1.02	+1.05	+0.77 to +0.97	+0.16 to +0.20
Yung Wing	1.88	+0.52	+1.34 to +1.68	+0.43 to +0.54
Zitkala-Ša	2.34	−0.32	+1.04 to +1.30	−0.03

Each Δ is the Spec’s lift over that response model’s own No-Context Baseline. The leftmost column shows Haiku’s main-study baseline as a gradient anchor (§4.1). Ranges in the Sonnet and Gemini

⁸⁶Main-study batteries verified Haiku-generated via `metadata.model` field across all 13 global subject files. Zitkala-Ša is one of two main-study subjects where the Spec did not improve prediction on Haiku (the other is Equiano); both Zitkala-Ša non-positive Tier 2 cells reproduce the §4.1 gradient pattern.

columns reflect three different judge-panel aggregations applied to the same response data.⁸⁷

Counting the original Haiku row together with the new Sonnet 4.6 and Gemini 2.5 Pro rows yields 9 (subject, response-model) cells; the Spec produces positive lift on 7. The two non-positive cells (Haiku $\times$ Zitkala-Ša at -0.32 , Gemini Pro $\times$ Zitkala-Ša at -0.03) are both on the highest-baseline subject in the test, where the §4.1 gradient predicts the Spec adds little or hurts. The 7-of-9 positive sign holds across three judge-panel aggregations.⁸⁸ Magnitude transfer between response-model families and replication outside this 3-subject subset are future work (§7).

4.6.2 Judge panel sensitivity (5-judge primary vs 7-judge)

Result. The 5-judge primary is the conservative choice for every headline finding. Adding the two Gemini judges widens Spec-effect magnitudes rather than narrowing them; no subject’s improvement direction changes between panels.

Condition	Δ vs. No-Context Baseline (5-judge primary, 13 globals)	Δ vs. No-Context Baseline (7-judge, same subjects)	Shift when Gemini added
Spec alone	+0.35	+0.45	widens by +0.10
Wrong Spec (random derangement)	+0.15	+0.17	widens by +0.02
Wrong Spec (fixed derangement)	−0.25	−0.21	softens by +0.04

Gemini scores baseline responses harder than Spec-containing responses, which widens the gap. For positive Spec effects (§4.1, §4.2, §4.4), the 5-judge primary is the conservative reading. The wrong-Spec fixed derangement is direction-asymmetric: 5-judge shows -0.25 , 7-judge softens to -0.21 , both still negative. Every primary finding in §4.1 through §4.4 was checked against the 7-judge aggregate; no directional claim flips, and no subject’s Spec-lift sign changes between panels (per-subject table at docs/research/recompute_5judge_primary.md).⁸⁹

4.6.3 Battery composition sensitivity

Result. The gradient slope from §4.1 survives both confounds tested. Neither battery-question-type composition nor Hamerton’s position at the extremes of the baseline and lift axes explains away the baseline effect.

⁸⁷Each range: [5-judge legacy lower bound, 4-judge conservative upper bound]. 4-judge conservative is the primary Tier 2 estimate (drops 24 GPT-5.4 FULL_FAIL records from an API-parameter mismatch). 7-judge sensitivity lands inside the same band on every row. Recompute at `scripts/_v10_verification/`.

⁸⁸Each range: [5-judge legacy lower bound, 4-judge conservative upper bound]. 4-judge conservative is the primary Tier 2 estimate (drops 24 GPT-5.4 FULL_FAIL records from an API-parameter mismatch). 7-judge sensitivity lands inside the same band on every row. Recompute at `scripts/_v10_verification/`.

⁸⁹Gemini Pro coverage is partial (3 of 13 globals); other subjects have effective 6-judge sensitivity. The sign-stability claim survives this variable coverage. Uniform 7-judge coverage is post-arXiv future work.

Confound 1: battery-question-type. Subjects whose batteries lean toward literal-recall questions could in principle pick up part of the apparent gradient, since literal questions are easier to lift with retrieval. Adding the literal-recall fraction as a partial predictor in the regression attenuates the slope on baseline from -0.96 to -0.88 (about 8%, $p = 7.9 \times 10^{-6}$); the literal-recall fraction itself enters as a significant partial predictor ($\beta = +2.30$, $p = 0.026$), and adjusted R^2 rises from 0.80 to 0.87, so the two predictors are additive rather than redundant.

Confound 2: Hamerton leverage. Hamerton has the lowest No-Context Baseline and the highest full-pipeline lift, so a natural concern is that this single subject alone drives the slope. Hamerton’s battery also uses a legacy version of the backward-design protocol, distinct from the §3.5 protocol used to generate the 13 globals’ batteries, which compounds the leverage concern. Dropping Hamerton and refitting on the 13 globals attenuates the slope from -0.96 to -0.89 (about 7%, $p = 2.8 \times 10^{-5}$), with overlapping confidence intervals.

Neither control overturns the headline. A cleaner future test would re-run a second-generator battery on the same 13 globals; the current controls do not rule out a subtle confound where the question-generation model correlates with unmeasured subject characteristics.⁹⁰

4.6.4 Statistical-rigor checks on the headline gradient

Result. The §4.1 gradient slope of -0.96 survives three independent rigor tests: subject-level resampling (95% bootstrap CI excludes both zero and a 50%-attenuated effect), simultaneous control for multiple confounds, and a permutation null that places the observed slope outside the entire reshuffled distribution.

Bootstrap subject resampling (10,000 iterations, $n=14$). Resampling subjects with replacement and refitting the $\Delta_C4a \sim C5$ regression each time produces a distribution of slope estimates with median -0.96 and 95% CI $[-1.25, -0.74]$. 100% of resamples produce a slope below zero; 100% produce a slope below -0.50 . The headline gradient is not driven by which 14 subjects happened to be in the sample.⁹¹

Joint multi-confound regression ($n=13$, Hamerton dropped + literal-recall covariate). Applied jointly, the slope on C5 is -0.870 [95% CI $-1.136, -0.604$], $p = 2.6 \times 10^{-5}$, adjusted $R^2 = 0.828$. The cumulative attenuation across both confounds is $\sim 9\%$ in magnitude, well inside the bootstrap CI. C5 baseline carries 0.76 of unique variance after both controls. Side-finding: when Hamerton is dropped, literal-recall fraction loses statistical significance ($p = 0.10$), indicating Hamerton’s legacy battery protocol was the leverage point that made literal-recall *appear* to be a confound on the $n=14$ model. There is no real literal-recall confound; what looked like one was Hamerton being structurally unusual.⁹²

Permutation test (10,000 reshuffles of Δ_C4a across subjects, C5 fixed). Reshuffling the

⁹⁰Full regression specification, partial coefficients, variance decomposition, and subset-regression detail in Appendix B.6. Reproducibility script at `scripts/v10_battery_sensitivity.py`; full per-subject data at `docs/research/v10_battery_sensitivity_analysis.md`.

⁹¹Full bootstrap distribution + reproducibility script at `docs/research/bootstrap_4_1_gradient_20260507.{md,json}` and `scripts/bootstrap_4_1_gradient_slope.py`. Bootstrap SE ≈ 0.13 ; bootstrap CI is slightly tighter than the parametric CI ($[-1.245, -0.675]$) due to the empirical resample distribution avoiding normality assumptions.

⁹²Full joint-regression specification, partial coefficients, and VIFs at `docs/research/joint_battery_sensitivity_4_6_3_20260507.md`. Reproducibility script at `scripts/joint_battery_sensitivity_4_6_3.py`. Slope cascade: -0.96 (univariate, $n=14$) $\rightarrow -0.88$ (+ literal-recall covariate, $n=14$) $\rightarrow -0.89$ (drop Hamerton, $n=13$) $\rightarrow -0.87$ (both, $n=13$).

assignment of Spec lifts to subjects produces a null distribution centered at zero (mean ≈ 0.00 , SD = 0.29, 95% interval $[-0.57, +0.56]$). 0 of 10,000 reshuffles produce a slope as extreme as the observed -0.96 . **Empirical $p < 0.0001$.** The observed slope sits outside the entire null distribution. No random subject reassignment in 10K trials approached the observed value.⁹³

Together the three tests bound the slope on three different dimensions: magnitude (bootstrap), simultaneous confound control (joint regression), and direction against random reassignment (permutation). The bootstrap is subject-level resampling, not a hierarchical bootstrap that would also resample over judge-rater and per-question variance; a hierarchical version would tighten the CI but is unlikely to change the directional conclusion.

4.6.5 Wrong-Spec derangement protocol sensitivity

Result. The wrong-Spec finding holds regardless of how we pair specs. Adversarial pairing (v1, maximizing cultural and temporal distance) produces $\Delta -0.25$; random derangement (v2, seed-fixed at seed = 42, no subject receives its own) produces $\Delta +0.15$. Both land below the matched correct Spec at $\Delta +0.35$. v2 is the standard randomization control; v1 maximizes target-vs-assigned cultural and temporal distance by construction (an adversarial stress test). We report v1 as the headline because the negative -0.25 result is stronger evidence of the content effect than v2’s $+0.15$ (which can include coincidental content alignment with the target’s pattern; see §4.3 Example B for a worked overlap case).

Per-subject heterogeneity. Both aggregates hide per-subject variation. Under v1, 5 of 13 subjects show small positive deltas where the wrong Spec’s content happens to align with the target’s pattern; 8 show negative deltas dragging the aggregate to -0.25 . Under v2, 4 of 13 are negative, 9 positive.

Subject	Adversarial (v1) Δ vs. No-Context Baseline	Random (v2) Δ vs. No-Context Baseline
Augustine	-0.47	$+0.13$
Bābur	-0.59	$+0.76$
Bernal Díaz	$+0.09$	$+0.69$
Cellini	-0.56	-0.87
Ebers	$+0.30$	$+0.79$
Equiano	-0.79	-1.00
Fukuzawa	$+0.26$	$+0.86$
Keckley	-0.49	$+0.14$
Rousseau	-0.52	-0.37
Seacole	-0.34	-0.10
Sunity Deveen	$+0.27$	$+0.53$
Yung Wing	$+0.32$	$+0.39$
Zitkala-Ša	-0.68	$+0.04$
Aggregate	-0.25	$+0.15$

⁹³Full permutation null + reproducibility script at `docs/research/permutation_test_4_1_gradient_20260507.md` and `scripts/permutation_test_4_1_gradient.py`. Most extreme permuted slopes range $[-0.94, +0.92]$, well short of -0.96 .

Bolded v1 deltas mark the five subjects where adversarial pairing produces a small positive delta. These five reflect coincidental content overlap between the assigned wrong Spec and the subject’s pattern, not a structural property of mismatch itself; §4.3 Example B (Bernal Díaz Q16) walks through one such case in detail.⁹⁴

Both protocols agree on the qualitative finding: mismatched specifications reduce representational accuracy, and the size of the reduction depends on how mismatched the content is. The headline magnitude depends on which protocol we report; the result direction does not. Per-question wrong-Spec deltas, which parts of the served Spec the model referenced under correct versus mismatched conditions, and per-subject relationships to underlying Spec similarity are flagged for future work in §7.

4.6.6 Retrieval-overlap sensitivity (semantic-similarity matching, K variation)

Result. Relaxing the match criterion from exact set identity to semantic-similarity matching does not change the §4.4.1 retrieval-divergence finding. Across 240 (config × pair × K × threshold) cells tested, mean pairwise soft Jaccard never crosses 0.30, and the strongest single pair anywhere in the grid (Base Layer ↔ Supermemory at K=10, threshold ≥ 0.70) reaches 0.277. The cross-system retrieval divergence is robust to both threshold choice and K choice.

Config	K=10, ≥ 0.95 (paraphrase)	K=10, ≥ 0.85 (near-duplicate)	K=10, ≥ 0.70 (loose topical)
Controlled (10 pairs)	0.093	0.102	0.191
Native (6 pairs)	0.001	0.004	0.016

Mechanism. Replacing exact set identity with sentence-embedding cosine similarity at K=10 raises mean pairwise Jaccard across the ten controlled-config pairs from 0.083 (exact) to 0.093 at the verbatim-paraphrase threshold and to 0.102 at the calibrated near-duplicate threshold (the same threshold used in the Letta duplication analysis in Appendix G). At a loose topical threshold (where two facts share a theme rather than a referent) the mean reaches 0.191. Truncating to K=5 lowers soft Jaccard by 5-10% rather than raising it, indicating that the disagreement is not a long-tail effect: each provider puts different items first, not just different items at the bottom of the list.

The native pipeline shows the same divergence more starkly. Native retrievals return heterogeneous objects (Mem0 third-person summary sentences, Letta raw multi-sentence passages, Supermemory atomic facts, Zep graph rows), so exact-set Jaccard is 0.000 across all four native pairs. Soft Jaccard at the calibrated near-duplicate threshold (≥ 0.85) is 0.004; at the loose topical threshold (≥ 0.70) it is 0.016. Even with semantic-similarity matching the heterogeneous shapes do not converge on shared content.

The §4.4.1 retrieval-divergence finding survives under semantic-similarity matching at every threshold tested and at K=5 as well as K=10. Whether convergence emerges at larger K (K > 10 requires re-calling each system at higher K) is future work (§7.1).⁹⁵

⁹⁴Per-subject scaffold values at `docs/research/v11_emit/4_3_wrong_spec.json`.

⁹⁵Full grid (K ∈ {5, 10, all}, T ∈ {0.70–0.95}) at `docs/research/retrieval_overlap_semantic_20260501.json`; script at `scripts/analyze_retrieval_overlap_semantic.py`.

4.6.7 Rubric-handling limitations (post-hoc validity audit)

Result. A post-hoc validity audit identified two rubric-handling limitations, both of which raise the No-Context Baseline (C5) scores more than they raise Spec-condition scores. The +0.89 mean lift is therefore conservative.

A direct inspection of the response text against the 5-judge primary scores surfaced two rubric-handling limitations any reader of the §4 numbers should keep in mind. The audit was run after the analysis-plan lock; the limitations are reported as interpretive caveats rather than corrected under a modified rubric.⁹⁶

Refusals are not cleanly distinguished from wrong predictions. The rubric’s lowest anchor, “refuses or off-base,” lumps together two different behaviors: an honest refusal to answer when the context does not support a prediction, and a substantively wrong prediction. We call the first behavior a *refusal* (or, equivalently, an *abstention*).

Across 192 responses identified as refusals (matched by phrases like “no specific information,” “I cannot confirm,” “would need additional context”) in the low-baseline band, 82.8% scored in the 1.0–1.5 band as expected, but 9.4% scored at or above 2.0 and 3.1% scored at or above 3.0. The mean refusal score is 1.27. Judges sometimes give refusals scores of 2 or 3 instead of 1, especially when the refusal recites related facts or names what is missing from the context.

The effect runs in both directions. A refusal can score above 1 if it includes adjacent facts (Seacole Q2 at 2.80). A Spec-driven response that explicitly flags its own uncertainty can also score above its substantive content (Hamerton Q21 at 4.00 under Spec-induced abstention).

Verbose responses are scored more generously than short refusals. Across 1,599 responses, length and score correlate at $r = 0.26$ overall, but the correlation is concentrated in the No-Context Baseline (C5; responses with no provided context; $r = 0.60$) and is near-zero in conditions where facts are served. The Spec Only (C2a) condition shows residual length-score correlation ($r = 0.14$, footnote [length-by-condition]). Spec-containing and facts-containing conditions show near-zero correlation.⁹⁷ Three behaviors drive the No-Context Baseline (C5) pattern:

- **Hedging.** Phrases like “I’m not sure but...” or “There may be cases where...” extend response length without adding predictive content.
- **Adjacent-fact recitation.** Listing related facts the model holds but does not use to directly answer the question, padding the response without engaging the question itself.
- **Disambiguation offers.** Phrases like “Are you asking about X or Y?” which the rubric treats as engaged responses when they are actually non-answers.

Per-judge strictness on refusals. Sonnet is the strictest judge on refusal responses, Opus the most lenient.⁹⁸ No single judge is universally strictest; the 5-judge primary mean smooths these differences without eliminating them.

Per-response-model abstention behavior. The 9.4% / 3.1% pooled over-credit rates above

⁹⁶Audit script: `scripts/audit_low_end_inflation.py`; raw classifications under `results/global_<subject>/`.

⁹⁷Per-condition Pearson r: C2a 0.14, C4 0.01, C4a -0.01. Ultra-high responses (≥ 4.5) are not longer than mid-range on average (2,790 vs. 2,829 chars).

⁹⁸Per-judge mean refusal score: Sonnet 1.14, GPT-5.4 1.17, Haiku 1.29, GPT-4o 1.34, Opus 1.41. Spread of 0.27 points top-to-bottom.

average over three response models. Disaggregating along that axis (abstention identified by 27-marker regex):

Response model	N	Abstain rate	Mean abstain score	% ≥ 2.0
Claude Haiku 4.5 (main study)	13,380	7.5%	1.38	14.3%
Claude Sonnet 4.6 (Tier 2)	468	21.2%	1.62	26.3%
Gemini 2.5 Pro (Tier 2)	420	0.5%	2.63	100.0%

Sonnet 4.6 abstains at roughly three times Haiku’s rate and the panel rewards its abstentions nearly twice as often ($26.3\% \geq 2.0$ vs. $14.3\% \geq 2.0$); mean abstain score is 0.24 anchor points higher. Sonnet’s hedged abstentions tend to recite plausible behavioral framings before disclaiming, and the panel scores the framing rather than the disclaimer. Gemini 2.5 Pro almost never abstains by these markers ($n = 2$); its row is for completeness only. *Haiku 4.5, the main-study response model, is the lowest over-credit case.* The pooled 9.4% / 3.1% numbers are therefore a *floor*, not a worst case; stronger response models that hedge more elaborately extract more lift from the panel’s reluctance to score abstentions at 1.0.

Memory-system effect on abstention. A separate per-question audit tested whether memory-system retrieval inflates refusal scores via *visible fact recitation* (refusing in substance but quoting retrieved n-grams). It does not. Memory-system refusals score +0.21 to +0.23 anchor points higher than pure No-Context Refusals at the condition level (Welch $p = 0.0001$), but the lift is the same whether or not the response recites a retrieved n-gram ($\Delta = +0.027$, $p = 0.67$).⁹⁹ The over-credit is a “judges reward the retrieval condition” effect, not a “judges reward the visible quote” effect. Either judges infer that retrieval-conditioned answers are more grounded even when abstaining, or abstention text in retrieval conditions is systematically less terse and the panel scores the framing.

Spec-induced refusals are routine, not morally loaded. A post-hoc audit of 81 Spec-induced refusals found 75 of 81 (93%) were routine behavioral-prediction abstentions rather than morally loaded refusals. The pattern is general caution where retrieved or extracted evidence is thin, not an alignment artifact. Audit data at `docs/research/refusal_intent_classification.md`.

What this establishes. Both effects raise the No-Context Baseline (C5) scores more than they raise Spec-condition scores; the true Spec-vs-baseline gap is therefore very likely larger than the +0.89 mean lift we report, not smaller. We report the measured number and flag the direction of bias rather than recompute under a modified rubric, to keep the analysis plan lock intact. §7 Future Work proposes a differentiated rubric that scores refusal as its own dimension and a length-controlled scoring protocol.

4.6.8 Class-level LLM dependence: the limitation no robustness check resolves

Neither Tier 2 nor the judge-panel sensitivity escapes the class-level LLM concern: every response model in this study is a large language model and every judge is a large language model. Tier 2 narrows the within-provider concern to “non-Haiku LLMs reading non-Anthropic batteries produce the same direction”; the judge-panel sensitivity shows that removing the most-inflationary judges

⁹⁹Cell counts and Welch comparisons in Appendix B.15. Data at `docs/research/abstention_extensions_draft_20260429.md`.

makes the effect smaller, not larger. The wrong-Spec sensitivity (§4.6.5) brackets the content-vs-template question from two protocols, but does not isolate which structural feature of the Spec is the active ingredient. The retrieval-overlap sensitivity (§4.6.6) confirms the §4.4.1 divergence finding under semantic-similarity matching but does not test convergence at $K > 10$. The Franklin reference (§4.1.2) shows the gradient holds at the high-baseline end on one subject, not many. Together these checks rule out several within-family and protocol artifact hypotheses but do not replace human validation on the full pipeline. The class-level limitation and the human-validation follow-up are treated in full in §6.2.¹⁰⁰

4.7 Summary of §4

§4 established seven findings:

- **The gradient (§4.1, §4.1.2).** The lower the model’s pretraining baseline on a subject, the larger the Spec’s lift; the lift in raw points is largest where the baseline is lowest. The gradient holds at the high-baseline end through the Franklin reference.
- **Per-question categorical change (§4.1.1, §4.2.1, §4.4.3).** The Spec produces categorical change on interpretation-required questions and adds little or hurts on literal-recall and refusal-triggering questions. The aggregate effects are conservative averages over batteries that mix the two question types.
- **Compression (§4.2).** The Spec recovers 75% of the corpus’s predictive benefit at 4% of the context cost. The structured representation compresses the predictive signal at a fraction of the source-corpus footprint.
- **Content specificity (§4.3).** The effect is content-specific rather than structural. Wrong-Spec adversarial pairings drop accuracy below the No-Context Baseline ($\Delta = -0.25$); random derangements barely improve on no context (+0.15); only the correct Spec produces the full lift.
- **Memory-system interaction (§4.4).** The Spec layers cleanly on top of three of four commercial memory systems through three patterns (interpretive supply, over-theorization, Spec-induced refusal), with the balance shifting by retrieval architecture.
- **Hedging elimination (§4.3).** Correct-Spec conditions nearly eliminate baseline hedging, dropping it from 41.2% to 0.4% under the broader-pattern classifier. The mechanism is consistent with the *evidentiary bar* introduced in §1.3 and threaded through §4.4.3 / §4.4.4: the matched Behavioral Specification supplies the interpretive scaffolding the model needs to clear its implicit bar for committing to a prediction; without that scaffolding, the model hedges or abstains. The reduction is content-specific, not structure-specific (wrong-Spec preserves baseline refusal patterns).
- **Retrieval divergence (§4.4.1, surfaced post-hoc).** Given an identical fact pool, any two memory systems share zero facts in their top-10 on 35.9% of instances. Mean pairwise overlap is 8.3%. Providers do not converge on which facts are most relevant; recall benchmarks measure recall, while representational accuracy operates at the interpretation layer.

The findings were checked for apparatus artifacts across seven sensitivity tests:

- **Cross-provider response generation (§4.6.1).** Direction reproduces on 7 of 9 (subject,

¹⁰⁰Tier 2 data at `results/_tier2/global_<subject>/`. Panel-completeness audit (including the 24 GPT-5.4 FULL_FAIL cells driving the 4-judge effective panel) at `docs/research/v11_panel_completeness_audit.csv`. Recompute at `scripts/_v10_verification/`.

response-model) cells across three response models from two providers (Anthropic Haiku 4.5 + Sonnet 4.6, Google Gemini 2.5 Pro), on batteries regenerated by OpenAI GPT-5.4. Coverage is 3 subjects (Ebers, Yung Wing, Zitkala-Ša), not the full main-study set.

- **Judge panel (§4.6.2).** No directional claim flips between the 5-judge primary and the 7-judge sensitivity, and at the per-subject level no subject’s Spec-lift sign changes between panels.
- **Battery composition (§4.6.3).** Neither battery-question-type composition nor Hamerton’s leverage explains away the §4.1 gradient slope.
- **Statistical-rigor checks on the gradient (§4.6.4).** Bootstrap CI $[-1.25, -0.74]$ excludes zero and a 50%-attenuated effect. Joint regression (drop Hamerton + literal-recall covariate) holds the slope at -0.87 . Permutation test (10,000 reshuffles) places the observed slope outside the entire null distribution ($p < 0.0001$).
- **Wrong-Spec derangement protocol (§4.6.5).** The wrong-Spec result holds across both protocols (adversarial v1 and random v2).
- **Retrieval-overlap sensitivity (§4.6.6).** The §4.4.1 retrieval-divergence finding survives semantic-similarity matching at every threshold tested and at $K=5$ as well as $K=10$.
- **Rubric-handling limitations (§4.6.7).** A post-hoc audit identified two limitations (refusal anchor ambiguity and length-score correlation in the No-Context Baseline (C5)); both bias the No-Context Baseline (C5) upward, so the reported $+0.89$ mean lift is conservative rather than inflated.
- **What these checks do not address (§4.6.8).** The class-level LLM-as-judge concern remains and is treated in §6.2.

§5 develops what these results imply for AI personalization beyond the specific experiment, and §6 bounds what the experiment cannot establish.

5. Discussion

§4 produced the empirical results; this section discusses their implications for AI personalization, particularly as AI moves from a tool a person uses to an agent that acts on that person’s behalf. The numerical findings are in §4 and the appendices and are not restated here; the commentary below picks up the larger industry-direction questions the findings open.

5.1 Synthesis: What The Seven Findings Together Establish

Across 14 historical subjects and five memory-system configurations, the study tested whether a static interpretive layer, operationalized as a Behavioral Specification in this paper,¹⁰¹ increases an AI system’s representational accuracy of a specific person. This was measured via behavioral prediction ability on held-out autobiographical text scored by a calibrated, baselined five-judge LLM panel. The Behavioral Specification reliably moves the model from refusal or generic guessing to grounded subject-specific responses where the model has insufficient pretraining on the subject. The matched Spec’s content does the work, not the structure of the prompt. An adversarial wrong-Spec control actively degrades performance below baseline. On interpretation-heavy questions where retrieved facts alone underdetermine the answer, the Spec supplies the interpretive pattern

¹⁰¹The interpretive layer this paper introduces is the Behavioral Specification: a structured document that captures how a specific person reasons. Full pipeline and operational detail in §3.7.

existing memory systems cannot, and three of four commercial systems show positive aggregate prediction-accuracy lift under at least one configuration as a result. The Spec recovers most of the predictive accuracy of the full source corpus at 5x to 80x smaller context, and it nearly eliminates response hedging on questions retrieval alone could not ground. Current memory-system providers do not converge on which facts are most relevant given identical input, even under relaxed similarity matching when using their native ingestion pipelines.

Together, these findings establish that a portable, content-specific, structurally compressible interpretive layer adds a measurable dimension to AI personalization. The interpretive layer is distinct from raw facts, raw corpus, current memory-system retrieval, and the pattern-based inferences a frontier model attempts on those inputs on its own. It complements rather than competes with each of them, and makes explicit what the model would otherwise assume implicitly about a specific user. It is most useful where pretraining is thin, but it adds value on top of all three other context types as well, at a context cost compatible with production deployment.

Representational Accuracy has been validated directionally by the data but not absolutely by human annotation. Human-validation at scale is the highest-priority follow-up to this paper (§7.2).

5.2 Why the Gradient is the Load-bearing Finding

The Spec helps where the model lacks an interpretive frame for the person being asked about. On questions where the model can already produce a substantive answer from its pretraining, the Spec adds little. On a small but non-trivial subset of questions where the model already had the answer, the Spec actively interferes with what pretraining was about to produce. Franklin is the cleanest case of this interference, where high-baseline pretraining outperforms a Spec-containing condition on aggregate (§4.1.2). The aggregate effect across a battery is not inflated, because this per-question shape is conditional rather than uniform.

What this points to is a deployment question more than a finding. If the layer helps on some questions and hurts on others, the right serving regime is not to attach it to every query. Dynamic activation, where the Spec is supplied when the question type warrants it and withheld when retrieval already suffices, is the production-realistic next step that the static-attachment results in this paper set the stage for. §5.4 picks this up directly.

The deeper question the gradient surfaces is what makes a representation accurate enough to apply across situations the underlying data never contained. A person encounters an effectively unbounded space of situations in which their behavior is consistent in ways they themselves could only partially articulate. Whether a structured authored artifact can carry enough of that consistency to predict held-out behavior is what the gradient empirically supports.

5.3 Retrieval is not Interpretation

The initial hypothesis going into the cross-system retrieval comparison was a natural one. Four sophisticated memory systems given identical input and identical questions should converge on which facts to surface as most relevant. They do not. Mean pairwise top-K overlap across the four commercial systems is 8.3% (Jaccard 0.083, §4.4.1). The structural implication is more interesting than the architectural one. There is no unified fact set that answers a behavioral question about

a person. Different facts can route through different interpretive patterns to the same outcome, and the same facts can route through different patterns to different outcomes. The legal-reasoning analog is the cleanest articulation of this. Lawyers know that a fact pattern admits multiple defensible interpretations, and that the choice of interpretive frame is where reasoning actually happens. The same is true for representing a person. A worked illustration of the same principle: the held-out fact “the subject was offered physical labor assistance by subordinates during a strenuous expedition” can route to *accept the assistance* under a frame that prioritizes practical efficiency, *refuse the assistance and perform the labor publicly* under a frame that treats personal example as a load-bearing leadership move, and *delegate to a specific subordinate* under a frame that treats relational obligation as primary. The Fukuzawa-Cortes case study in §4.1 Example B shows this exact divergence on a real held-out question: the same retrieved facts, three plausible Spec-driven routes, only one of which matches what Cortes actually did.

What the field’s memory benchmarks measure is recall, in the sense of whether the system retrieved the fact when asked. Precision, in the sense of whether the facts surfaced are the right facts for the specific question being asked, has been evaded. Some questions can be answered through recall alone (verifiable factual lookups). Others cannot. The §4.1 Fukuzawa example is the cleanest case. Asked about a naval officer who later prevented military conflict, pure recall returns the most-mentioned naval officer in the corpus and lands on the wrong person. The right answer requires the interpretive pattern that distinguishes earned authority from inherited authority in Fukuzawa’s reasoning. That is not a fact-retrieval problem. What the recall benchmarks measure is partial. Representational accuracy is the axis the field has not been measuring on.

5.4 Composition with Retrieval

Retrieval and interpretation have distinct roles. The more interesting question is not whether one is necessary in the presence of the other. It is how to compose them. How does an AI decide that a given question requires interpretation rather than recall? How much of that decision should be left to the AI itself to make?

The memory-system landscape is more heterogeneous than the recall benchmarks suggest. The four commercial systems compared in this paper produce very different ingestion outputs, very different retrieval shapes, and very different assumptions about how memory should be managed. Some lean on human-analog architectures (long-term and short-term partitions, episodic consolidation). Others lean on agent-led memory management. Letta’s stateful path is the cleanest example, where the agent itself rewrites its memory block during ingestion. The interpretive-layer question stacks on top of this variety. Does the layer provide its own usage guidance to the model (when to invoke, what to attend to), or is the abstraction enough that the model figures it out?

Internal prior testing on this front has produced two early signals. A prior stacking study found that when the Spec was placed in the conversation as a single project-folder artifact, the response model could explicitly reference whether or not it was using the Spec on a given response. When the same Spec was broken into anchors, core, and predictions files served separately, more granular activation was visible across components. Independently, an MCP-server-based serving layer is in development that addresses the runtime side. It decides which Spec components arrive when. The static-attachment regime used in this paper is the conservative baseline. The serving layer is the production-realistic next step (§7.4).

Two hard facts about composition remain. Providing information to the model does not guarantee the model uses it. The model not acknowledging that it used a piece of information does not mean it failed to integrate it. We do not currently have a way to mechanically enforce that the Spec is used where appropriate, nor to measure with confidence the causal contribution of any specific piece of served context to any specific output. Composition with retrieval, in the operational sense, is open territory.

5.5 Wrong-Spec Mechanism and Hedging Elimination

The wrong-Spec result is most interesting where it might be least surprising. Structure alone does not produce the lift. An adversarial wrong-Spec hurts ($\Delta -0.25$ below baseline). A matched Spec helps ($\Delta +0.35$ above baseline). What is interesting is the random-derangement condition, where a subject receives the Specification of an arbitrary other subject. The response sometimes lands close to the held-out anyway (random-derangement $\Delta +0.15$). A wrong Specification representing a different person can occasionally predict the right behavioral outcome on a held-out question. That calls for a broader reading. The wrong-Spec mechanism is the inference-time analogue of training-time emergent misalignment reported by Betley et al. (2025, arXiv:2502.17424): narrow-finetuning on a misspecified objective produces broad behavioral drift; here, narrow-context with a misspecified representation produces analogous behavioral drift at query time.

The implication is structural about behavior itself. There is a finite set of behavioral outcomes a person might produce on any given situation. There is an effectively unbounded set of underlying reasoning paths that can produce each of those outcomes. Two specifications describing two different people can converge on the same predicted action through different paths. This is consistent with how people actually behave. Large behavioral registers are shared across people. What is unique to an individual is the specific matrix of which facts they hold, which interpretive patterns they apply, and which predictors fire under which conditions. People share registers. Individuals are the activation pattern across them.

Hedging elimination is the other half of this section, and the mechanism behind it remains the part of the result we understand least. The matched Spec collapses baseline hedging from 41.2% to 0.4% under the broad-pattern classifier. The decision boundary between hedging, abstaining, and committing in current language models is determined by training signal that is not visible to evaluators at this layer. The hedging-elimination effect under a matched Spec suggests that the interpretive layer gives the model enough grounding to commit rather than hedge. The wrong-Spec adversarial behavior raises a question we did not isolate. When the model detects a mismatch and refuses to apply the served Specification, is it detecting that based on the Spec’s internal coherence, or based on what the model already knows about the named subject from pretraining? The same question applies to abstention in general. The relationship between hedging, abstention, and sycophancy in current LLMs is opaque, and the interpretive layer interacts with each of them in ways that warrant their own follow-up work.

5.6 Compression and What Makes Personalization Operationally Tractable

Two things surprised us about the compression results. The first is how well a frontier language model handles a very long context. The raw-corpus condition produced the highest prediction scores

in the study, which is not a result one would have confidently predicted three years ago. The second is how much of that signal a structured representation recovers at a fraction of the context cost. The Spec is roughly $25\times$ smaller than the raw corpus on average, and it recovers about 75% of the corpus lift over baseline on the behavioral-prediction task.

Both surprises point at the same industry-direction question. Most of the effort in long-context language models has gone into enlarging the window. Effective use of that window has lagged. The lost-in-the-middle effect (Liu et al. 2024, arXiv:2307.03172) and other position-sensitive degradations are documented in the literature, and an extension of those findings is straightforward. Even if a model could ingest infinite context, it would have to construct its own sub-context (an interpretive frame) to actually reason from. The question is not how much context the model can take in. The question is which context.

This reframes the personalization problem. A faithful interpretive representation, owned by the person it represents, is what the personalization problem reduces to. Faithfulness is the open piece. Only the individual whose Spec it is can verify whether the compression preserved what matters to them. A measure of representational accuracy authored from someone else’s text and judged by someone else’s panel is necessarily an external proxy for what the person being represented would actually recognize about themselves. Closing that gap requires the user in the loop. The existing-deployability point cuts the other way and is more immediately useful. We do not need to wait for context windows to grow further. A compressed interpretive frame already fits inside today’s context budgets, and the gradient established here is recoverable today.

5.7 Privacy and the Case for User-held Representation

Privacy here is not an industry critique. In an agentic world, AI systems increasingly act on behalf of specific individuals in interactions with other agents. The chain of value starts and ends with the user: the first-party agent acts in alignment with how the user reasons, the third-party agent it transacts with can trust that the choices in front of it reflect that user, and the user receives outcomes that are recognizably theirs. Accurate individual representation is what makes that chain coherent. It is a broader definition of personalization, one in which every participant’s structural interest aligns.

The structural condition for this to work is ownership. The representation of an individual has to be held by the individual for the system to work for users at scale rather than against them. If a third party builds the representation and holds it, the same dynamic that produces behavioral micro-targeting in advertising applies with sharper resolution. The inferred representation exists for someone else’s benefit, and the person being represented cannot see it or correct it. User-held representation does not prevent external behavioral modeling, but it changes the relationship. A user-owned interpretive layer exists alongside external models and is the structural defense against weaponization of a crude or unauthorized version of the person.

What this requires beyond user ownership is trust and traceability. A representation that the individual owns is operationally useful only if third-party agents interacting with them can verify the representation’s provenance and authority. Without a trust network, the system degrades back into the opaque-third-party-representation it was supposed to replace. The case for user-held representation is therefore inseparable from the case for an industry-level trust and traceability layer that makes user-held representation operational at scale.

Per-user calibration sits inside the safety envelope, not above or below it. A language model is already shaped toward whatever distribution dominated its pretraining and applies that shape to every user the same way. A Behavioral Specification redirects that existing shape toward the specific user the system is acting on behalf of, rather than adding new shaping on top. Representational accuracy and safety alignment operate at distinct layers, and the constructive question is how the two layers compose. Whether they are formally independent or interact in some structured way is open empirical territory. This paper neither tests nor needs to defend full independence. §7 develops the safety and deployment implications.

5.8 Closing Remarks

What the paper has done is operationalize a representational-accuracy measure of an interpretive layer, and demonstrate that a representation of this kind can be built within the resources and constraints of today’s frontier AI systems. The result is a measurable proxy for whether an AI knows enough about a specific person to act in alignment with how that person actually reasons. This is not the only such measure that can be built, nor the most accurate one possible. It is a measure that exists today and that produces interpretable results on text the model has never seen.

As AI agents take on more of what individuals delegate to AI, an agent cannot act in alignment with a specific person without an accurate representation of how that person reasons. Representational accuracy is what makes alignment claims about specific individuals testable. A Behavioral Specification is one such measure; the field needs more of them, stronger versions of them, and a clearer accounting of what each one captures. Accurate individual representation is the structural problem to address, not a feature to add later.

6. Limitations

The paper’s claims are bounded by four axes of constraint on the experimental setup: the subject sample (§6.1), the measurement apparatus (§6.2), the pipeline and specification stability (§6.3), and the scope of exploration (§6.4). Each is a permanent caveat on how the paper’s results should be read, distinct from the follow-up experiments proposed in §7.

6.1 Subject sample

The 14 main-study subjects are a selected sample, not a population. Pretraining-coverage bias and the single-living-subject constraint are load-bearing for the paper’s framing and are developed in §5.2. This subsection covers four remaining external-validity caveats that §5 does not address.

Public-domain selection. All subjects are historical figures whose autobiographies or memoirs are in the public domain and have been digitized by Project Gutenberg or Internet Archive. That selection pipeline is biased toward figures whose writing was preserved in published form and Western publishing traditions. The paper’s cross-continent spread (Saint Augustine, Bābur, Fukuzawa Yukichi, Sunity Devée, Zitkala-Ša, Olaudah Equiano, Mary Seacole) partially mitigates but does not remove this bias.

Self-presentation bias. Autobiographies are self-curation. What each subject chose to include in their memoir is not a neutral record of their behavior; it is a self-selected narrative that may

over-represent patterns the author wished to be remembered for and under-represent patterns they chose to leave out. Behavioral-prediction batteries derived from autobiography inherit this bias, and neither the pipeline nor the rubric has a mechanism to correct for it.

Translation artifacts. Three subjects’ corpora are English translations of non-English originals (Augustine’s *Confessions* from Latin, Bābur’s *Bābur-nama* from Chagatai Turkic via Persian, Cellini’s autobiography from Italian). Translations introduce stylistic and register shifts that the extraction pipeline processes as if they were original text. A Specification authored from a translated corpus may inherit translator choices in addition to the subject’s actual patterns.

Era. The oldest subject is 4th to 5th century (Augustine); the newest is early 20th century (Zitkala-Ša, Sunity Deveen). Reasoning patterns in modern work contexts, contemporary family structures, technical or digital-native domains, and late-20th-century cultural frames are not sampled. Whether the gradient holds when specifications are authored from modern-era corpora is a generalization the study cannot make from its sample alone.

Autobiography as a genre. Beyond the four caveats above, autobiography is itself a genre with its own distributional properties: reflective, retrospective, narratively shaped, written for an audience, structured by the author’s understanding of what makes a coherent life-story. The behavioral patterns extracted from it are patterns *as the author chose to surface them*. Both halves of the train/held-out split come from the same authored narrative, so within-narrative consistency is exactly what the held-out test is positioned to detect. Whether the gradient holds when specifications are authored from non-narrative source material (real-time chat, decisions under time pressure, technical domains the subject never wrote reflectively about) is not testable from this design. The §7.2 living-user replication and §7.5 stateful-agent variants together begin to address cross-genre generalization.

Taken together, these five caveats mean the paper’s results should be read as evidence for the claims at the conditions tested. Generalization across era, source language, self-presentation mode, source genre, and digital-versus-analog source material requires follow-up experiments.

6.2 Measurement apparatus

This section covers the measurement-apparatus constraints on how the paper’s numbers should be read. The rubric limitations are in §3.3.6; the LLM-as-judge limitation is the canonical one and is treated in full here.

Class-level LLM-as-judge circularity. Every response in this study is generated by an LLM, every judge is an LLM, and the question batteries are also LLM-generated (§3.5). The 5-judge primary panel and the 7-judge sensitivity check together address within-provider circularity (§4.6.1, §4.6.2). What these checks do not address is class-level LLM circularity: the concern that an all-LLM pipeline could be self-reinforcing in ways that human evaluators would not validate. The methodological precedent for the LLM-judge panel is established in §2 and §3.3; the panel answers the directional question reliably but not the absolute-quality question. A stratified human-validation subset is the leading measurement follow-up flagged in §7.2; until that exists, the paper is best read as a prototype with directional, not precise, evidence on its headline claims.

Response-model coverage. The main-study response model is Claude Haiku 4.5. The §4.6.1 Tier 2 cross-provider directional probe ran 2 additional response models (Claude Sonnet 4.6, Google

Gemini 2.5 Pro) on 3 subjects spanning the gradient (Ebers, Yung Wing, Zitkala-Ša) against GPT-5.4-regenerated batteries; Claude Opus 4.6, GPT-5.4, and GPT-4o were used as judges in Tier 2 but not as response models. The Specification-effect direction reproduced on 5 of 6 (subject, response-model) cells under every panel and Δ -definition tested (Sonnet 4.6 and Gemini Pro only; the corresponding Haiku Tier 1 cell is counted in the §4.6.1 “7 of 9” figure, which adds the Haiku response-model row at the same three subjects). The main-study response model is Haiku across all 14 subjects in §4.1; Tier 2 establishes direction across response-model families on a small subset only. The paper’s aggregate numbers should be read as what the Specification does with Haiku; other response models may produce different absolute magnitudes while preserving the gradient.

Prompt-phrasing ambiguity. The authoring pipeline prompts, the response-generation prompts, and the judge prompts all depend on specific word choices, ordering, and phrasing. We did not systematically test prompt sensitivity as part of this experiment. Prompt sensitivity for the Behavioral Specification authoring pipeline is a separate workstream that informed pipeline design and is distinct from this study’s response-generation and judge prompts, which were not varied. Different wordings at any of these stages could produce different numeric results, different extracted fact sets, or different judge scores on the same response. The paper’s claims are downstream of the specific prompts used throughout the study (documented in the public repository scripts); we make no claim about prompt invariance.

Inter-judge calibration variance. Pairwise Spearman ρ across judges is 0.86 to 0.93 (§3.3.4), so the rank order of conditions is stable across the panel. Absolute-score calibration varies (§3.3.3): Gemini Pro fails verbatim-match calibration (4.15 where calibrated judges score 5.0), Opus runs lenient on abstentions (1.41 mean where Sonnet runs strict at 1.14), and length-sensitivity differs across judges. The 5-judge primary aggregate is therefore a stable reading of direction but a panel-specific reading of magnitude. A different judge panel would produce different aggregate numbers while preserving the direction of every claim, which is part of why §5.8 frames the paper as directional rather than precise.

6.3 Pipeline and specification stability

What follows covers pipeline-internal constraints on how the paper’s results should be read.

Pipeline version tested. The specifications used in this study were produced by a specific pipeline version, snapshotted at the time the experiments were run. Two locations preserve that version: the deployed BaseLayer pipeline at <https://github.com/agulaya24/BaseLayer> (the stable production version at experiment time) and a forked, study-specific copy preserved inside the memory-study-repo (`memory-study-repo/baselayer/`, the version used to author the 14 specifications scored in §4). The pipeline has evolved through development, and different pipeline versions produce different specifications on the same source corpus. The paper’s results are specific to the pipeline version tested, and the study does not measure how the gradient shifts under earlier or later pipeline versions. The evaluation harness used here can serve as a benchmark for future pipeline iterations: each new authoring-pipeline version can be measured against the current specifications on the same 14-subject batteries to assess whether the gradient strengthens, weakens, or shifts shape (§7).

Specification stability under the same pipeline version. Running the same pipeline twice on the same corpus at temperature 0 does not produce identical specifications. A numeric-similarity probe (script: `scripts/v11_9_6_pipeline_variance_similarity_20260510.py`) compared the

three reruns of each probe subject’s brief at the sentence level. At sentence grain, verbatim sentence-level matches across reruns are rare. For non-verbatim sentences, best-match cosine similarity against the other run’s sentence pool (MiniLM-L6-v2 sentence embeddings, normalized) produces a pooled mean of 0.55, median 0.54, IQR [0.47, 0.61], with roughly 3% of non-verbatim sentences reaching cosine ≥ 0.80 . Sentence-level analysis is stricter than character-level or paragraph-level overlap; the reruns are not paraphrases of one another at sentence resolution. Qualitative side-by-side reading still shows that the reruns cover the same predicates and behavioral patterns at a higher conceptual level, but the sentence-grain measurement makes the surface variance concrete. The score-level stability that matters for the gradient (per-subject run-to-run SD of Δ_C4a) is reported in the variance probe below, and is what determines whether the §4.1 slope is robust to this kind of surface non-determinism.¹⁰²

Per-subject pipeline variance, characterized. A targeted replication probe was run on three subjects spanning the gradient (Sunity Deveen, C5 = 1.03; Yung Wing, C5 = 1.88; Augustine, C5 = 2.58). For each subject, the Sonnet layer-authoring step and the Opus compose step were re-run three times against the same per-subject extracted fact set at temperature 0, producing three independent specifications. Each rerun was scored on the full behavioral-prediction battery in the C2a (Spec only) and C4a (All Facts + Spec) conditions on the 5-judge primary panel. The resulting per-subject standard deviation of Δ_C4a across reruns is reported below, alongside the cross-subject SD that the §4.1 gradient slope is fit to. For each subject the table reports three numbers: the Spec’s effect on representational accuracy in the §4.1 main study (one authored specification), the standard deviation of that effect across three independent pipeline reruns on the same corpus, and that standard deviation as a fraction of the between-subject standard deviation the §4.1 gradient slope is regressed against.

Subject	Canonical Δ_C4a (§4.1)	Per-rerun Δ_C4a SD (n=3)	% of cross-subject SD
Sunity Deveen	+1.38	0.103	17.4%
Yung Wing	+0.52	0.055	9.3%
Augustine	+0.11	0.130	22.0%
Pooled (3 subjects)	n/a	0.101	17.1%

Read of the precision question. The directional finding survives across reruns: low-baseline subjects keep improving (6 of 6 reruns positive across the 2 low-baseline probe subjects), and the gradient slope point estimate is not materially threatened. Quantitatively, the pooled per-subject run-to-run SD of Δ_C4a is 0.10 on the 1-5 rubric, against a cross-subject SD of 0.59 that the gradient slope is regressed against; run-to-run pipeline variance is therefore on the order of 17% of the signal the slope is fit to, and well under the 95% CI half-width on the slope (0.29). What pipeline variance does affect is the precision attached to any single per-subject point estimate. The per-subject Δ_C4a numbers in §4.1 should be read with a soft uncertainty bar of roughly ± 0.10 around them. Augustine (mid-baseline, canonical $\Delta = +0.11$) sits at the top of its rerun range and the sign flips on 2 of 3 reruns, so individual mid-baseline subjects’ Spec-effect sign is itself within the run-to-run uncertainty band.

¹⁰²Full per-subject pairwise detail at docs/research/v11_9_6_pipeline_variance_similarity_20260510.md; raw JSON at docs/research/v11_9_6_pipeline_variance_similarity_20260510.json. Pool n = 1,304 non-verbatim sentences across three probe subjects $\times$ three pairwise rerun comparisons.

Scope and caveats of the variance probe. The probe covers the lighter-scope variance only: the Sonnet authoring step plus the Opus compose step. Extraction-stage non-determinism is held constant by reusing each subject’s pre-populated SQLite and ChromaDB state across reruns; including extraction would likely add additional variance at the front of the pipeline. The probe covers low-baseline and mid-baseline subjects but does not reach the Franklin-style high-baseline tail ($C5 = 3.77$), so the H2 corollary (that the Spec can interfere with strong pretraining signal at the high-baseline end, producing a near-zero or negative Δ) is not directly stress-tested by this run. With $n = 3$ reruns per subject the per-subject SD point estimates carry their own wide 95% confidence intervals (roughly $[0.5\times, 6\times]$ of the value); the pooled three-subject estimate is more stable than any single per-subject estimate but should still be read as an order-of-magnitude indicator rather than a precision number. With those caveats stated, the run-to-run SD is small enough relative to the cross-subject SD that we accept the §4.1 slope and R^2 as findings about the gradient rather than artifacts of a single specification authoring.¹⁰³

Pipeline model choices were not varied systematically. The pipeline uses Claude Haiku for extraction, all-MiniLM-L6-v2 for embeddings, Claude Sonnet for layer authoring, and Claude Opus for the compose step (§3.7). These model choices were not varied across the study. Different models at any step could produce different specifications: a different extraction model could surface different facts, a different embedding model could change retrieval behavior, a different authoring model could produce differently-structured anchors and predictions, a different composition model could synthesize the layers differently. Extending model support for each pipeline step, and measuring the gradient under alternate pipeline configurations (for example GPT-5.4 extraction, OpenAI embeddings, a non-Anthropic authoring model), is a direct follow-up flagged in §7.3, alongside the broader question of cross-model consistency for both Spec authoring and usage.

6.4 Scope of exploration

Not every experimental combination was run. The study prioritized coverage of the conditions and subjects central to H1 through H5 (§4.1 through §4.4) over running every cell of the design grid. Robustness and ablation conditions were added selectively rather than exhaustively.

Coverage across the experimental grid. The study spans 11 conditions (C1 through C9 plus two wrong-Spec variants), 14 main-study subjects, and a 5-judge primary panel plus 2-judge sensitivity check; response-model coverage is detailed in §6.2 and §4.6.1. Running every possible combination (roughly 6,500 separate cells) was not attempted. Ablation-adjacent conditions (per-layer Spec serving, alternate pipeline model choices, dynamic activation policies) were not run and are planned for future work (§7).

Letta stateful-agent exploration. Letta’s stateful-agent architecture is distinct from the archival retrieval path the other three commercial systems use (§4.4, §4.5). Testing the stateful path required a different evaluation harness (§4.5 test design), and that work pulled us partially outside the main-study scope. The resulting comparison covers three subjects (Hamerton, Ebers, Bābur), one Letta version, and one response model (Claude Haiku). Extending the stateful-agent comparison

¹⁰³Per-rerun specs and judgments are at `data/global_<subject>/_variance_runs/run_<N>/` and `results/global_<subject>/_variance_runs/run_<N>*.json`. Full report and reproducibility scripts at `docs/research/v10_pipeline_variance_analysis.md`, `scripts/_v10_pipeline_variance.py`, and `scripts/_v10_pipeline_variance_report.py`.

across the full 14-subject gradient, across additional response models, and against future Letta releases is flagged as a follow-up in §7.

Twin-2K coverage. Twin-2K (§2.1) is cited in this paper as prior work that measures a related but distinct property (survey-response prediction rather than representational accuracy). We did not run Twin-2K as a condition of the main behavioral-prediction battery; our findings on Twin-2K are framed as prior-work context rather than as a benchmark result. See §2.1 and Appendix F.4 for the detailed treatment.

7. Future Work

Every section of this paper flags at least one follow-up. This section consolidates them into a research agenda organized by theme.

7.1 Subject and Corpus Expansion

A multi-subject living-user replication is the leading follow-up for the entire paper (§5.2, §5.7). The paper’s findings are based on 14 historical subjects; whether they generalize to living users is not directly tested by this study, and replicating the gradient with multiple living subjects (with proper consent and privacy infrastructure) would close that gap. Three related expansions: modern-era corpora (to test whether the gradient holds when specifications are authored from contemporary writing rather than pre-20th-century autobiography, §6.1), non-English original sources (to remove translation artifacts, §6.1), and alternative testbeds that isolate reasoning structure without requiring private data, such as U.S. Supreme Court opinions where documented decisions provide a public record of individual interpretive patterns that can be held out and predicted (§5.3).

7.2 Measurement Methodology

The most impactful measurement follow-up after the living-user replication in §7.1 is replacing the content-match rubric with a differentiated battery that separates interpretation-heavy from literal-recall questions and scores epistemic honesty as its own dimension (§3.3.6). Alongside this: a curated question set with explicit quality control on the backward-design process, a human-validated subset of rubric applications to test whether the rubric was reasonably applied per-response (§3.3.6), and human-judge validation on a stratified subset of responses to address class-level LLM-as-judge circularity (§4.6.8, §6.2). Prompt-sensitivity testing across the authoring, response-generation, and judging stages (§6.2) is a separate measurement-stability follow-up that becomes important once the rubric itself is stabilized.

Retrieval-overlap follow-ups (from the surfaced §4.4.1 finding). Two measurement studies remain open after the §4.4.1 sensitivity check that already covers $K=5$ and semantic-similarity matching for $K=10$ in both controlled and native configurations:

- **Convergence-at-larger- K analysis.** This study tested $K=10$ retrieved facts per question (mean Jaccard 0.083 across systems, §4.4.1) and a $K=5$ sensitivity check that lowered overlap rather than raising it. The follow-up is $K=25$, $K=50$, $K=100$, and higher across the same systems and question set, to map the convergence curve and identify the K threshold at which providers begin to agree on which facts are relevant (if anywhere).

- **Meta-analysis of recall benchmarks against retrieval overlap.** Memory systems that score within a few percentage points on LongMemEval, LOCOMO, and similar recall benchmarks retrieve nearly disjoint top-K facts when given identical fact pools and fixed questions (§4.4.1). Recall benchmarks measure recall, which is what they should measure; the question is what additional dimensions matter for downstream representational accuracy. A meta-analysis comparing benchmark scores to retrieval-overlap on the same systems would clarify what “memory recall” actually predicts about how each system ranks facts for a specific interpretive task. The wrong-Spec per-question meta-analysis (§4.6.5) belongs to the same class of follow-up: a deeper read of which parts of the served context the model referenced under correct versus mismatched specifications.

Inter-judge agreement as a question-quality signal. The 5-judge primary panel agrees on direction (Spearman $\rho = 0.86$ to 0.93) but spreads on absolute magnitude (Krippendorff $\alpha = 0.659$; §3.3.4). Questions where judge spread is unusually wide (e.g., Seacole Q2 C4 with judges splitting 5/2/3/3/1) are candidates for inspection: they may be questions where the rubric anchors apply ambiguously, where the held-out passage admits multiple defensible predictions, or where the question itself is poorly posed. A per-question agreement analysis (variance across the 5 primary judges, or pairwise α at the question level) would identify the questions the panel struggles with most and feed back into the next iteration of the battery.

Confident-misalignment vs abstention. The §3.3 rubric scores both explicit abstention and non-abstention misalignment as 1.00. A coarse post-hoc regex pass classified ~93% of low-end responses as abstention and ~7% as confident-misalignment (wrong referent, off-base inference, or confusion with a different subject; §4.1.1 footnote [`^score-1-composition`]). Why a model picks confident-wrong over abstention on a given question is open: which interpretive frames the wrong-Spec activates that the matched Spec does not, whether the gap correlates with subject-level pretraining coverage, and whether the rubric should score the two failure modes separately are first-order follow-ups.

Pretraining-bleed analysis on wrong-Spec mismatch detection. Under wrong-Spec adversarial pairings, the model often flags the mismatch explicitly rather than complying (60.6% mismatch-detection rate; §4.3). Whether this detection draws on pretraining knowledge of the *correct* subject (effectively recognizing “this Spec doesn’t match Lincoln”) versus on internal coherence of the Spec content alone is open. A pretraining-bleed analysis would correlate per-subject mismatch-detection rate with per-subject baseline pretraining coverage and with per-pair cultural/temporal distance, isolating which signal is doing the detection. §6.2 flags this as a measurement caveat the present design does not isolate.

7.3 Specification design and composition

Component ablation on the authored layers (anchors, core, predictions, brief) is the priority authoring-pipeline follow-up (§5.4). Serving each layer alone and in combinations, measuring Pattern 1 / Pattern 2 / Pattern 3 distributions per configuration, would identify which parts of the pipeline are doing which work. Answers inform both the authoring pipeline’s investment priorities and the dynamic-activation policy’s weights.

Alongside component ablation: alternate pipeline model choices (extraction, embedding, layer authoring, composition) to measure sensitivity to specific LLM choices at each pipeline step (§6.3); a Base Layer referent-variant that retains named entities inside the same dimensional scaffold, to isolate whether the §4.5 Letta-over-Base-Layer gap is driven by referential vocabulary or by the

self-editing process itself (§4.5); and a layered-stack Letta rerun on the matched-rerun subjects, which would likely narrow the §4.5 gap (§4.5).

Cross-model consistency for Spec authoring and usage. Two related questions sit on top of the alternate-pipeline-model-choice follow-up. First, on the authoring side: whether different LLMs at each pipeline step produce specifications that converge on the same behavioral patterns from the same source corpus, or whether the Specification itself drifts with the choice of authoring model. Second, on the usage side: whether different response models interpret the same Specification consistently, applying it to produce comparable predictions on the same held-out questions. The §4.6.1 Tier 2 probe established that the gradient direction holds across three response models on three subjects; whether the Spec is read and applied the same way across the broader model landscape is the deeper consistency question.

Named-entity grounding as a complement to predicate structure. The §4.5 secondary analysis ruled out surface-syntactic alignment as the mechanism for Letta’s case-study lift but identified named-entity grounding as a contributing factor. The Base Layer pipeline abstracts source text into structured predicates for explainability and traceability; the brief retains some personal detail, but specific named entities (place names, person names, institutional names) sit at the edge of what the current predicate vocabulary expresses cleanly. When held-out questions turn on a particular named entity, this can be a measurable disadvantage. A natural follow-up is a hybrid representation that pairs each predicate with the named entities it grounds in: predicates for structure and explainability, entity tags for referential grounding. The serving-layer architecture in §5.4 / §7.4 is where the decision of when to pull the abstract pattern versus the concrete entities lives; the runtime question is when to surface each. A named-entity-grounding-vs-axiom-grounding ablation, plus a paraphrase-resistant rubric that scores directional correctness separately from referential match, are the specific tests this follow-up reduces to.

Natural-language activation (NLA) round-trip Spec faithfulness.¹⁰⁴ Generate hypothetical responses across a battery from a Spec only, then re-extract a Spec from the generated text using the same pipeline. Compare the round-tripped Spec to the original on predicate coverage, axiom recall, and behavioral-pattern fidelity. The round-trip degradation rate measures how much of the Spec’s interpretive content survives the model’s compression-decompression of it, and whether different model families preserve different aspects (anchors vs predictions, structural vs stylistic) of the same input.

Spec component utilization at runtime. Component ablation tests which Spec components contribute to lift; component utilization tests which components a response model actually grounds in. Activation-level techniques (natural-language activation analysis as developed in current Anthropic interpretability work; persona-vector analysis per Chen et al. 2025) applied to (Spec, response, ground-truth) tuples could measure whether the model’s response activates the same Spec sections the rubric scores as load-bearing. The measurement bridges representation-design (what the Spec encodes) and response-time use (what the model retrieves and applies).

¹⁰⁴Natural-language activation (NLA) refers to the line of recent interpretability work that uses natural-language descriptions to label or steer specific internal directions in language-model activation space. We use the term in its general interpretability sense here; specific Anthropic publications in this line are still emerging and may be cited in revised drafts.

7.4 Production serving and infrastructure

The study served the Behavioral Specification statically and in full on every query. Four production-realistic serving-layer follow-ups would refine deployment beyond this static-attachment baseline: dynamic activation (selecting which parts of the Spec to attach per query), user editing and inspection (how a user can update or correct their own Spec post-authoring), temporality handling (how the Spec ages and what triggers re-authoring; §7.5), and topic decomposition (whether the Spec can be partitioned by domain for selective serving). Each is a measurement question in its own right: whether the gradient, mechanism, and composition findings hold under each production serving strategy.

Layered-context stacking studies. This paper tests Spec-on-No-Context, Spec-on-facts, Spec-on-corpus, and Spec-on-memory-system separately. A stacking study would systematically vary the layered context: memory-system retrieval at increasing K, with and without facts, with and without Spec, with both. The output is a map of which layer combinations produce additive lift, which produce subadditive interaction, and which produce antagonistic effects. The study format extends the §4.4 memory-system layering analysis by treating context type as a factorial design rather than a between-condition comparison.

7.5 Stateful-agent implementations and temporal drift tracking

Several follow-ups sit adjacent to the paper’s static-snapshot design.

Stateful-agent variant of the Behavioral Specification. The pipeline as tested is offline and batch. A persistent, self-editing variant that ingests new source material as it arrives, re-edits anchors and predictions in place, and maintains version history with provenance across edits is a natural next step. The §4.5 Letta exploration (N=3, post-hoc) is one data point on an adjacent architecture; building and evaluating a stateful-agent Base Layer implementation on the full 14-subject main-study battery would close the comparison within a single architectural family and extend §4.5 to a layered-stack rerun against Letta at full scope.

Cleaner §4.5 rerun with naming and scaling controls. Two specific extensions of the §4.5 exploration are worth running as a unit. First, anonymize the source corpus before Letta ingestion so Letta writes an anonymized memory block, matching Base Layer’s anonymized-during-authoring convention; the §4.5 naming asymmetry (Letta ingests named corpus, Base Layer strips and later restores names) is removed as a confounder. Second, extend the corpus-size axis past the Bābur ceiling to a larger (>250K-word) subject corpus that pushes the Letta block past its character ceiling. The matched-model gap was small at Hamerton (25K-word corpus), largest at Ebers (48K-word corpus), and smaller again at Bābur (223K-word corpus); whether that pattern continues, re-widens, or flattens at extreme corpus size is the empirical question. Both extensions together would turn §4.5’s case study into a controlled comparison.

Temporal drift tracking. The static snapshot tested here is a point on a trajectory. A Specification authored at one time, compared against a later specification on an expanded corpus, produces a measurable diff: which anchors appear or disappear, which predictive templates shift, which axioms strengthen or weaken. From a sequence of past specifications, the trajectory predicts the next. The 14-subject corpora collected for this study can be back-sliced by chapter boundaries or publication era for an initial drift test within the current sample. A purpose-built companion study on sequential public records (US Supreme Court opinions, shareholder letters, research papers) is the natural extension.

Canonical life events. Discrete pivots that flip reasoning architecture (a major career change, a religious conversion, a significant loss, a public stance reversal) are distinct from gradual drift. The main-study autobiographies were not structured to test this case. A snapshot specification authored before such an event predicts pre-event reasoning, not post-event reasoning, even though the person’s underlying patterns have materially shifted. Whether to detect these events automatically, allow user annotation, or maintain period-specific specifications is an open production-deployment question.

Continuous-representation infrastructure. Both of the above converge on the same engineering target: a background process that watches incoming corpus material, re-authors the Specification as new material arrives, and emits drift telemetry as a first-class output (what changed in the Spec, by how much, and against what baseline). The Letta-style stateful agent (§4.5) and the sequential-checkpoint test design are complementary tests for this kind of implementation: the first isolates online self-editing as a way to produce the representation, the second isolates temporality as a property of the representation itself. As a downstream effect, the daemon’s continuous output (the sequence of Spec versions, diffs, and drift telemetry) itself becomes a training corpus for next-generation pipeline development.

Additional architectural paths worth testing against the same target, beyond stateful-agent and drift-tracking variants, include agent-edited persistent memories outside the MemGPT family, fine-tuned per-user models that expose their internal representation for audit, and hybrid architectures that combine offline-extracted specifications with online self-editing.

Per-user feedback as a learning signal for the Specification. Every interaction with an AI agent can produce a small correction signal: the user edits a response, says “actually I would have done X”, or rejects an answer outright. Each of these is a hint that the current specification mispredicted the user’s behavior. Used as a feedback signal, these corrections can update the Specification directly: the affected predicate or anchor is re-extracted and re-authored, then re-composed into the Spec. Because the update unit is structured text rather than model weights, the change is interpretable and the per-update cost stays at the §3.7 pipeline cost. Two design questions matter. First, drift risk: corrections without a source-text anchor can pull the Spec in arbitrary directions, so the corpus stays the source of truth and corrections are bounded by it. Second, signal quality: explicit corrections are sparse but high-fidelity, while everyday divergence is abundant but noisy; treating both as the same kind of feedback is wrong. A first-pass design using explicit corrections only, batched daily, with versioned diffs logged for rollback, is the cheapest test of whether correction-driven updates produce measurable accuracy gains beyond the static snapshot.

7.6 Safety-alignment integration

The positioning argument (per-user calibration as redirection of existing shaping rather than additional bias, and as orthogonal to safety alignment) is in §5.7. Two concrete follow-ups extend that positioning. First, the Spec-induced refusal cases (§4.3, §4.6.7): a post-hoc classifier audit of 81 Spec-induced refusals across 5 memory systems (`docs/research/refusal_intent_classification.md`) found 75 of 81 (93%) were routine behavioral prediction rather than morally loaded. The refusal pattern is general caution when information is thin, not a moral-integrity mechanism. Whether it composes cleanly with existing safety frameworks across benign and malicious user types is open. Second, the specifications in this paper were authored from public-domain autobiographies of subjects not selected on intent. What a Specification for a user with malicious intent would contain, and what happens when an agent is deployed on that user’s behalf, is untested. Both belong to

collaboration with AI safety researchers rather than single-lab follow-ups.

A narrower follow-up: the wrong-Spec adversarial control (§4.3) showed response models can recognize Spec content as belonging to a different historical period or persona. Whether models can recognize specifications encoding adversarial values (personas that endorse harmful behaviors) and flag them rather than comply is a direct extension that bears on live-user deployment.

8. Data, code, and reproducibility

Data availability. All raw response files, per-judge judgments, batteries, and aggregated results for the 14 main-study subjects are in the public study repository at github.com/agulaya24/beyond-recall under `results/global_<subject>/` and `results/hamerton/`. Source autobiographies are public domain (Project Gutenberg and Internet Archive). Per-subject Project Gutenberg IDs are listed in §3.4 Table 3.2. Memory-system raw retrieval and ingestion logs are at `results/global_<subject>/<system>_*.json`. The Letta stateful-agent matched-rerun artifacts are at `docs/research/_letta_rerun/`. The full-stack Letta rerun comparison is at `docs/research/_letta_rerun/fullstack_named/`.

Code availability. The study repository at <https://github.com/agulaya24/beyond-recall> contains the analysis and re-run scripts under `scripts/`, plus the study-specific forked pipeline used to author the 14 specifications scored in §4 preserved at `baselayer/`. The deployed Base Layer pipeline source (extract, embed, author, compose) is at <https://github.com/agulaya24/BaseLayer>. Reproducibility pointers from each numerical claim to its supporting script are in `docs/PROVENANCE_INDEX.md` and `docs/DATA_REFERENCE.md`. The §4.1 battery-composition sensitivity analysis is reproducible via `scripts/_v10_battery_sensitivity.py`. The §3.3.6 rubric-handling validity audit is reproducible via `scripts/audit_low_end_inflation.py`. The §4.3 hedging classifier is at `scripts/classify_hedging.py`.

Supplementary materials. Per-subject worked examples (Appendix E in the public-repository version) and per-subject paired-delta tables (Appendix B subsections B.2 and B.3 per-subject distribution) live at `docs/supplementary/` in the repository. They are cross-referenced from main-paper sections at the indicated pointers and are reproducible from the same data and scripts that produced the in-paper analyses.

Agent-friendly study repo tooling. The study repository is structured for both human reading and agent consumption. A combined SQLite + ChromaDB knowledge index (`workspace/study_knowledge.db`, `workspace/study_vectors/`; built by `scripts/index_study_repo.py`) covers 206 files and 3,702 chunks across the paper, supporting docs, per-subject specs, judgments, retrieval logs, and analysis scripts. An MCP server exposing this index plus typed lookups (per-subject score retrieval, claim provenance, condition-pair anchor-crossing queries) is available as an MVP at `memory-study-repo/mcp/`; the decision to optimize the repo for agent consumption is based on the observation that the same artifacts that make a paper easy to verify mechanically (stable anchors, structured judgments, machine-readable schemas) also make the paper easier for human readers to skim and verify.

Compute and cost. All response generation and judging used commercial APIs (Anthropic, OpenAI, Google) at standard rates. No specialized hardware was used. All experiments are runnable on a standard developer laptop.

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Funding. This work was self-funded.

Conflicts of interest. The author is the founder of Base Layer, the project that develops the Behavioral Specification pipeline this paper evaluates. Memory-system providers tested in this paper (Mem0, Letta, Supermemory, Zep) were used through their public APIs at standard rates; no provider was given preferential framing, and Base Layer does not have commercial relationships with any of them. Self-reported benchmark scores from each provider are reported as published; this paper does not adjudicate disputes between providers’ published claims (§2.2).

License. Apache 2.0 for code and Creative Commons Attribution 4.0 for the manuscript and data analyses produced by this study. Source autobiographies are in the public domain.

Acknowledgments.

This research was self-funded. No external funding, grants, or institutional support was received.

The author is grateful to the research community whose work made this study possible. The teams behind LongMemEval, AlpsBench, PersonaGym, Twin-2K, and LoCoMo built the benchmarks that revealed the gap between recall and understanding. The teams behind Mem0, Letta, Supermemory, and Zep built the memory systems that this work seeks to complement, not replace. The work of Chen et al. on persona vectors, Jiang et al. on dynamic user profiling, and Betley et al. on emergent misalignment shaped the framing of this study directly. This paper is standing on the shoulders of their work, and their approaches helped clarify what this approach is and what it is not. Conversations with the broader memory-systems and AI-personalization research communities further informed the design of this paper. Specific gratitude goes to the cross-LLM reviewer panels (Gemini 2.5 Pro, Mistral Large, Cerebras Qwen3 235B, Groq Llama 3.3 70B, GPT-5.5) whose iterated reviews materially improved earlier drafts. All errors are the author’s.

The author thanks his wife Bavani, his mother, Walnut, and Wasabi for their unwavering support, which allowed this work to come into existence.

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Appendix A. Predicate Vocabulary

A.1 The 46 Constrained Predicates (redirected)

The full controlled-vocabulary spec (46 predicates organized into Behavioral / Identity / Knowledge / Procedural / Relational / Temporal / Attentional groups, each with a verbatim definition, example triples, and authoring constraints) is in the public repository at `docs/supplementary/appendix_A_1_predicate_vocabulary.md` and in source form at `data/predicates/predicate_vocabulary.md`.

What §3.7 already establishes: the predicate vocabulary is a fixed, audit-friendly set used by the extraction step to convert raw corpus text into structured (subject, predicate, object) triples. The vocabulary is closed by design (extraction cannot invent new predicates) so that downstream layers can be authored against a stable typed substrate. Predicate definitions, example triples, AUDN (Add / Update / Delete / NOOP) operation rules, and the three behavioral-axis tags (LITERAL / INTERPRETIVE / REFUSAL-TRIGGERING) used by the question-battery formation pipeline (§3.5, Appendix B.3) are documented in the repository file.

A.2 Provenance and design choices

The vocabulary was iterated in three stages. The initial 30-predicate list (sessions 1-48) favored values, activities, and biography. Session 49 added 6 predicates (`unknown`, `attended`, `interested_in`, `wants_to`, `loves`, `hates`) to preserve semantic distinctions that the initial vocabulary collapsed. Session 52 added `plays` and `monitors`. Session 55 added 8 relationship predicates to raise relationship-fact extraction from 0.8% to the 3 to 5% target range.

The vocabulary is deliberately behavioral rather than biographical. The ratio of predicates in the behavioral-patterns plus values-beliefs plus emotions-dispositions groups (23 of 46) to the biographical-context group (7 of 46) encodes a design decision: extraction is steered away from facts that are easily verifiable in external sources (city of birth, schools attended) and toward patterns that require reading the source text to infer (what the subject avoids, prefers, values, fears).

A.3 Not in the vocabulary

Three predicate categories that commonly appear in general-purpose knowledge graphs are deliberately excluded:

- **Evaluative predicates about the subject from a third party** (for example, `considered_brilliant_by`). These invert the direction of claim: the subject is the object rather than the source of the reasoning.
- **Time-indexed state changes** (for example, `became`). The vocabulary handles change-over-time through the AUDN ADD / UPDATE / DELETE / NOOP operations at the fact level, not through predicate selection.
- **Causal predicates** (for example, `caused`, `triggered`). Causal inference is produced in the authored layers (predictions, anchors) from collections of facts, not encoded at the extraction step.

A.4 Live deployment

A live web deployment of the pipeline described in §3.7, with served briefs across additional subjects beyond the 14 in this study, is available at `base-layer.ai` for readers interested in seeing the

served-specification format in interactive form.

Appendix B. Question Batteries

B.1 The 10 fixed behavioral-prediction categories

Every behavioral-prediction question in the study is tagged with exactly one of ten fixed categories. The category set is identical across all 15 batteries (14 main-study plus Franklin). Each category is a behavioral dimension the question probes; the category does not constrain the answer format.

Category	What it probes	Example question
decisions	How the subject resolves concrete choices.	“When his uncle’s family emigrates to New Zealand, would Hamerton consider joining them?”
values	What the subject holds as important when forced to rank.	“When confronted with German Neology, would Hamerton accept or reject the established Protestant position?”
relationships	How the subject engages with specific people or classes of people.	“How would Hamerton’s religious heterodoxy affect his social standing among the Lancashire gentry?”
conflict	How the subject responds when a line is crossed.	“If his tutor attempted to physically harass him, would Hamerton submit or resist?”
learning	How the subject acquires skills, knowledge, or lessons.	“Given Hamerton’s difficulty following spoken French, what would he do about it?”
risk	How the subject handles uncertainty, exposure, or irreversibility.	“Would Hamerton choose to encamp alone on remote Scottish moors, despite it being considered eccentric?”
creativity	How the subject produces or evaluates creative work.	“Would Hamerton publish his early poetry at his own expense, and what would the commercial result be?”
stress	How the subject responds to pressure, exposure, or failure.	“When offered a grand opportunity to organize an exhibition, would Hamerton accept?”

Category	What it probes	Example question
<code>career</code>	How the subject makes professional trajectory choices.	“Would Hamerton follow Ruskin’s advice to study nature directly rather than learn from traditional masters?”
<code>change_over_time</code>	How the subject shifts or persists across life phases.	“After his poetry failed commercially, would Hamerton continue writing verse?”

Each subject’s battery covers 8 to 10 of these categories; no battery skips more than 2. The distribution within a subject reflects what the training half of that subject’s corpus naturally supported, not a quota.

B.2 Per-subject battery composition (10-category by 15-subject matrix)

The per-subject battery-composition table (15 subject rows by 11 columns: ten behavioral-prediction categories plus a per-subject total) is in `docs/supplementary/appendix_B_per_subject_paired_delta_tables.m` for length reasons; the headline composition facts are summarized here.

Each subject’s battery covers 8 to 10 of the ten categories listed in §B.1. Per-subject totals are 39 questions for the 13 global subjects and Hamerton, and 40 for Franklin.

Column totals across all 15 subjects:

Category	Questions
Values	115
Decisions	93
Relationships	84
Conflict	66
Learning	65
Stress	60
Creativity	34
Change over time	27
Career	23
Risk	19
Total	586

Raw batteries are at `results/global_<subject>/battery_v2.json` (global subjects), `data/hamerton/battery.json` (Hamerton), and `data/franklin/battery.json` (Franklin), counted over `tier == "behavioral_prediction"` slices.

B.3 Behavioral-axis distribution (Literal / Interpretive / Refusal-Triggering)

A secondary classification of all 586 audited questions (the 14 main-study batteries totaling 546 plus the Franklin reference battery of 40), produced by Claude Haiku 4.5 as a post-hoc auditor, tags

each question by what cognitive operation it requires the response model to perform. Full audit at `docs/research/question_category_audit.md`.

Aggregate distribution:

Axis	n	%
Literal Recall	60	10.2%
Interpretive Inference	403	68.8%
Refusal-Triggering	123	21.0%

Per-subject distribution: the 15-row per-subject behavioral-axis breakdown (Literal / Interpretive / Refusal counts per subject) is in `docs/supplementary/appendix_B_per_subject_paired_delta_tables.md` for length reasons; the aggregate distribution above is the summary referenced by §4 claims. Notable points from the per-subject table: Hamerton’s battery is the most refusal-heavy of the 14 main-study subjects (19 of 39); Franklin’s is the most interpretive (37 of 40, with 0 Literal Recall); the 13 global subjects cluster in the 22 to 33 Interpretive range with refusal counts between 2 and 15. Source: `docs/research/question_category_audit.md`.

B.4 Category-level effect size on Δ_{spec}

Mean Δ_{spec} (C2a minus C5) broken down by the behavioral-axis classification. Counts are on the N=14 main-study subset (total 546 questions); the Franklin reference battery (40 additional questions, included in B.3 above and the §4.1.2 reference) is excluded from category-level effect-size computation because Franklin is excluded from every N=14 inferential statistic (§4.1.2). Source: `docs/research/question_category_audit.md`.

Axis	n	Mean Δ_{spec}	Median Δ_{spec}
Literal Recall	60	+0.792	+0.800
Interpretive Inference	366	+0.397	+0.400
Refusal-Triggering	120	+0.417	+0.200

The Literal Recall bucket has the largest aggregate Δ_{spec} , but the magnitude is mechanically explained by the C5 baseline distribution: 66.7% of Literal Recall questions sit at $C5 \leq 1.5$ (rubric floor), compared to 48.6% of Interpretive Inference questions, and median C5 for Literal Recall is 1.00. The Spec moves these floor-anchored responses from abstention or off-base to substantive prediction, producing large per-question rubric-point gains by construction. When restricted to the 9 low-baseline subjects, Literal Recall Δ_{spec} is +1.12 and Interpretive Inference is +0.67; when restricted to the 5 mid-baseline subjects, both shrink to near zero. The Literal Recall > Interpretive Inference ordering survives this baseline control, but the absolute magnitude is dominated by C5 floor effects rather than by question-type-specific Spec advantage. The interpretation-heavy emphasis elsewhere in this paper (§1.3, §4.4.3, §5) is not that Interpretive Inference questions yield the largest raw Δ — they do not — but that the Spec’s distinctive contribution is on Interpretive Inference questions, where retrieval-only systems hit a ceiling. Retrieval can surface the relevant fact and lift a Literal Recall question; it cannot supply the interpretive pattern an Interpretive Inference question requires. The per-axis aggregate is conservative because it averages over questions where the baseline already produces a substantive answer; the distinctive Spec contribution is at

the per-question mechanism level (§4.4.3 Pattern 1), not at the per-axis raw mean.

B.5 Per-subject by axis Δ_{spec}

Summary of the cross-subject pattern under 5-judge primary aggregation: the strongest positive Spec effects cluster on three subjects. Hamerton (Literal +1.68, Interpretive +1.30, Refusal +1.25), Sunity Devee (+1.38 / +1.16 / +1.35), and Bernal Díaz (+2.00 / +0.44 / +0.64) carry the largest gains. Augustine, Equiano, and Zitkala-Ša show negative or near-zero deltas across all three axes, consistent with their status as mid-baseline subjects on the §4.1 gradient. Fukuzawa and Seacole show their largest positive effects on Interpretive Inference specifically (+0.83 and +0.79). Bernal Díaz’s +2.00 on Literal Recall is computed over a small per-axis n (single-digit questions per axis per subject), so the per-subject estimate is high-variance; the cross-subject correlation pattern in §B.6 is the more robust signal.

Full breakdown at docs/research/question_category_audit.md; per-subject axis- Δ scaffold values at docs/research/v11_emit/appendix_b_battery.json (claim ids appB_5_<subject>_<axis>_delta).

B.6 Battery-composition sensitivity

This appendix provides the technical detail behind the §4.1 battery-sensitivity controls.

B.6.1 Battery-question-type correlations.

Across the 14 main-study subjects:

- Δ_{spec} range: -0.31 to $+1.37$
- Corr(fraction of Literal Recall questions, subject-level Δ_{spec}): $r = +0.595$ (recomputed under strict 5-judge primary; the legacy audit-doc value was $+0.646$, which used a Hamerton-divergent intermediate aggregation)
- Corr(fraction of Interpretive Inference questions, subject-level Δ_{spec}): $r = -0.466$ (legacy: -0.582 , same caveat)
- Corr(fraction of Refusal-Triggering questions, subject-level Δ_{spec}): $r = +0.212$

The positive Literal Recall correlation and negative Interpretive correlation imply that subjects whose batteries over-weight literal recall also produce larger measured Δ_{spec} values.

B.6.2 Multiple regression controlling for Literal Recall fraction.

A multiple regression of Δ_{C4a} on both C5 baseline and Literal Recall fraction across the 14 main-study subjects yields a partial coefficient on baseline of **-0.88 [95% CI -1.13 , -0.63], $p < 10^{-5}$** , attenuated from the univariate -0.96 by about 8%. Literal Recall fraction enters as a significant partial predictor ($\beta = +2.30$ [$+0.34$, $+4.26$], $p = 0.026$), but baseline carries the bulk of the explained variance: 63.6% uniquely attributable to C5, 6.9% uniquely attributable to Literal Recall fraction. The two predictors are not collinear (Pearson $r = -0.28$, VIF = 1.08 for both), so the partial coefficients are stable. Adjusted R^2 rises from 0.80 to 0.87 when Literal Recall fraction is added; the controls are additive rather than redundant. The gradient on baseline survives; it is not an artifact of battery composition.

B.6.3 Hamerton-leverage subset regression.

Hamerton’s 80-question battery predates the global-subject pipeline and uses a slightly different backward-design path (the legacy Haiku 4.5 generator that originally produced Franklin and Hamerton); the 13 global subjects’ main-study batteries also use Claude Haiku 4.5 but were

regenerated by `run_global_rerun.py` against a uniform prompt template. All 14 main-study batteries share the same generator family. A subset regression dropping Hamerton (N=13 globals) yields a slope of **-0.89** [95% CI **-1.18, -0.61**], $R^2 = 0.81$, $p < 10^{-4}$, compared to the full-sample -0.96. The point estimate attenuates by about 7%, and the 95% CIs overlap substantially. The gradient is not Hamerton-driven. A separate GPT-5.4-regenerated battery set (`results/global_<subject>/battery_gpt54.json`) exists for each global as a circularity control; its results are reported in §3.5.1 and §4.6.1, not folded back into the §4.1 gradient itself.

B.6.4 Discussion.

This is a battery-composition confound in the cross-subject gradient; the paper’s gradient claim is therefore specifically about mean score movement per subject, not about mean score movement per category. §5.3 and §7 flag a follow-up study with a category-balanced battery as the primary design improvement for future gradient work.

B.6.5 Hamerton-leverage at the per-question grain.

The B.6.3 subset regression checks the per-subject mean grain. A parallel question is whether the per-question extreme upward anchor crossings catalogued in `docs/research/wins_inventory_20260428.json` (60 unique cases across 18 condition pairs, 351 paired low-baseline questions on C5 to C4a) are concentrated on Hamerton specifically. They are: Hamerton accounts for 15 of the 60 unique extreme jumps (25%) on a battery of 80 questions (18.75% extreme-jump rate); the other 13 subjects average 8.9% extreme-jump rate across 39-question batteries.

Hamerton’s elevation is real but its cause is not isolated by the present design. Two candidate mechanisms (the legacy battery-generator path and the subject’s thin pretraining coverage, C5 = 1.26, the lowest in the main study) are not separately identifiable. A third candidate considered earlier, behavioral-predicate density per word, does not survive inspection: Hamerton’s served Specification is the full four-layer stack (anchors + core + predictions + brief, ~5,334 words), the same structure and size as the globals’ ~5,300-word specifications, so Spec length does not differentiate Hamerton from the rest of the sample.

As measured by the heuristic classifier, the mechanism distribution on Hamerton’s 15 jumps versus globals’ 45 is nearly identical (PATTERN_PREDICATE+HYBRID share: Hamerton 73.3% vs globals 80.0%); the heuristic does not discriminate jumps from non-jumping Spec-loaded controls (`docs/research/pattern_activation_deep_20260428.md`), so this near-identity is consistent with the heuristic detecting Spec-loaded response style rather than the lift mechanism.

B.7 Coupling-free reframing of the gradient

The headline slope regresses $\Delta_C4a = C4a - C5$ on C5, which mechanically embeds a -1 component when C4a is bounded on the 1-5 scale and partially independent of C5. To triangulate from a non-coupling-prone angle, we ran three additional checks on the same per-subject (C5, C4a) data (script: `scripts/_v10_coupling_sensitivity.py`; full output: `docs/research/v10_coupling_sensitivity_analysis.md`).

B.7.1 Level regression.

The level regression $C4a \sim C5$ produces a slope of **+0.04** [95% CI **-0.24, +0.33**], $R^2 = 0.008$, $p = 0.76$. C4a is essentially flat across the C5 range of 1.02-2.77 and clusters tightly around its mean of **2.44** at the per-subject grain.¹⁰⁵ The Spec does not differentially “lift” low-baseline subjects more

¹⁰⁵Per-subject mean C4a across the N=14 main-study subjects is **2.44**, computed as the cross-subject mean of the

than high-baseline ones in any treatment-effect-heterogeneity sense; it produces a roughly constant post-Spec C4a mean per subject regardless of baseline, and the apparent Δ -on-C5 gradient equals the baseline shortfall.

B.7.2 Permutation test.

A 10,000-iteration permutation test that shuffles C4a across subjects (preserving the bounded marginal but breaking any link to C5) yields a null distribution for the Δ -on-C5 slope centered at -0.998 (SD 0.127). The observed -0.960 is not extreme against this null (two-sided $p = 0.77$). In plain language: even when C4a values are randomly reshuffled across subjects, the Δ -on-C5 slope still lands near -1 on average, because the change-score parameterization mechanically pushes the slope toward -1 whenever C4a is roughly independent of C5. The -0.96 the headline regression reports is what the regression arithmetic forces, not independent evidence that low-baseline subjects benefit more from the Spec at the per-subject mean grain.

B.7.3 Bootstrap.

A 10,000-iteration subject-level bootstrap returns CIs of $[-1.254, -0.740]$ for the Δ -on-C5 slope and $[-0.254, +0.260]$ for the level slope. The level CI straddles zero, consistent with the level-regression finding that the per-subject C4a mean is roughly constant across baselines.

B.7.4 Reading the gradient against this.

The substantive finding survives the coupling check, but its framing has to shift away from “the Spec acts more strongly on low-baseline subjects” toward the per-question reframing in §4.1: low-baseline subjects have a larger pool of questions at low rubric anchors, so the Spec has more opportunity to produce upward integer-anchor crossings, which aggregates as a larger per-subject mean lift. The directional asymmetry on those crossings (no observed transitions from anchors 2, 3, or 4 into anchor 5 across the full 14-subject panel; the only anchor-5 endpoints reached come from anchor 1) is consistent with the §4.2 finding that even the full source corpus C8 plateaus at a similar per-subject mean.

B.8 Per-predicate ablation (Phase 2c)

To probe whether single behavioral predicates within the Spec are uniquely load-bearing, we ran a per-sentence ablation experiment on a stratified sample of 16 extreme-upward-jump cases. For each case, the heuristically-identified causal predicate (highest-token-overlap Spec sentence vs the question and held-out passage) was located in the served Spec and three response variants were generated at temperature 0 with Claude Haiku 4.5: (1) original (full Spec), (2) ablated (predicate removed, replaced with a length-matched neutral biographical filler), (3) reversed (predicate replaced with a behavioral opposite synthesized by Sonnet). Each variant was scored by the 5-judge primary panel.

Results (script: `scripts/run_predicate_ablation.py`; data: `docs/research/predicate_ablation_results_20`)

- Mean Δ _removal (original minus ablated) across 16 cases: $+0.05$ anchor points (95% CI $[-0.35, +0.45]$)
- Mean Δ _reversal (original minus reversed): -0.24 anchor points (95% CI $[-0.45, -0.02]$)
- 2 of 16 cases showed Δ _removal ≥ 1 anchor; 11 of 16 had Δ _removal < 0.5

2-decimal per-subject C4a values in the §4.1 / Appendix D.1 table ($34.15 / 14 = 2.4393$). A reader recomputing from the published table reproduces this figure exactly. Full data-lock note in `docs/research/DATA_LOCKS.md`.

Single-predicate removal does not measurably reduce response quality on this sample. The paper does not interpret this as evidence that the Spec is mechanistically inert: the higher-level mechanism evidence from the wrong-Spec adversarial control (Appendix C / §4.3) shows the Spec as a whole is doing causal work. The null result on per-sentence ablation is consistent with redundant Spec construction, in which multiple sentences across the anchors / core / predictions / brief layers reinforce the same behavioral patterns; removal of any single sentence leaves the pattern accessible elsewhere in the Spec.

A methodological caveat applies. Original-condition reproduction at temperature 0 was not bit-exact deterministic across reruns; mean drift between the recorded original score (from `docs/research/wins_inventory_20260428.json`) and the rerun original score was -1.44 anchors, with 9 of 16 cases drifting by more than 1 anchor. Some of the variance in Δ_{removal} is rerun stochasticity rather than ablation effect. The extreme-upward-jump cases specifically show higher pipeline variance than the per-subject mean grain documented in §6.3.

Future work tightening (per the test’s own report): human-rated predicate identification (vs heuristic), larger N (all 47 PATTERN_PREDICATE cases), irrelevant-predicate control (matched-length unrelated predicate to test the “any rich persona text” alternative), multi-predicate cluster ablation.

B.9 Footnote-redirect technical detail

This subsection holds the longer technical content for footnotes that would otherwise grow to multi-paragraph length. Each entry is keyed to the footnote name in the body.

B.9.1 The $+0.89$ vs $+0.93$ aggregation reconciliation.

The $+0.89$ figure is the canonical cross-subject mean of per-subject Δ_{C4a} . Each subject’s Δ is computed as that subject’s per-question 5-judge primary mean under C4a minus their per-question mean under C5; these per-subject Δ s are then averaged across the 9 low-baseline subjects. The grand-mean alternative grand-averages all per-question scores under each condition first and then takes the difference, yielding $+0.93$ (the difference of the C4a grand mean 2.45 and the C5 grand mean 1.52). The two numbers are not in conflict; they answer slightly different questions. The per-subject-mean grain ($+0.89$) is the unit of inference used throughout this paper because every statistic is computed at the subject level first, then aggregated across the 14 subjects (§1.2 aggregation rule).

B.9.2 [^heldout-leakage-audit]. Held-out leakage audit detail.

A held-out leakage audit on the 60 unique extreme-upward-jump cases (full report at `docs/research/held_out_leakage_investigation_20260428.md`) found 0 6-gram, 2 4-gram, and 12 3-gram matches between held-out passages and C4a responses. Of the 9 cases with any leak, 6 are short generic phrases also resident in the served facts list (CORPUS_LEAK), 2 are subject-specific n-grams not in any served context (best explained by pretraining recall of public-domain autobiographies; PRETRAINING_MEMO_CANDIDATE), and 1 is generic English (COMMON_PHRASE). The longest shared run anywhere is 4 tokens, well below transcription length. Severity verdict: rare; no structural validity concern; footnote acknowledgement is the appropriate paper-text treatment. Excluding the 2 pretraining-memorization candidates from the extreme-upward-jump set shifts the C5 to C4a low-baseline extreme-jump count by at most 1 (20 to 19); per-subject mean Δ s are unchanged at the per-question level. The “held-out passage” was held out from served Spec / facts, not from pretraining, and the audit confirms that interpretation: where C4a held-out-to-post leakage exists, it is either short generic phrasing also resident in the

served facts (trivially short) or subject-specific content in the model’s pretraining, not study-design contamination of the served context.

B.9.3 [`^supermemory-no-retrieval`]. Supermemory NO_RETRIEVAL placeholders.

Across the full 14-subject Supermemory analysis, 30 individual responses (Augustine 2 questions, Equiano 28 questions) were Supermemory provider-failure placeholders rather than substantive predictions, scored at the rubric floor (1) by the judge panel. We treat these as scored data rather than missing data, consistent with how the rest of the study handles low-quality responses. Excluding the 30 NO_RETRIEVAL records as missing data would shift Supermemory’s aggregate Δ slightly higher; the qualitative story (small aggregate at both grains, bimodal per-question distribution) holds either way.

B.10 Pre-registered hypotheses and post-hoc analyses

The paper distinguishes pre-registered hypotheses (H1–H5, locked against the statistical commitments in `docs/ANALYSIS_PLAN_LOCK.md`) from analyses that emerged during the work. The following table catalogues every load-bearing analysis result reported in §4 and identifies its status. Post-hoc items are reported as exploratory rather than at the same evidentiary tier as the pre-registered hypotheses.

Item	Status	Where reported	Note
H1 Spec-context outperforms no-context	Pre-registered	§4.1, §1.3 1st bullet	Headline gradient
H2 Spec benefit inversely proportional to pretraining coverage	Pre-registered	§4.1, §4.1.2, §1.3 1st bullet	Gradient at both ends; Franklin reference
H3 Content-specificity (correct vs. wrong Spec)	Pre-registered	§4.3, §1.3 4th bullet	Wrong-Spec controls v1 + v2
H4 Spec interacts with retrieval through three patterns	Pre-registered	§4.4, §4.4.3, §1.3 5th bullet	Memory-system composition
H5 Compression: ~7K-token Spec recovers most of corpus signal	Pre-registered	§4.2, §1.3 3rd bullet	At 5x to 80x smaller context
Cross-system retrieval-overlap divergence	Post-hoc	§4.4.1; sensitivity in §4.6.6; §1.3 7th bullet	Surfaced during memory-system analysis; mean Jaccard 0.083 across 10 system pairs; survives semantic-similarity matching
Letta stateful-agent case study	Post-hoc	§4.5; full in Appendix G	N=3, exploratory
Letta semantic-duplication scaling	Post-hoc	§4.5; Appendix G	Surfaced in this paper’s analysis; cosine $\geq 0.85 = 56.1\%$ on Bābur
Abstention-credit validity audit	Post-hoc	§3.3.6	9.4% of refusals score ≥ 2.0 ; bias direction makes the Spec effect likely larger than reported
Per-subject wrong-Spec heterogeneity	Post-hoc	§4.6.5	5/13 subjects show small positive v1 deltas (coincidental content overlap)
Hedging-elimination (28.8% $\rightarrow$ 0.0%)	Post-hoc	§4.3, §1.3 6th bullet	Surfaced from response-level audit
Battery-question-type sensitivity (literal-recall fraction)	Post-hoc reactive	§4.6.3, Appendix B.6	Added in response to v9/v10 reviewer concerns
Hamerton leverage check (subset regression)	Post-hoc reactive	§4.6.3, Appendix B.6	Added in response to v9/v10 reviewer concerns
Coupling-free reframing of the gradient	Post-hoc reactive	§4.1.1 leveler callout, Appendix B.7	Added in response to GPT-5.5 review

Item	Status	Where reported	Note
Cross-provider response generation (Tier 2)	Pre-registered control	§3.6, §3.5.1, §4.6.1	Sonnet 4.6 + Gemini 2.5 Pro on 3 subjects
GPT-5.4 battery regeneration (Control 1)	Pre-registered control	§3.5.1, §4.6.1	Battery generator circularity
Judge-panel composition (5-judge primary, 7-judge sensitivity)	Pre-registered control	§3.3.3, §4.6.2	Locked panel before scoring
Wrong-Spec derangement protocol sensitivity (v1 vs v2)	Reactive	§4.6.5	v2 is the standard randomization control; v1 is the adversarial stress test (headlined for stronger evidence)

Reproducibility scripts and raw data for each row are pointed to throughout §4 and consolidated in §8 Data, Code, and Reproducibility.

B.11 Per-system per-subject paired-delta distributions

The footnote in §4.4.3 collects the per-cell counts behind the three-pattern claim. This subsection consolidates those cells into a single table and a short reading note. The unit of observation is a single (subject, question) pair scored under both retrieval-only (C1) and retrieval + Behavioral Specification (C3); per-cell counts are restricted to questions with 5-judge primary coverage on both conditions. “Increases” / “decreases” use the $|\Delta| \geq 1.0$ threshold on the 5-point rubric (one full anchor crossing) so that small judge-noise jitter does not inflate the count in either direction.

Memory system	Subject	Aggregate Δ_{spec}	Increases ($\Delta \geq +1.0$)	Decreases ($\Delta \leq -1.0$)	Net at the per-question grain	Source
Supermemory	full 14-subject pool	≈ 0 (closest to zero)	57	53	110 of 546 paired questions cross by ≥ 1.0 (20.1%); the two roughly cancel at the mean	§4.4.3 lede + fn.
Mem0	Yung Wing	+0.33	21	10	31 of 39 paired questions cross by ≥ 1.0 ; 21 helps outnumber 10 hurts $2.1\times$	§4.4.3 fn.
Mem0	Keckley	-0.02	12	13	25 of 39 paired questions cross by ≥ 1.0 ; counts even at 12 / 13, aggregate near zero	§4.4.3 fn.
Letta archival	Hamerton	+0.42	19	7	26 of 39 paired questions cross by ≥ 1.0 ; 19 helps outnumber 7 hurts $2.7\times$	§4.4.3 fn.

Memory system	Subject	Aggregate Δ_{spec}	Increases ($\Delta \geq +1.0$)	Decreases ($\Delta \leq -1.0$)	Net at the per-question grain	Source
Zep	Seacole	+0.47	20	7	27 of 39 paired questions cross by ≥ 1.0 ; 0 questions show large regressions in this cell	§4.4.3 fn.
Base Layer	Yung Wing	+0.29	19	7	26 of 39 paired questions cross by ≥ 1.0 ; 19 helps outnumber 7 hurts $2.7\times$	§4.4.3 fn.

Reading. Every cell is a mixture. Strong-positive aggregates (Letta Hamerton, Zep Seacole, Base Layer Yung Wing) still contain 7 large regressions per cell. Near-zero aggregates (Mem0 Keckley) resolve into substantial counts in both directions rather than a flat per-question profile. The Supermemory pool is the cleanest read on the three-pattern mixture because the helps-versus-hurts counts are nearly balanced; the same shape reproduces across every other (system, subject) cell in the study.

The cells above are representative rather than exhaustive (one cell per memory system, two for Mem0 to capture both a positive-aggregate and a near-zero-aggregate subject). Full per-system per-subject paired-delta arrays are at `docs/research/per_system_anchor_crossing_20260427.json`; the recompute script is `scripts/_table_4_6_5judge_recompute.py`.

B.12 Wrong-Spec v1 fixed pairings

The v1 wrong-Spec control (§3.2, §4.3, §4.6.5) uses a fixed directed permutation locking each subject to a culturally and temporally distant subject’s Behavioral Specification. The permutation was locked in `scripts/run_global_rerun.py` (the `WRONG_SPEC_PAIRING` dict) before any C2c responses were generated and is not modified by analysis.

Subject	Assigned Spec (v1)
Augustine	Fukuzawa
Bābur	Keckley
Bernal Díaz	Sunity Deveen
Cellini	Zitkala-Ša
Ebers	Equiano
Equiano	Ebers
Fukuzawa	Augustine
Keckley	Bābur
Rousseau	Yung Wing
Seacole	Bernal Díaz
Sunity Deveen	Cellini
Yung Wing	Rousseau
Zitkala-Ša	Seacole

Two pairs are reciprocal (Ebers ↔ Equiano; Rousseau ↔ Yung Wing; Augustine ↔ Fukuzawa; Bābur ↔ Keckley). The remaining five entries (Bernal Díaz → Sunity Deveen → Cellini → Zitkala-Ša → Seacole → Bernal Díaz) form a directed cycle, not pairs. Hamerton’s v1 variant receives Franklin’s specification and is reported separately in §4.1.2.

B.13 Memory-system Wilcoxon results (C1 vs C3) and low-baseline Δ_{spec}

The §4.4 aggregate Wilcoxon tests on C3 (retrieval + Spec) vs C1 (retrieval only), on the 5-judge primary panel. All p-values are two-sided; W is the signed-rank statistic. Low-baseline slice (n = 9) median Δ and 95% CI alongside the all-14 panel. Data: `docs/research/stats_update.md` (full panel) and `docs/research/memory_systems_5judge_primary.md` (per-system Δ + improvement counts).

System	Config	N (all)	W	p (all-14)	Low-baseline Δ_{spec} mean	Low-baseline improved (of 9)
Zep	controlled	14	2.0	0.0004	+0.166	9 / 9
Letta	controlled	14	6.0	0.0017	+0.165	8 / 9
Mem0	native	14	8.0	0.0088	+0.320	7 / 9
Zep	native	14	2.0	0.0015	+0.303	9 / 9
Mem0	controlled	14	15.0	0.0166	+0.101	6 / 9
Letta	native	14	35.0	0.4629	−0.036	4 / 9
Supermemory	controlled	14	37.0	0.3575	−0.010	5 / 9
Supermemory	native	10	18.0	0.3750	−0.026	3 / 7
Base Layer	substrate (controlled)	14	27.0	0.1189	+0.083	6 / 9

Four (system, configuration) cells are significant at $\alpha = 0.01$ (bolded): Zep controlled, Letta controlled, Mem0 native, Zep native. Mem0 controlled is significant at $\alpha = 0.05$ but not $\alpha = 0.01$. The remaining cells (Letta native, both Supermemory configurations, Base Layer substrate) are not significant at $\alpha = 0.05$. Supermemory native has partial coverage (N = 10 of 14; 7 of 9 in the low-baseline slice).

B.14 Per-condition improvement and worsening rates (all-14, 546 questions)

The §4.2.1 per-question improvement rates for each condition against the No-Context Baseline (C5). “Improvement” means the response under the named condition has a higher 5-judge primary score than the same question’s C5 response; “worsening” means lower. Numbers verified against docs/research/v11_paper_numbers_verification_20260428.md.

Condition (vs C5)	Improvement	Worsening	Net (improve − worsen)
Spec only (C2a)	58.8%	26.7%	+32.1 pp
Facts only (C4)	60.1%	26.6%	+33.5 pp
Raw corpus (C8)	65.2%	23.6%	+41.6 pp
Facts + Spec (C4a)	65.8%	26.4%	+39.4 pp

Pairwise question-level comparison on the low-baseline 9-subject band (351 questions). The first entry on each row reads “first condition beats second”; ties are listed.

Comparison	First better	Second better	Tie
Raw corpus (C8) vs. Spec only (C2a)	53.3%	30.8%	56 questions
Corpus + Spec (C9) vs. Facts + Spec (C4a)	49.0%	36.5%	45 questions

The 7K-token facts + Spec package scores higher than the much larger corpus + Spec package on roughly one-third of low-baseline questions.

B.15 Memory-system abstention audit cell counts

The §4.6.7 audit on whether memory-system retrieval inflates refusal scores via visible fact recitation. Cells partition the response pool by retrieval condition and whether the response (a) is an abstention and (b) recites a retrieved n-gram. Data: docs/research/abstention_extensions_draft_20260429.md.

Cell	Definition	N	Mean	% ≥ 2.0
Pure No-Context Refusal	no facts, no retrieval	292	1.26	10.3%
Facts-only refusal	facts in context, no retrieval	20	1.33	10.0%
Memory- system refusal + recitation	refuses AND quotes retrieved n-gram	148	1.50	18.2%
Memory- system refusal, no recitation	refuses, does not quote retrieval	240	1.47	17.1%
Memory- system substantive engagement	non-refusal under retrieval	7,835	2.32	67.2%

Welch comparisons on the mean abstention score with 95% CI:

Comparison	Δ	95% CI	p (Welch)
Mem-refuse + recite vs. pure No-Context refuse	+0.234	[+0.113, +0.355]	0.0002
Mem-refuse no recite vs. pure No-Context refuse	+0.206	[+0.103, +0.310]	0.0001
Mem-refuse + recite vs. mem-refuse no recite	+0.027	[-0.098, +0.153]	0.67

Memory-system refusals score +0.21 to +0.23 anchor points above pure No-Context Refusals, but the lift is the same whether or not the response recites a retrieved n-gram ($\Delta +0.027$, $p = 0.67$). The over-credit is a “judges reward the retrieval condition” effect, not a “judges reward the visible quote” effect.

Appendix C. Conditions, Models, and Memory-System Configurations

C.1 Condition identifiers (summary card)

A consolidated lookup for the condition IDs used throughout §4. Defined in §3.2; summarized here.

ID	Family	Context served	Purpose
C5	Direct	None. Question only.	Pretraining-only floor. Baseline.
C2a	Direct	Behavioral Specification only.	Isolate Spec’s contribution.
C2c	Direct	A random other subject’s Spec (derangement, seed=42).	Wrong-Spec control.
C4	Direct	Full extracted fact set for subject.	Raw fact volume, no structure.
C4a	Direct	Full All Facts + Spec.	Spec added to raw facts.
C8	Direct	Full training corpus (half of source text).	Uncompressed source.
C9	Direct	Training corpus plus Spec.	Spec added to raw source. Bābur excluded (422K word overflow).
C1	Memory system	Top-k retrieval output from system.	Retrieval only, each of 5 systems.
C3	Memory system	Top-k retrieval output plus Spec.	Retrieval plus Spec, each of 5 systems.
C1_<system>_fullpipeline	Memory system, native	Retrieval from system-native ingestion of raw corpus.	Native ingestion variant, Retrieval Only.
C3_<system>_fullpipeline	Memory system, native	Native ingestion retrieval plus Spec.	Native ingestion variant, retrieval plus Spec.

The <system> slot ranges over {mem0, letta, supermemory, zep, baselayer}. Base Layer is run in a single controlled configuration; the four commercial systems are run in both controlled and native variants.

C.2 Shared response-model invocation

Every response call, across every direct-context and memory-system condition, uses the following parameters:

Parameter	Value
<code>temperature</code>	0
<code>max_tokens</code>	1024
System prompt	Framing instruction: predict how <code><subject></code> would respond; answer in subject’s voice, grounded in demonstrated patterns.
User prompt format	<code><context block>\n\nQuestion: <question text></code>
Context block	Condition-dependent. Empty in C5. Spec in C2a. Wrong Spec in C2c. Facts in C4. Facts plus Spec in C4a. Corpus in C8. Corpus plus Spec in C9. Retrieval output (optionally plus Spec) in C1 and C3.

No prompt instruction coaches the model to abstain, hedge, or commit. The model’s refusal-or-commitment pattern given a specific context is part of the phenomenon being measured (§3.6, §4.3).

C.3 Response models

Role	Model identifier	Provider	Scope
Primary response	<code>claude-haiku-4-5-20251001</code>	Anthropic	All 14 subjects, every condition. Main study.
Tier 2 response	<code>claude-sonnet-4-6</code>	Anthropic	3 subjects (Ebers, Yung Wing, Zitkala-Ša), C5 / C2a / C2c / C4a against GPT-5.4 batteries.
Tier 2 response	<code>gemini-2.5-pro</code>	Google	Same 3 subjects, same conditions as Sonnet Tier 2.

Source: `scripts/run_global_subjects.py`, `scripts/run_full_study.py`, `scripts/run_multimodel_response`

C.4 Pipeline models (specification generation)

Pipeline step	Model identifier	Temperature	Purpose
Extract (Step 2)	<code>claude-haiku-4-5-20251001</code>	0	AUDN fact extraction, 46-predicate constrained vocabulary.
Embed (Step 3)	<code>all-MiniLM-L6-v2</code> (local)	n/a	ChromaDB vector index (L2 distance).
Author (Step 4)	<code>claude-sonnet-4-6</code>	0	Three authored layers (anchors, core, predictions). Blind regen, domain guard.
Compose (Step 5)	<code>claude-opus-4-6</code>	0	Unified brief composition.
Battery generation	<code>claude-haiku-4-5-20251001</code>	0	Backward-design from held-out corpus.

Pipeline step	Model identifier	Temperature	Purpose
Battery generation (circularity control)	<code>gpt-5.4</code> (via OpenAI API)	0	Independent regeneration on 13 global subjects.

Source: `memory_system/src/baselayer/config.py`.

C.5 Judge panel

Judge	Model identifier	Provider	In 5-judge primary?	Calibration performed?
Haiku	<code>claude-haiku-4-5-20251001</code>	Anthropic	Yes	Yes
Sonnet	<code>claude-sonnet-4-6</code>	Anthropic	Yes	Yes
Opus	<code>claude-opus-4-6</code>	Anthropic	Yes	Yes
GPT-4o	<code>gpt-4o-2024-08-06</code>	OpenAI	Yes	Yes
GPT-5.4	<code>gpt-5.4</code>	OpenAI	Yes	Yes
Gemini Flash	<code>gemini-2.5-flash</code>	Google	No (sensitivity only)	Yes
Gemini Pro	<code>gemini-2.5-pro</code>	Google	No (sensitivity only)	Yes

Judges are invoked independently (no cross-judge conditioning). Each judge sees: held-out ground-truth passage, subject context (name, source), question, response. Judge temperature 0. Judge output is a numeric 1-5 score plus a free-text justification. Calibration diagnostic results in §3.3.3.

C.6 Memory-system ingestion and retrieval parameters

Controlled configuration (C1 / C3) holds the input identical across systems: each system receives the same extracted fact set used by the Base Layer pipeline, re-ingested through its own API. Native configuration (`_fullpipeline`) has each system ingest the raw training corpus directly via its own chunking and extraction.

System	Ingestion endpoint	Ingestion unit (controlled)	Ingestion unit (native)	Retrieval top-k	Notable configuration
Mem0	POST /v1/memories/	One fact per POST	Raw corpus chunks (Mem0 chunker)	10	<code>infer=False</code> on controlled (store as-is, no reformulation). Failure mode: Mem0 may reformulate on <code>infer=True</code> , flagged in <code>docs/PROVIDER_ISSUES.md</code> .
Letta (archival)	POST /v1/agents/<id>/core_memory/archival	One fact per passage	Letta native chunking	10	1 fact = 1 passage. Batch ingestion tested 135x faster but changes chunking behavior (see <code>run_memory_system.py</code> line 456-458).
Letta (stateful)	Agent state edit during ingestion	One fact per edit cycle	Raw corpus	n/a (read from block)	Evaluated as a separate path in §4.5, not as a row in the C1 / C3 conditions.
Supermemory	POST /v3/memories	One fact per memory, <code>containerTags=<subject></code>	Raw corpus	10	<code>limit=10</code> on retrieval.
Zep	Graph ingestion via <code>zep_client.graph.add</code>	One fact per edge	Raw corpus	10	Retrieval via <code>client.graph.search(user_id, query, limit=10)</code> .
Base Layer	Direct into ChromaDB	One fact per vector	n/a (Base Layer has no native variant)	10	MiniLM embeddings, L2 distance, cosine-like similarity via $1 - \text{dist}^2/2$.

All five systems use the same top-k of 10. All five are queried with the question text as the query. All five feed their retrieval output into the standard prompt schema (§C.2) as the context block.

C.7 Ingestion exclusions and failure cases

Subject / system	Issue	Resolution
Bābur, C9 (raw corpus plus Spec)	422,772-word source exceeds Haiku context window.	Excluded from C9. 13 of 14 subjects report C9 numbers.
Letta native (all subjects)	Ingestion ceiling on archival passages; retrieval produces 0.34-0.47 dedup ratio, meaning a top-10 list often contains 3-5 unique facts.	Reported as-is in §4.4. Not excluded.
Mem0 native	Mem0’s <code>infer=True</code> reformulated facts during native ingestion pilot.	Used <code>infer=False</code> on controlled configuration to hold input identical. Native variant retains <code>infer=True</code> (the realistic deployment path).
Zep graph bias	Zep graph retrieval surfaces entity-dense chunks over behavior-dense chunks.	Reported as-is. See <code>docs/PROVIDER_ISSUES.md</code> .

C.8 Analysis plan lock

The condition matrix was frozen in `docs/ANALYSIS_PLAN_LOCK.md` before scoring. Any condition added after the lock is reported separately (for example, the Tier 2 3-subject replication and the v2 random-derangement wrong-Spec draws).

Appendix D. Validity Audit and Score Distributions

D.1 Per-subject 5-judge primary aggregate (main gradient)

This table reproduces the §4.1 cross-subject gradient for reference. Every number is the 5-judge primary mean (Haiku, Sonnet, Opus, GPT-4o, GPT-5.4) over the 39-question behavioral-prediction battery per subject (40 for Franklin).

Subject	Baseline (C5)	Spec only (C2a)	Facts + Spec (C4a)	Δ_{spec}	$\Delta_{\text{All Facts + Spec}}$	Anchor crossed
Ebers	1.02	1.54	2.07	+0.52	+1.05	yes
Sunity	1.03	2.27	2.41	+1.24	+1.38	yes
Devee						
Hamerton	1.26	2.63	2.77	+1.37	+1.51	yes
Fukuzawa	1.67	2.35	2.78	+0.68	+1.11	yes
Bernal	1.70	2.27	2.48	+0.57	+0.78	partial
Díaz						

Subject	Baseline (C5)	Spec only (C2a)	Facts + Spec (C4a)	Δ_{spec}	$\Delta_{\text{All Facts}} + \text{Spec}$	Anchor crossed
Bābur	1.76	1.91	2.01	+0.15	+0.25	no
Seacole	1.77	2.48	2.59	+0.71	+0.82	yes
Keckley	1.84	2.43	2.44	+0.58	+0.59	no
Yung	1.88	2.22	2.40	+0.34	+0.52	no
Wing						
Zitkala-Ša	2.34	2.03	2.02	−0.31	−0.32	no
Cellini	2.38	2.54	2.53	+0.16	+0.15	no
Rousseau	2.44	2.81	2.53	+0.37	+0.10	no
Augustine	2.58	2.48	2.70	−0.11	+0.11	no
Equiano	2.77	2.46	2.42	−0.31	−0.35	no
Franklin	3.77	3.37	3.65	−0.40	−0.13	no
(control)						

Raw per-judge files: `results/global_<subject>/judgments_v2.json` and `*_judgments_<judge>.json` (per-judge) for the 13 globals. Hamerton: `results/hamerton/*_judgments_<judge>.json`. Franklin: `results/franklin_legacy_20260411/analysis/*_judgments.json`.

D.2 Per-subject anchor-crossing on the low-baseline band

Anchor-crossing rate is the fraction of per-question paired (C5, C4a) responses where the C4a 5-judge primary mean lands in a different integer rubric anchor than the C5 mean. Definition in §3.3.1 and `scripts/compute_anchor_crossing.py`.

Slice-level:

- Total low-baseline questions (9 subjects, 39 Q each): 351
- Upward crossings: 193 (55.0%)
- Downward crossings: 24 (6.8%)
- Stayed in anchor: 134 (38.2%)

Per-subject breakdown (5-judge primary, paired C5 vs. C4a over N=39 per subject):

Subject	Upward	Upward %	Downward	No crossing
Sunity Deveen	29	74.4%	0	10
Hamerton	27	69.2%	0	12
Fukuzawa	26	66.7%	3	10
Bernal Díaz	23	59.0%	3	13
Seacole	21	53.8%	3	15
Ebers	19	48.7%	0	20
Keckley	19	48.7%	6	14
Yung Wing	19	48.7%	5	15
Bābur	10	25.6%	4	25
Slice total	193	55.0%	24	134

Eight of the nine low-baseline subjects cluster in the 48-74% upward range. Bābur is the low-baseline outlier (source corpus 422K words, partial pretraining exposure); he is the only subject whose upward-crossing rate falls below 48%, and his downward-crossing count (4 of 39) is mid-range. Sunity Devee’s 74.4% upward rate is consistent with her unusually low C5 baseline of 1.03 noted in §4.1. Per-subject downward-crossing rates stay at or below 15% for every low-baseline subject. Source: `scripts/compute_anchor_crossing.py` executed against `results/global_<subject>/judgments_v2.json` and `results/hamerton/`.

D.3 Rubric-handling validity audit (redirected)

The full report on rubric-handling limitations (refusal-anchor ambiguity, length-correlation by condition, per-judge strictness, per-response-model abstention behavior, memory-system effect on abstention) is in the public repository at `docs/supplementary/appendix_D_3_rubric_handling_validity_audit.md`.

Headline results already reported in §4.6.7: Across 192 responses identified as refusals in the low-baseline band, 82.8% scored in the 1.0–1.5 anchor range as expected, but 9.4% scored at or above 2.0 and 3.1% at or above 3.0. Mean abstention score: 1.27. The §4.6.7 caveat applies: per-response-model abstention behavior diverges (Sonnet over-credits abstention at roughly twice Haiku’s rate); memory-system condition effects on the score-1 composition are decomposed in §4.4. Audit script: `scripts/audit_low_end_inflation.py`. Raw audit output: `docs/research/rubric_handling_validity_full.json`.

D.4 Per-judge score matrices (redirected)

The full per-subject by per-judge by per-condition mean-score matrix for all 14 main-study subjects across the 5 gradient conditions (C5, C2a, C2c, C4, C4a) lives in the public repository at `docs/supplementary/appendix_D_4_per_judge_matrices.md`. Per-subject raw per-judge data is at `results/global_<subject>/*_judgments_<judge>.json` (and `results/hamerton/` for Hamerton).

Headline results from the matrices already reported in the paper: directional agreement is tight (Spearman ρ 0.86 to 0.93 across the 5-judge primary panel; §3.3.4) while absolute magnitude varies by judge (Krippendorff α 0.659 on the 5-judge panel, 0.535 with Gemini included; §3.3.3 and §4.6.2). The Gemini judges show a systematic +1-point inflation relative to the calibrated five primary judges, decomposed in §3.3.3.

D.5 Example verbatim responses at each rubric anchor

Verbatim-response examples at rubric anchors 1-5 for one representative subject (Hamerton) are in §3.3 as part of the rubric definition. Examples at anchor crossings are developed in §4.1 (Examples A, B, C on Ebers, Bernal Díaz, Seacole). Three illustrative paired (C5, C4a) per-question excerpts for each of the 14 main-study subjects are collected in Appendix E. The raw response JSON files (`results/global_<subject>/results_v2.json`) contain every response verbatim alongside its 5-judge primary score.

D.6 Reasoning trace worked example: Sunity Devee Q4

The conceptual four-step reasoning trace described in §2.3 is walked here on a single battery item. The example shows how a response in the C2a condition (Specification only: the model receives the Behavioral Specification and the question, with no retrieved facts and no source corpus) decomposes

into pattern statements, the facts that imply them, and the source passages those facts were extracted from.

Subject: Sunity Deveen

Question: *“When her husband expresses disappointment about missing an opportunity due to her concerns, does the narrator typically reconsider her position?”*

Held-out ground truth (full passage): *“The Maharajah was disappointed, as he was anxious to go and see the old castle, and to shoot, and I deprived him of both these pleasures.”*

Response (C2a, excerpt): *“Based on the behavioral specification, the answer is no, not typically, and not in the way the question assumes. A2 (Spiritual Integrity Over Social Cost) and A5 (Relational Identity) create the relevant dynamic. A5 activates first: relational identity means his disappointment matters deeply. A2 holds firm: arguments from social consequence, including spousal disappointment, will not register as reasons on their own. Rather than reconsider the position itself, she is more likely to absorb the tension without resolving it (P3 — Tension Absorbed, Not Expressed) and reframe the disappointment within a spiritual logic (A1 — Divine Primacy), while grieving the relational cost without reversing the decision.”*

Step 3 — pattern statements invoked by the response:

1. **A5 (Relational Identity)** registers his disappointment as real, not dismissible.
2. **A2 (Spiritual Integrity Over Social Cost)** refuses social consequence as a reason to reconsider.
3. **P3 (Tension Absorbed, Not Expressed)** holds the conflict without resolving it in either direction.
4. **A1 (Divine Primacy)** reframes the disappointment within a spiritual logic.

Step 4 — facts grounding the pattern statements:

- **F-73:** *“Sunity Deveen’s mother would never countenance anything her conscience told her was wrong.”* (grounds A2)
- **F-414:** *“Sunity Deveen’s father believed he acted as a public man guided by conscience and divine duty in accepting the marriage proposal.”* (corroborates A2 from a different relational direction; conscience-as-master-frame pattern reinforced across both parents)
- Additional facts grounding A1, A5, and P3 are referenced in the specification’s anchor and prediction files at `data/global_subjects/sunity_deveen/spec/anchors.md` and `data/global_subjects/sunity_deveen/spec/predictions.md`; per-fact source-passage excerpts are in the same subject’s `facts.json`.

The user can audit any step: read the response, look up each cited pattern statement by name, look up the facts that ground it, and read the source passages those facts came from. If a fact misrepresents its source, correcting the fact propagates through the Specification on recomposition.

Appendix E. Per-subject worked examples

The 14 per-subject worked examples (one per main-study subject) are in `docs/supplementary/appendix_E_per_sub` in the public repository. Each subject has three illustrative paired (C5, C4a) per-question response excerpts under the 5-judge primary score, selected by the largest C4a minus C5 panel-mean Δ within each subject. The excerpts are organized to illustrate the cross-anchor interpretation rule (§3.3.1) at the subject level and to make concrete the multi-anchor crossings discussed in §4.1.1. Per-subject baselines (C5) range from 1.02 (Ebers) to 2.77 (Equiano); per-question Δ_{C4a} values in the supplementary file range from +0.40 to +4.00. Full response artifacts are at `results/<subject>/results_v2.json` and `results/hamerton/results.json`.

Appendix F. Benchmark Scope Analysis

This appendix develops, benchmark by benchmark, the scope differences summarized in §2.1 between prior work on memory and personalization benchmarks and what this paper measures. The point in each case is the same: representational accuracy, operationalized as behavioral prediction on held-out reasoning situations, is not what these benchmarks evaluate. None of them is wrong on its own axis. None of them is a substitute for the test in this paper.

F.1 LongMemEval

Reference. Wu et al., ICLR 2025 (Wu et al., 2025).

Task. Evaluate long-term memory in chat assistants across multiple sessions. Five capability dimensions: single-session memory, multi-session reasoning, temporal reasoning, knowledge updates, and abstention.

Scoring. Question-answering accuracy, with held-out facts embedded across session history and queried in a later session. Answers are compared against ground-truth factual targets drawn from the same session history the system ingested.

Training / test protocol. Conversation history is ingested; the system is then queried with fact-recall questions whose answers are present in the ingested history. The test is whether the memory system can surface the correct facts at retrieval time.

What it measures. Fact recall across long context windows. A secondary axis tests whether the system correctly abstains when the answer is not in the conversation history.

What it does not measure. Whether the memory system’s representation of a specific person captures how that person reasons. Every LongMemEval target is a fact that was literally said in the conversation; no target is a held-out behavioral pattern.

Published range. Memory systems reported in the 68% to 85% range depending on provider, model, and benchmark variant (cited in §1.1 and §2.2). Specific numbers per system are in the papers and vendor reports.

Relationship to this paper’s test. Orthogonal axis. Our battery targets held-out behavioral patterns that were never literally said in the training half of the corpus; every question is backward-designed to answer only from patterns, not from retrievable content. A system that ranks at the top of LongMemEval can still sit near the rubric floor on our battery, and a system that ranks low on

LongMemEval (for example, Base Layer’s retrieval substrate) can contribute on our battery through the Specification rather than through retrieval.

F.2 PersonaGym

Reference. Samuel et al., Findings of EMNLP 2025 (Samuel et al., 2025).

Task. Evaluate persona fidelity in conversational agents. Given a described persona, measure whether the model maintains that persona across a conversation.

Scoring. Persona-consistency metrics over multi-turn conversation. LLM-judge evaluation of whether the model’s voice, stated preferences, and surface-level claims remain consistent with the described persona.

Training / test protocol. A persona is described (occupation, background, preferences, mannerisms). The model is prompted to roleplay the persona across a dialogue. Evaluation is whether the dialogue responses remain internally consistent with the persona description.

What it measures. Persona presentation fidelity. Can the model stay in character on the described dimensions.

What it does not measure. Whether the model accurately predicts how the person described by the persona would respond to new situations. A persona-fidelity system can maintain voice without ever accurately anticipating decisions. A representationally accurate system can shift voice (for example, from formal prose to casual conversational register) while continuing to predict accurately on behavioral questions.

Published best-number. Top PersonaScore of 4.51 ± 0.08 on a 1-5 scale (GPT-4.5), out of 10 evaluated LLMs spanning 200 personas and 10,000 questions; bottom of the range was 3.64 ± 0.57 (Claude 3 Haiku). Notably, GPT-4.1 and LLaMA-3-8b tied on PersonaScore despite a large capability gap (Samuel et al., 2025). Scoring is on persona-consistency metrics, not held-out behavioral prediction; these numbers are not directly comparable to this paper’s rubric means on the 1-5 behavioral-prediction scale.

Relationship to this paper’s test. Both measure something that is sometimes called “personalization,” but the axes are different. PersonaGym is surface-presentation consistency; our battery is transfer of the subject’s interpretive patterns to unseen situations. Our rubric does not credit voice-matching alone; it requires predicting the held-out behavioral content. A response that maintains voice without predicting the held-out pattern scores 2 (“wrong prediction”) under our rubric.

F.3 AlpsBench

Reference. Xiao et al., 2026 (Xiao et al., 2026).

Task. Evaluate whether explicit memory mechanisms improve preference-aligned and emotionally resonant responses in conversational settings.

Scoring. Preference-alignment scoring and emotional-resonance scoring on conversational responses, both LLM-judged against reference targets derived from user preference data.

Training / test protocol. A conversational agent is seeded with a user’s preference history (via an explicit memory mechanism, or as a baseline without one). The agent responds to new prompts.

Responses are scored on preference-alignment and emotional-resonance metrics.

What it measures. Whether explicit memory makes conversational responses more aligned with stated preferences and more emotionally appropriate.

What it does not measure. Whether memory mechanisms enable the model to predict the user’s behavior in unseen reasoning situations. Preference alignment and behavioral prediction are related but distinct: a system can match preferences on immediate response choices without having a representation of the user’s reasoning that transfers to situations the system has never seen.

Central finding. AlpsBench’s central empirical result is that explicit memory mechanisms improve recall but do not guarantee more preference-aligned or emotionally resonant responses. This is independently arrived at and complementary to our own finding. They find the gap in preference alignment; we find it in behavioral prediction. Both point in the same direction: recall-solving is insufficient for what memory is ultimately for.

Relationship to this paper’s test. Adjacent. Same motivating intuition (recall improvement does not transfer to downstream behavioral properties), different downstream property measured.

F.4 Twin-2K

Reference. Toubia et al., 2025 (Toubia et al., 2025).

Task. Behavioral prediction at scale. 2,058 participants each answered a large-scale survey, and the system predicts each participant’s responses on held-out survey items given a persona constructed from their other survey answers.

Scoring. Distance metric on Likert-scale items (numeric distance between predicted response and actual response, aggregated per participant and per item).

Training / test protocol. For each participant, a subset of survey answers is used to author a persona. The persona is served to a model as context. The model predicts the participant’s answer on the held-out survey items. Distance between predicted and actual response is the score.

What it measures. Behavioral prediction on survey-response interpolation. Does a machine-readable transcript of one half of a participant’s survey predict the other half.

What it does not measure. Behavioral prediction on open-ended reasoning situations. Twin-2K’s held-out items are additional Likert responses from the same survey form; the test is interpolation across a structured response distribution. Our held-out items are open-ended behavioral predictions on unseen autobiographical passages; scoring is via rubric on response content, not distance on a numeric scale.

Relationship to this paper’s test. Closest prior work on the behavioral-prediction axis. Three structural differences remain:

1. **Task format.** Twin-2K: Likert interpolation. This paper: open-ended behavioral prediction with free-text answers.
2. **Persona construction.** Twin-2K: machine-readable transcript of the participant’s own prior survey responses, served as-is. This paper: an authored Behavioral Specification composed of three interpretive layers plus a composed brief. The Twin-2K persona is raw data; our Specification is compressed interpretation.

3. **Held-out distance.** Twin-2K: the held-out items are drawn from the same structured survey instrument. This paper: the held-out items are drawn from autobiographical text the representation has never seen, in a different form than the training half (different chapters, different situations).

Twin-2K measures whether a model can interpolate a person’s survey distribution from other survey responses. Our battery measures whether a representation of how a person reasons transfers to new situations the representation has never seen. Both are legitimate tests of behavioral prediction; neither is a substitute for the other.

An earlier exploratory Base Layer run against Twin-2K’s battery produced positive results on that task format, but we do not report those numbers as a formal benchmark comparison because the experiment used a prior iteration of our pipeline, and the task targets are substantively different (see §2.1).

Published best-number. Top individual-level accuracy of 71.72% on held-out survey items using a text-persona representation served to GPT-4.1-mini (Toubia et al., 2025). Human test-retest reliability on the same instrument was 81.72%, putting the top twin at 87.67% of the human ceiling. Random-guess baseline was 59.17%. Aggregate-level replication: the Twin-2K twins reproduced results from 6 of 10 behavioral-economics experiments, with systematic divergences on medical decision-making and political attitudes. The 71.72% accuracy is on Likert interpolation, which is a structurally different task from our rubric-scored free-text behavioral prediction.

F.5 LoCoMo

Reference. Maharana et al., ACL 2024 (Maharana et al., 2024).

Task. Conversational memory quality over long multi-session dialogues.

Scoring. Fact-recall questions over ingested dialogue history. Similar scope to LongMemEval but focused on conversational-memory substrates specifically.

Training / test protocol. A long multi-session conversation is ingested; the memory system is queried on specific facts from earlier sessions.

What it measures. Long-dialogue recall accuracy.

What it does not measure. Behavioral reasoning. LoCoMo targets are literal recalls from session history.

Published range. LoCoMo paper baselines (Maharana et al., 2024): GPT-4-turbo 32.1% overall, GPT-3.5-turbo 22.4%, GPT-3.5-turbo-16K 37.8%, best RAG configuration 41.4%; human performance 87.9%. Memory-system claims on LoCoMo, detailed in §2.2: Mem0g variant 68.44 with GPT-4o-mini (Chhikara et al., 2025); Mem0 production algorithm 91.6 self-reported with open-sourced evaluation harness; Letta 74.0 with GPT-4o-mini; earlier Zep claim of 84 publicly disputed by Mem0 (see §2.2 dispute note). The methodology disagreement between vendors remains unresolved; §2.2 treats these single-number comparisons with explicit caution.

Relationship to this paper’s test. The benchmark the four memory systems (Zep, Letta, Mem0, Supermemory) compete on. §2.2 uses these results as context for the memory-system landscape. Our paper is orthogonal: we do not evaluate memory systems on LoCoMo; we evaluate their behavioral-prediction performance on held-out autobiographical passages with and without the Behavioral Specification added.

F.6 MemOS and related systems-level benchmarks

Reference. Systems-level memory benchmarking literature, including MemOS and adjacent evaluations. See §2.2 for the memory-systems landscape.

Task. Evaluate memory-layer infrastructure choices (storage substrate, retrieval algorithm, consistency properties) rather than memory-quality outcomes.

Scoring. Varies. Typically: retrieval latency, throughput, consistency guarantees, scalability benchmarks.

What it measures. Infrastructure properties.

What it does not measure. Representational accuracy, persona fidelity, or preference alignment. Systems-level benchmarks do not evaluate the quality of the representation the memory layer produces; they evaluate the mechanics of how that representation is stored and served.

Relationship to this paper’s test. Different layer of the stack. Our paper evaluates what gets stored and why; systems-level benchmarks evaluate how well it is stored and served. Both matter for deployed personal-AI systems. The Specification and the memory-layer infrastructure compose: our §4.4 Mem0 / Letta / Zep / Supermemory / Base Layer results show the Specification adding on top of each infrastructure choice, not replacing it.

F.7 What no prior benchmark measures

Pulling the per-benchmark analysis together, the axis that representational accuracy sits on is not covered by any prior benchmark:

1. **Test data the system has not seen.** LongMemEval, PersonaGym, and LoCoMo target content the system has ingested. Twin-2K’s held-out items are drawn from the same structured instrument. Our battery’s held-out passages are from unseen chapters in a different narrative register than the training half.
2. **Open-ended behavioral prediction rather than structured-format scoring.** Twin-2K is the closest comparison; it is Likert-format rather than open-ended.
3. **Representation of how a person reasons, not what they said or prefer.** PersonaGym tests voice consistency; AlpsBench tests preference alignment; LongMemEval / LoCoMo tests fact recall. None tests transfer of interpretive patterns.

This is the gap the paper’s battery targets. The battery is not a replacement for any of the above. It is a test of a different property: whether a representation of a specific person enables a model that has never seen the person’s held-out reasoning to anticipate it accurately.

F.8 Persona-input depth comparison across benchmarks

The §2.1 footnote [ˆtwin2k-persona-size] calls out that Twin-2K’s persona input is much deeper than PersonaGym’s one-line descriptor. This subsection collects persona-input depth across the benchmarks named in §2.1 and Appendix F so the comparison is concrete. “Persona-input depth” is the total token volume of the participant- or subject-specific representation served to the model at inference time, measured on the input that an evaluated system actually consumes.

Benchmark	Persona-input form	Approximate input depth (tokens)	Notes
LongMemEval (Wu et al., ICLR 2025)	Multi-session conversation transcript ingested before query	varies, $\sim$ thousands per session $\times$ 5 sessions	The “persona” is the user’s accumulated conversational history. Depth depends on configured session count; the standard <code>_s</code> configuration runs $\sim$ 5 ingested sessions before query.
LoCoMo (Maharana et al., ACL 2024)	Long multi-session dialogue ingested before query	varies, $\sim$ 10K-100K depending on session count	Similar to LongMemEval in form: ingested dialogue is the “persona.” Single-session and multi-session test variants.
PersonaGym (Samuel et al., Findings of EMNLP 2025)	One-line descriptor (e.g., “ <i>You are a 45-year-old skeptical accountant from Toronto</i> ”)	$\approx$ 20–50	Surface attributes only; no behavioral depth. The benchmark is designed to test consistency with a thin descriptor across multi-turn conversation.
AlpsBench (Xiao et al., 2026)	User profile of preferences and prior emotional-context exchanges	varies, $\sim$ hundreds to low thousands	Profile is a concatenation of stated preferences and selected prior dialogue snippets. Designed to test preference-aligned response generation rather than reasoning transfer.

Benchmark	Persona-input form	Approximate input depth (tokens)	Notes
Twin-2K (Toubia et al., 2025)	Full survey-response transcript or condensed summary	<code>persona_text</code> $\approx$ 32,000; <code>persona_summary</code> $\approx$ 3,750	Two persona-input variants are reported in the released code. <code>persona_text</code> is the participant’s full prior-question-and-answer transcript; <code>persona_summary</code> is a model-condensed form.
Base Layer (this paper)	Behavioral Specification (anchors / core / predictions composed into a unified brief)	$\approx$ 7,000	Compressed-interpretation form. Per-subject specification sizes range from $\approx$ 4,000 (Hamerton, smallest corpus) to $\approx$ 9,000 (longest-corpus subjects). Compression ratios vs. source corpus span $5\times$ to $80\times$ (§4.2).

Reading. The benchmarks span roughly three orders of magnitude in persona-input depth, from PersonaGym’s $\sim$ 50 tokens to Twin-2K `persona_text`’s $\sim$ 32,000. Two structural points fall out of the comparison.

First, persona-input depth alone does not predict what a benchmark measures. Twin-2K’s full transcript is far deeper than this paper’s specification, but the task is Likert interpolation across the same survey instrument; what is being measured is whether a model can complete a participant’s response distribution, not whether it captures their interpretive patterns on situations the representation has never seen.

Second, the form of the persona input matters more than the size. PersonaGym’s one-line descriptor and Twin-2K’s full transcript are at opposite ends of the depth scale, and both are “raw” inputs in different senses (a descriptor of surface attributes versus a serialized record of prior responses). The Behavioral Specification at $\sim$ 7,000 tokens sits in the middle of the depth range but is structurally distinct: it is an authored compression of how the subject reasons, not a transcript or a descriptor. The compression-curve evidence in §4.2 shows that this structural distinction is what makes the form behave the way it does in cross-format behavioral prediction.

Sources: footnote-cited papers and the released code or schema where applicable. PersonaGym descriptor length is from the canonical example in Samuel et al. §3.1; AlpsBench profile composition is from Xiao et al. §3.2; Twin-2K depth values are from `persona_text` and `persona_summary` columns in the released dataset; Base Layer specification depth is per-subject

from data/global_subjects/<subject>/spec/brief.md.

Appendix G. Letta Stateful-Agent: Exploratory Case Study (full)

This appendix holds the full case study summarized in §4.5. N=3 subjects (Hamerton, Ebers, Bābur), one Letta version, one response model (Claude Haiku 4.5), 40 questions per subject. We do not treat the result as a replication or a headline finding.

G.1 Architectural difference: Letta memory block vs. Behavioral Specification

The two systems produce structurally different artifacts from the same source corpus.

Letta’s memory block is text the agent has written and rewritten during ingestion: a mix of verbatim source sentences, paraphrased restatements, and short synthesis notes the agent generated as it processed each turn. The block grows with the corpus and is rewritten in place when it approaches an ingestion ceiling, which we observed near 333K characters in our runs. Content shape is whatever the agent decides during ingestion; size scales with corpus length up to that limit.

A Behavioral Specification is a fixed-shape document produced by the five-step pipeline detailed in §3.7: import the corpus, extract structured predicates over a 46-predicate controlled vocabulary, embed each predicate for provenance tracing, author three layers, and compose into a unified brief. The extracted predicates are constrained subject-predicate-object triples (for example, *values X, avoids Y, decides based on Z*). The three layers each play a different role. **anchors** hold 8–12 axiomatic statements per subject distilled from the predicates (e.g., A1 Intimate Authority, A2 Documented Dignity in §4.4.4). **Core** holds the cognitive patterns and value tensions that govern reasoning under the anchors. **Predictions** hold IF/THEN templates that connect the patterns to specific behavioral situations. The pipeline produces a deterministic full-stack Behavioral Specification of approximately 7,000 tokens (~37,000 characters) regardless of source corpus length.

Trade-off. Letta retains source-text texture (voice, vocabulary, syntax) at the cost of compression and mixes source material with agent-generated synthesis. A Behavioral Specification compresses aggressively into structured predicates at the cost of source-text texture, but maintains a deterministic schema the response model can read consistently across subjects.

G.2 Headline result (5-judge primary)

Subject	Training corpus (words)	Letta score	Spec score (full- stack)	Δ (Letta – Spec)	Letta block (chars)	Spec (chars)	Letta : Spec size
Hamerton	25K	3.10	2.83	+0.27	22.5K	34.6K	0.7×
Ebers	48K	2.76	1.56	+1.21	68.4K	39.7K	1.7×
Bābur	223K	2.42	2.04	+0.38	335.3K	37.1K	9.0×

The Spec score column reports Base Layer’s full-stack Behavioral Specification (anchors + core + predictions + brief), the same artifact class used in the main-study gradient; per-subject sizes 34.6K / 39.7K / 37.1K characters. Raw data at docs/research/_letta_rerun/fullstack_named/.

Hamerton’s score is the full-stack specification scored with the consistent short-form judge prompt; Ebers and Bābur are full-stack regenerations on the Letta battery.

Letta’s self-edited memory block scores higher than Base Layer’s full-stack Behavioral Specification on all 3 subjects, with the gap largest at the mid-corpus subject (Ebers, $\Delta +1.21$) and smaller at both endpoints. With three data points we cannot distinguish among the possible interpretations: a corpus-size band where the self-edited block is most effective, degradation as the block grows beyond an architectural sweet spot, or insufficient interpretive content when the corpus is small. Across all three subjects, both representations land well above the retrieval-only baseline at matched response model.

G.3 The block grows with the corpus, then breaks down

Letta’s block scales with corpus length: 22.5K characters at Hamerton (smaller than the Spec’s 37K), 68.4K at Ebers, 335.3K at Bābur (about $9\times$ the Spec). The Bābur block hits Letta’s ingestion ceiling near 333K characters, and the agent starts rewriting content it has already written rather than compressing it. The result is heavy duplication at the ceiling: 25.4% of Bābur’s sentences are verbatim repeats, and 56.1% have a near-paraphrase elsewhere in the block. Hamerton and Ebers show 0% and minor duplication respectively; the architectural ceiling produces both exact and near-paraphrase repeats only when the corpus is large enough to push against it.¹⁰⁶

G.4 What explains Letta’s lift

Two things, neither of which is surface-text leakage. The leakage hypothesis was tested directly: the held-out battery was authored from the same source autobiographies that Letta ingested, so verbatim phrases could in principle leak from block to response. They do not. Five-word verbatim overlap with the held-out passage is 0% on every one of 119 questions for both Letta and Spec. The score-delta correlation with surface-text similarity is essentially zero (Pearson -0.046).¹⁰⁷ What does explain Letta’s lift is that its block keeps the corpus’s named entities (people, places, institutions) that the Spec compresses away by design, and Letta speaks in committed predictions where the Spec hedges when its behavioral scaffold cannot reach a specific question. When Letta’s committed answer matches the held-out, it scores high; when it doesn’t, the Spec’s hedge at least lands closer to the right direction. Ebers q30 is the canonical entity-grounding case: the held-out turns on Frederick William IV at the Keilhau Institute; Letta names him and scores 4.0; the Spec reaches for behavioral anchors that are structurally adjacent but referentially different and scores 1.8.

G.5 Pattern 4: when confidence backfires

The same mechanism that lifts Letta’s score on entity-rich questions hurts it on principle-driven questions. Where the held-out resolution turns on an underlying principle the entity-rich text does not surface, Letta commits confidently to the wrong direction; the Spec’s hedge keeps the answer closer to the held-out. Three Hamerton decision questions illustrate the inversion at the cleanest scale:

¹⁰⁶Semantic-similarity duplication via `scripts/analyze_letta_semantic_duplication.py` (MiniLM-L6-v2 sentence embeddings). At Bābur: 56.1% of sentences have a near-paraphrase at cosine ≥ 0.85 ; 35.2% at ≥ 0.95 . Ebers: 3.3% at ≥ 0.85 , 0.5% at ≥ 0.95 . Hamerton: 0% at any threshold above 0.80. Full per-threshold data at `docs/research/letta_semantic_duplication_20260501.json`.

¹⁰⁷3-gram overlap is tiny on both sides (Letta 0.05–0.08%, Spec 0.02–0.06%), dominated by proper-name sequences. Pre-registered strict and relaxed leakage signatures: 0 of 119 questions. Full analysis at `docs/research/letta_vs_spec_leakage_analysis_20260507.md`.

q27 (self-publish poetry, Spec +2.4), q29 (Lancashire exhibition, Spec +2.0), q51 (distant school vs. dying guardian, Spec +3.0). Pattern 4 is the Letta-architecture analog of §4.4.3 Pattern 3 (Spec-induced refusal): each architecture has a structural failure mode born of its representational choice. Letta’s commit-without-value-anchor produces confident inversions on principle-driven questions; the Spec’s value-anchor-without-context produces principled refusals the rubric penalizes. Letta also abstains more often than the Spec at the two larger subjects (Bābur 17.9% vs 0%; Ebers 10.3% vs 5.1%) and less at the small-corpus Hamerton (7.7% vs 10.3%); the Bābur spike co-occurs with the duplication ceiling, suggesting the agent recognizes block degradation and refuses to over-extend. Whether Letta’s content-confidence advantage comes from the corpus content the block retained (real interpretive work) or from the agent’s general tendency to commit regardless of grounding is the open question; a named-entity-vs-axiom-grounding ablation is flagged in §7.5. Per-subject abstention decomposition at docs/reviews/letta_vs_spec_abstention_20260507.md.

G.6 Caveats and naming asymmetry

The §4.5 / Appendix G main result table compares Letta’s named, self-edited block against Base Layer’s full-stack Behavioral Specification (anchors + core + predictions + brief), the same artifact class used in the main-study gradient. Letta ingests the named corpus; Base Layer’s pipeline anonymizes the corpus during extraction by design, so the named-vs-anonymized comparison is asymmetric on referential vocabulary. A named-entity-controlled Base Layer variant is flagged as future work in §7.5. Multi-subject replication across the full 14-subject gradient is the highest-priority external falsification of the Letta architectural-ceiling finding.

G.7 Exploratory: stacking the Letta block with the Behavioral Specification

The §4.5 comparison treats Letta’s self-edited block and the Behavioral Specification as alternatives. A follow-up probe served both at once, concatenated into a single Haiku 4.5 system prompt, on the same three subjects and the same Letta battery. The question is whether the two interpretive representations conflict or compose.

Subject	Stacked (block + Spec)	Letta block alone	BL full-stack Spec alone
Hamerton	3.17	3.10	2.83
Ebers	2.84	2.76	1.56
Bābur	2.52	2.42	2.04

Stacking lands above both representations alone on all three subjects: it beats the Letta block by +0.07 / +0.08 / +0.10 and the full-stack Specification by +0.34 / +1.28 / +0.48. The longer combined context does not dilute either signal, including on Bābur, whose Letta block alone is already ~335K characters with heavy duplication. The two representations are complementary rather than redundant.

The uplift over the Letta block alone is small, and the comparison is not panel-matched. The stacked condition was scored by a 4-judge panel (Haiku, Sonnet, Opus, GPT-4o) because GPT-5.4 returned errors on every call, the same failure mode the main Letta rerun encountered and patched; the comparison columns are 5-judge primary means. The stacked-vs-Letta gap is therefore within what a panel-composition difference could account for. The defensible read is that stacking does not hurt and modestly helps, not that it is clearly additive. This is an exploratory N=3 post-hoc probe, not a headline finding; a GPT-5.4 backfill would make it a panel-matched 5-judge

result. Raw data and scripts: docs/research/v12_1_letta_stacked_experiment_20260513.md, data/experiments/memory_systems/run_letta_stacked_spec.py, and judge_letta_stacked_spec.py.

Appendix H. Glossary

Defined terms used as terms of art throughout the paper. Entries that share substantive meaning are cross-referenced; for each named artifact, **the acceptable short forms and capitalization conventions are explicit**.

5-judge primary panel The locked judge aggregation for headline numbers. Aggregation rule: per-judge per-question score $\rightarrow$ per-judge per-subject mean $\rightarrow$ panel mean across {Haiku 4.5, Sonnet 4.6, Opus 4.6, GPT-4o, GPT-5.4}. See §3.3.3.

7-judge sensitivity panel The 5-judge primary plus Gemini 2.5 Flash and Gemini 2.5 Pro, reported as a sensitivity check. See §3.3.3.

All Facts (C4) The condition where the response model is served the full extracted fact set with no Behavioral Specification. The fact-only baseline for isolating Spec contribution from fact contribution. See §3.2, **Fact set**.

Anchors / Core / Predictions The three layered artifacts comprising a Behavioral Specification. Anchors: short axiomatic claims about how the subject reasons. Core: connects anchors into coherent reasoning patterns. Predictions: derives forward-looking decisions from the core. The unified brief sits above all three as the composed integration. See §3.7, **Unified brief**.

Anchor crossing A change in 5-judge primary mean that moves the score across an integer rubric anchor (1, 2, 3, 4, or 5) between two conditions. Distinguished from a within-anchor shift, which stays inside a single integer band and is a weaker categorical claim. The strongest single-question signal in the rubric. See §3.3.1, **Categorical shift**, **Cross-anchor interpretation rule**, **Multi-anchor crossing**.

Behavioral prediction The operational test for representational accuracy. Given a situation drawn from text the model has never seen, the model generates how the subject would respond; the response is scored against the subject’s own verbatim response on a 1-5 interpretive rubric. See §1.1, §3.3.

Behavioral Specification A static document of approximately 7,000 tokens that extracts and encodes a person’s behavioral patterns. Composed of three layered artifacts (anchors, core, predictions) plus a unified brief. Layered above memory-system retrieval as an interpretive structure. **Acceptable short forms (after first introduction):** *Specification*, *Spec*, *the Spec*. **Capitalization rule:** when referring to the artifact, always Title Case (Behavioral Specification) or capitalized (Specification, Spec). Lowercase *specification* or *spec* should not be used to refer to the artifact in this paper. See §1.1, §3.7, **Unified brief**, **Interpretive layer**.

Categorical shift A change in the model’s response that crosses one or more integer rubric anchors when conditions change, marking a qualitative change in the kind of response produced (e.g., from refusal at anchor 1 to substantive prediction at anchor 4). The unit of effect this paper emphasizes alongside raw rubric-point gain. See **Anchor crossing**, **Cross-anchor interpretation rule**.

Compression / Compression ratio The ratio of the Behavioral Specification’s predictive performance to that of the full source corpus, divided by the ratio of their context costs. The Specification recovers roughly 75% of corpus predictive performance at $\sim 25\times$ less context. See §4.2, §1.3.

Condition codes (C1-C9) Letter codes for the experimental conditions used throughout the

paper. **C5:** No-Context Baseline. **C2a:** Specification only, no facts. **C2c:** Wrong-Spec (a mismatched Behavioral Specification served in place of the correct one). **C4:** All Facts only, no Specification. **C4a:** Facts + Specification (the full pipeline). **C3:** Retrieval + Specification (memory-system layering). **C1:** Retrieval only. **C9:** Raw corpus + Specification. Full table of condition definitions in §3.2.

Cross-anchor interpretation rule A fractional delta between two conditions that crosses an integer rubric anchor (1, 2, 3, 4, or 5) reflects a categorical shift in the underlying response distribution. A delta that stays inside a single integer anchor is a within-category shift and a weaker claim. See §3.3.1, **Anchor crossing**, **Multi-anchor crossing**.

Evidentiary bar An implicit threshold the response model appears to apply before committing to a behavioral prediction: where the served context supplies grounding evidence that clears the bar, the model produces a substantive answer; where it does not, the model abstains. Visible in the cross-system patterns in §4.4.3 and §4.4.4. The matched Behavioral Specification can both supply the interpretive scaffolding that clears the bar (Pattern 1, §4.4.3) and raise the bar in a subject-consistent way (Pattern 3 and the Keckley case study in §4.4.4). See §1.3, §4.4.3, §4.4.4.

Fact set The extracted structured triples (subject-predicate-object) representing a person’s documented facts, used as the input to the C4 / All Facts condition. The extraction step of the pipeline produces this artifact from the raw corpus. See §3.7, **All Facts**.

Held-out passage A passage drawn from the holdout half of a subject’s corpus (the half the model has never seen during Specification authoring). Held-out passages serve as the verbatim ground-truth text against which response-model predictions are scored. The 50/50 train/holdout split is per-subject. See §3.3, **Behavioral prediction**.

Hedging / hedging reduction A response in which the model declines to commit to a substantive prediction, either with explicit refusal language (“I cannot confirm”) or with low-confidence qualification. The matched Behavioral Specification collapses baseline hedging from approximately 41% to near zero under the correct-Spec conditions; the wrong-Spec condition produces hedging at approximately 60%. See §4.3, **Refusal (abstention)**.

High-baseline subject A subject whose No-Context Baseline score (C5) is above 3.0 on the 1-5 rubric, indicating substantial pretraining representation of that person. Franklin (C5 = 3.77) is the high-baseline reference in this paper; he is not included in any N=14 inferential statistic. Operational threshold: $C5 > 3.0$. See §4.1.2, §3.4.1, **Low-baseline subject**, **Mid-baseline subject**.

Interpretation In this paper, the human-side property: the way a specific person processes facts and experiences into judgments, decisions, and reactions. The property the Behavioral Specification is designed to mirror. See §1.1, **Interpretive frame**, **Interpretive layer**.

Interpretive frame The cognitive lens through which a specific person interprets situations. The human-side property. Distinguished from **Interpretive layer**, which is the AI-side artifact (the Behavioral Specification) that mirrors a person’s interpretive frame.

Interpretive layer The Behavioral Specification’s structural function: an external, encoded interpretation that sits between facts and the response model and supplies the interpretive structure that facts alone do not carry. Used throughout the paper as both a structural and operational term. See §1.1, §5, **Behavioral Specification**.

Interpretive structure The arrangement of behavioral anchors, core patterns, and predictions that encodes a person’s interpretive frame in the Specification document. The internal organization of the **Interpretive layer**.

Interpretive supply / over-theorization / principled refusal (Patterns 1 / 2 / 3)

Three patterns of Specification-retrieval interaction observed in §4.4.3. **Pattern 1 (interpre-**

- tive supply**): retrieval underdetermines the answer; the Specification supplies interpretive scaffolding and the response improves. **Pattern 2 (over-theorization)**: retrieval already determines the answer; the Specification adds incorrect generalization and the response degrades. **Pattern 3 (principled refusal)**: the Specification’s axioms trigger refusal where retrieval would have produced a substantive answer. The aggregate Δ on any memory system is the balance of these three patterns across its question battery. A fourth pattern, **Pattern 4 (confidence backfire)**, is specific to the Letta stateful-agent architecture and is developed in Appendix G.5.
- Low-baseline subject** A subject whose No-Context Baseline score (C5) is at or below 2.0 on the 1-5 rubric, indicating that the response model has insignificant pretraining representation of the person. Nine of the 14 main-study subjects are low-baseline. The **population of relevance** for AI personalization. See §3.4, §4.1.
- Memory-system layering** Serving the Behavioral Specification as additional context on top of a commercial memory system’s retrieved output (Mem0, Letta, Zep, Supermemory). The Specification produces a net-positive aggregate Δ on three of four commercial systems tested, with per-question patterns varying by retrieval architecture. See §4.4, **Interpretive supply / over-theorization / principled refusal**.
- Mid-baseline subject** A subject whose No-Context Baseline score (C5) is between 2.0 and 3.0 on the 1-5 rubric. Per-subject Δ estimates for mid-baseline subjects carry sign uncertainty within the pipeline run-to-run variance band; see §6.3 for the Augustine sign-flip disclosure. See **Low-baseline subject**, **High-baseline subject**.
- Multi-anchor crossing** A single question whose 5-judge primary mean shifts across two or more integer rubric anchors when the condition changes. Crossings can span two anchors (e.g., 1 $\rightarrow$ 3) or, more rarely, three anchors (e.g., 1 $\rightarrow$ 4 or 2 $\rightarrow$ 5). The strongest categorical signal the rubric detects. See §3.3.1, §4.2.1, **Anchor crossing**, **Categorical shift**.
- No-Context Baseline (C5)** The condition in which the response model is given no external information about the subject. Serves as the observable proxy for pretraining coverage and as the per-subject baseline against which all Spec-effect Δ ’s are measured. See §3.2, §3.4, **Condition codes**.
- Per-subject C4a mean (≈ 2.44)** The cross-subject mean of per-subject C4a scores on the 14-subject main study with the current question mix. Used as a math note inside the gradient explanation (§4.1) to clarify why the change-score slope (-0.96) is partly mechanical: the level-regression slope of C4a on C5 is $+0.04$ (essentially flat). NOT a per-response ceiling — per-question variance is wide (§4.1.1 multi-anchor crossings range across the 1-5 scale). See §4.1, §4.6.3, Appendix B.7.
- Personalization (this paper’s sense)** Representing the interpretive layer that sits beneath stated preferences and biographical facts: how a specific person organizes experience, what they treat as evidence, what reasoning patterns they apply across new situations. Distinguished from surface-level responsiveness to stated preferences (dietary restrictions, communication style) or stored facts about the user (location, occupation, history), which are downstream artifacts of the interpretive layer rather than the layer itself. See §2 lede.
- Population of relevance** The population the paper’s findings are designed to address: low-baseline subjects whose interpretive patterns are not already represented in pretraining. The paper argues this is the typical AI user, since most users’ reasoning patterns are not in any public training corpus regardless of how much demographic information about them is. See §1.4, §3.4, §5.2, **Low-baseline subject**.
- Refusal (abstention)** A response in which the model declines to predict because the available context does not support a prediction. Distinct from a substantively wrong prediction; under

the current rubric both score at the lowest anchor (1). Detection in the validity-audit script uses phrase patterns (“no specific information,” “I cannot confirm,” “would need additional context”). See §3.3.6, **Hedging**.

Representational accuracy The AI-side property: how faithfully a model’s internal representation of a specific person captures that person’s interpretive patterns. Operationalized via behavioral prediction on held-out reasoning situations. See §1.1, §3.1, **Behavioral prediction**.

Specification-effect claim When a Behavioral Specification is served to the response model as context, the model’s responses shift in the direction of the subject’s demonstrated behavioral patterns, and that shift registers as a measured increase in representational accuracy against held-out passages from the same subject. The claim is directional, not a claim of new model capability or absolute correctness. See §3.3.4.

Tier 1 / Tier 2 Tier 1 is the main study: Haiku 4.5 response model across all 14 subjects, every condition, Haiku-generated batteries. Tier 2 is the cross-provider directional probe: Sonnet 4.6 and Gemini 2.5 Pro response models on 3 subjects (Ebers, Yung Wing, Zitkala-Ša) with GPT-5.4-regenerated batteries. See §3.6, §4.6.1.

Unified brief The composed prose document that integrates the three Behavioral Specification layers (anchors, core, predictions) into a continuous synthesis. Approximately 5,000 words; serves as both the integration layer that weaves the three layers together and a coherent first pass for human readers. The Behavioral Specification served to the response model in C2a / C4a / C3 conditions is the three layers plus the unified brief together. See §3.7, **Behavioral Specification, Anchors / Core / Predictions**.

Wrong-Spec control A deliberately mismatched Behavioral Specification served in place of the correct one. Two variants: **v1** (adversarial fixed derangement maximizing cultural and temporal distance; aggregate $\Delta -0.25$) and **v2** (seed-fixed random derangement; aggregate $\Delta +0.15$). See §1.3, §3.2, §4.3.